Computer Intensive Statistics and Applications
Chapter 1 and 2: Random Variable Generation

Shaobo Jin

Department of Mathematics

Statistical Model

In statistical inference,

- the possible observations are from a sample space $\mathcal{X}$,
- observation x is a realization of a random variable/vector X.

Statistical inference is concerned with methods of using the observations to obtain information about the distribution of X.

A statistical model is a class of possible probability measures $\mathcal{P}$ on the sample space $\mathcal{X}$, parameterized by parameter(s) θ:

$$\mathcal{P} = \{\mathrm{P}_\theta : \theta \in \Theta\},$$

where Θ is the parameter space.

Simulate Data

A sample $X = (X_1, ..., X_n)$ is a collection of independent random variables where X_i is distributed according to a distribution $\mathrm{P}_{i,\theta}$.

- We can view data as "simulated" from P_θ and we would like to know how data are generated by P_θ.
- This is similar to conducting many experiments in order to collect data.

Once we can simulate data from P_θ, we will be able to perform various simulation-based inference:

- estimate unknown quantity (e.g., Monte Carlo methods),
- quantify uncertainty (e.g., bootstrap).

Some Examples

Some examples that you should be able to solve by the end of the course:

- approximate expectations $\mathrm{E}[X] = \int x f(x)\, dx$, if the closed form expression is difficult/impossible to find,
- approximate integrals $\int f(x, y)\, dx$,
- simulate a stochastic process,
- approximate distributions of functions of random variables $h(X)$,

among others.

The focus of our course is neither 100% programming nor 100% math. We will show you

- how simulations are performed,
- why simulations work or do not work.

Uniform Distribution

We start with the fundamental component: random number generation.

Definition (Definition 1.2)

A uniform pseudo-random number generator is an algorithm which, starting from an initial value u_0 and a transformation D, produces a sequence with $u_i = D^i(u_0)$ in $[0, 1]$. For all n, $(u_1, ..., u_n)$ reproduce the behavior of an i.i.d. sample of uniform random variables when compared through a set of tests.

Congruential Generators

To generate Uniform $[0, 1]$, we can use the classic congruential generator (a, b, M) on $\{0, ..., M-1\}$.

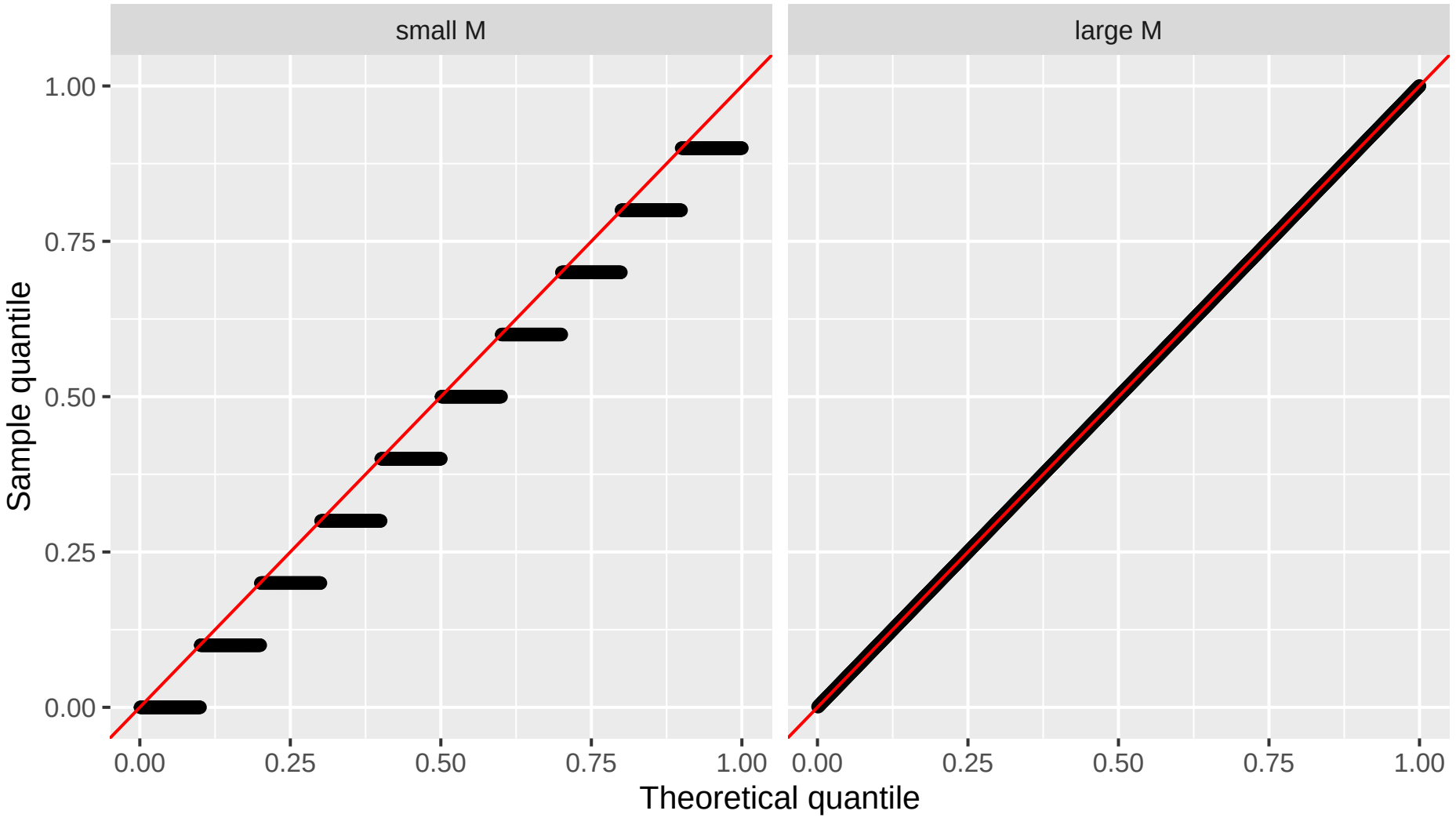
Algorithm 1: Generate pseudo random numbers from U $[0, 1]$

1 Choose a large $M > 0$ and (a, b) from $\{0, ..., M-1\}$;
2 Choose a seed x_0 ;
3 **for** $i = 1$ *in* $1 : n$ **do**
4 $\quad x_i = (ax_{i-1} + b) \mod M$;
5 **end**
6 Dividing the sequence by M gives a sequence that are regarded as random numbers from Uniform $[0, 1]$.

The operator $y = z \mod M$ means that, for an integer z, there is an integer k such that $y = kM + z$ but $0 \leq y < M$.

- If z is positive, then it is just the remainder.

Illustration of QQ plot



Expectation and Variance

Suppose that we generated $(x_1, ..., x_M)$ from a congruential generator (a, b, M). Then, the ordered outcome must be

$$\left[0 \quad \frac{1}{M} \quad \cdots \quad \frac{M-1}{M}\right].$$

The sample mean is

$$\frac{1}{M}\sum_{i=1}^{M} x_i \quad = \quad \frac{1}{M}\sum_{j=0}^{M-1} \frac{j}{M} = \frac{M-1}{2M} \to \frac{1}{2},$$

and the sample variance with denominator M is

$$\frac{1}{M}\sum_{i=1}^{M} x_i^2 - \left(\frac{1}{M}\sum_{i=1}^{M} x_i\right)^2 \quad = \quad \frac{M^2-1}{12M^2} \to \frac{1}{12},$$

as M increases.

Quantile

In this course, we will mostly assume that we can generate Uniform $[0, 1]$ and study how we can generate random numbers of other distributions.

Definition (Definition 1.1)

For a distribution function F on $\mathbb{R}$, the generalized inverse F^- of F is the function defined by

$$F^-(u) = \inf\{x;\ F(x) \geq u\} \quad \text{for } 0 \leq u \leq 1.$$

The generalized inverse is often called the quantile function, denoted by Q.

Example

Let $X \sim \text{Exp}(\lambda)$ with mean $1/\lambda$. Find F^-.

Quantile Method

Lemma (Lemma 1.1)

If $U \sim$ *Uniform* $[0,1]$, *then the random variable* $F^{-}(U)$ *has distribution* F.

This lemma means that as long as we can generate $U \sim \text{Uniform}[0,1]$, we can obtain $X \sim F$ by letting $X = F^{-}(U)$.

Example

For exponential distribution, we can generate data via

$$X = -\frac{\log(1-U)}{\lambda}.$$

Transformation Method

The quantile method is a special case of the transformation method that means that, if we are interested in Y and we know that $Y = h(X)$, then we can generate random numbers for X and $h(X)$ will be the random numbers for Y.

Example

Box-Muller algorithm for normal distribution: $U_1 \sim \text{Uniform}[0,1]$, $U_2 \sim \text{Uniform}[0,1]$, and U_1 is independent of U_2. Then,

$$\begin{aligned} X_1 &= \sqrt{-2\log(U_1)}\cos(2\pi U_2) \sim N(0,1), \\ X_2 &= \sqrt{-2\log(U_1)}\sin(2\pi U_2) \sim N(0,1), \end{aligned}$$

and X_1 is independent of X_2.

Conditional Method

Suppose that we can partition X into $X = (X_1, X_2, ..., X_d)$ such that

$$p(x_1, ..., x_d) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_1, x_2) \cdots p(x_d \mid x_1, ..., x_{d-1}).$$

Then, we can generate X_1, $X_2 \mid X_1$, $X_3 \mid (X_1, X_2)$, ..., $X_d \mid (X_1, ..., X_{d-1})$ in order to generate X.

Example

Suppose that $Y \sim \text{Bernoulli}(\theta)$, where $\log\left(\frac{\theta}{1-\theta}\right) = a + bX$ with $X \sim N(0,1)$. To generate (X, Y), we generate

$$\begin{aligned} X &\sim N(0,1), \\ Y \mid X = x &\sim \text{Bernoulli}\left(\frac{\exp\{a+bx\}}{1+\exp\{a+bx\}}\right). \end{aligned}$$

Dominance

Suppose that we want to generate a sample with target density $p(x)$ but we know that it is easier to generate random numbers from another instrumental density $g(x)$. Let M be a constant such that

$$p(x) \leq Mg(x), \quad \text{for all } x.$$

Example

Let $X \sim \text{Beta}(a, b)$ with density $p(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1}$, where $x \in (0, 1)$. We focus on $a > 1$ and $b > 1$, and let $g(x)$ be the uniform density.

Rejection Algorithm

Algorithm 2: Rejection algorithm

Choose a seed (omitted in all following algorithms) ;
for $i = 1$ *in* $1 : n$ **do**
  while *new trial is needed* **do**
    Draw $X = x$ from $g(\cdot)$;
    Compute the ratio $r(x) = \frac{p(x)}{Mg(x)}$;
    Draw $U = u$ from Uniform $[0, 1]$;
    If $u > r(x)$, start a new trial. Otherwise, let $x^{(t+1)} = x$;
  end
end

The algorithm works even when x is multidimensional.

But why does the rejection algorithm work?

Effect of M

- The probability of accepting a candidate is

$$\mathrm{P}\left(U \leq r\left(X\right)\right) \quad = \quad \frac{1}{M}.$$

  For large M, the proportion of useless draws that are discarded is large.
- But if M is not big enough, then we cannot guarantee $p\left(x\right) \leq Mg\left(x\right)$ for all x.

A Variant

Sometimes it is not easy to get the exact expression of $p(x)$, but it is easy to get $f(x)$ where $p(x) = cf(x)$.

- That is, the normalizing constant c is often hard to obtain, such as in a posterior distribution.

Instead of using $p(x)$ (hard to get) in the accept-reject algorithm, we can use $f(x)$ (easy to get) instead.

Algorithm 3: Rejection algorithm

```
for i = 1 in 1 : n do
    while new trial is needed do
        Draw X = x from g(·) ;
        Compute the ratio r(x) = f(x) / (M g(x)) ;
        Draw U = u from Uniform[0, 1] ;
        If u > r(x), start a new trial. Otherwise, let x^(t+1) = x ;
    end
end
```

Another Variant: Envelope Accept-Reject Method

Suppose that $p(x)$ can be computed by can be time consuming. We can find an extra lower bound $g_l(x)$ such that

$$g_l(x) \leq p(x) \leq Mg(x), \quad \text{for all } x.$$

- In the accept-reject algorithm, we accept the candidate value if

$$u \leq \frac{p(x)}{Mg(x)}.$$

- In the envelop accept-reject algorithm, we first check if

$$u \leq \frac{g_l(x)}{Mg(x)}.$$

Envelope Accept-Reject Method

Algorithm 4: Envelope accept-reject algorithm

for $i = 1$ *in* $1 : n$ **do**
  while *new trial is needed* **do**
    Draw $X = x$ from $g(\cdot)$;
    Draw $U \sim \text{Uniform}[0, 1]$;
    if $U \leq \frac{g_l(x)}{Mg(x)}$ **then**
      Accept X ;
    else
      if $U \leq \frac{p(x)}{Mg(x)}$ **then**
        Accept X;
      else
        Start a new trial ;
      end
    end
  end
end

Markov Property and Markov Chain

Another idea of generating random number is through a Markov chain.

Definition

Let $X_i \in \Omega$ for all i, where the set Ω is the state space. The random variables satisfy the Markov property if

$$\mathrm{P}\left(X_{i+1} \in A \mid X_j = x_j, 0 \leq j \leq i\right) \quad = \quad \mathrm{P}\left(X_{i+1} \in A \mid X_i = x_i\right).$$

Definition

A Markov chain is a sequence of random variables $X_i \in \Omega$ that satisfy the Markov property.

Transition Kernel

Consider a sequence of discrete random variables $\{X_n\}$. The Markov property means that

$$\mathrm{P}\left(X_n = y \mid X_{n-1} = x, X_{n-2}, \cdots X_0\right) \quad = \quad \mathrm{P}\left(X_n = y \mid X_{n-1} = x\right).$$

In this case, a transition kernel is a matrix $\boldsymbol{K}$ with elements $\mathrm{P}\left(X_n = y \mid X_{n-1} = x\right)$.

Consider a sequence of continuous random variables $\{X_n\}$. The Markov property means that

$$\mathrm{P}\left(X_n \in A \mid X_{n-1} = x, X_{n-2}, \cdots X_0\right) \quad = \quad \int_{y \in A} K\left(x, y\right) dy,$$

$$p\left(X_n = y \mid X_{n-1} = x, X_{n-2}, \cdots X_0\right) \quad = \quad p\left(X_n = y \mid X_{n-1} = x\right) = K\left(x, y\right),$$

where the transition kernel $K\left(x, y\right)$ is the conditional density of Y given $X = x$.

Discrete Marginal Distribution: Time 1

Take the sequence of discrete random variables $\{X_n\}$ as an example. Suppose that each X_n takes values from 1, ..., M. Then,

$$\mathrm{P}\left(X_1 = j\right) \quad = \quad \sum_{i=1}^{M} \mathrm{P}\left(X_0 = i\right) \mathrm{P}\left(X_1 = j \mid X_0 = i\right).$$

Expressing it as a matrix, we have

$$\boldsymbol{p}_1 \quad = \quad \boldsymbol{p}_0 \boldsymbol{K},$$

where $\boldsymbol{p}_k$ is a row vector with tth element $\mathrm{P}\left(X_k = t\right)$, and the transition kernel $\boldsymbol{K}$ is a $M \times M$ matrix with (i, j)th element $\mathrm{P}\left(X_1 = j \mid X_0 = i\right)$.

Discrete Marginal Distribution: Time n

If we consider a homogenous Markov chain, where

$$\mathrm{P}\left(X_i = y \mid X_{i-1} = x\right) \quad = \quad \mathrm{P}\left(X_1 = y \mid X_0 = x\right)$$

for all i, then

1. At time $t = 2$,

$$\begin{aligned} \mathrm{P}\left(X_2 = j\right) \quad &= \quad \sum_{i=1}^{M} \mathrm{P}\left(X_1 = i\right) \mathrm{P}\left(X_2 = j \mid X_1 = i\right) \\ \Rightarrow \quad \boldsymbol{p}_2 \quad &= \quad \boldsymbol{p}_1 \boldsymbol{K} = \boldsymbol{p}_0 \boldsymbol{K}^2. \end{aligned}$$

2. In general,

$$\begin{aligned} \mathrm{P}\left(X_n = j\right) \quad &= \quad \sum_{i=1}^{M} \mathrm{P}\left(X_{n-1} = i\right) \mathrm{P}\left(X_n = j \mid X_{n-1} = i\right) \\ \Rightarrow \quad \boldsymbol{p}_n \quad &= \quad \boldsymbol{p}_{n-1} \boldsymbol{K} = \boldsymbol{p}_0 \boldsymbol{K}^n. \end{aligned}$$

Stationary Distribution

Definition

The distribution π on Ω is a stationary distribution (or invariant distribution) of the Markov chain with the transition kernel K, if

$$\pi(y) = \sum_{x \in \Omega} \pi(x) K(x, y), \quad \text{discrete case,}$$

where $\pi(x) = \mathrm{P}(X = x)$, or

$$\pi(y) = \int_{x \in \Omega} \pi(x) K(x, y) \, dx, \quad \text{continuous case,}$$

where $\pi(x) = p(x)$.

- The stationary distribution means that if the initial state $X_0 \sim \pi$, then $X_n \sim \pi$ for all $n \geq 0$, the same distribution.

Stationary Distribution: Existence

The stationary distribution does not always exists. Our goal in MCMC is to design the algorithm such that the stationary distribution is the distribution that we want to sample random numbers from.

Example

1. The stationary distribution of $\boldsymbol{K} = \begin{bmatrix} 0.5 & 0.5 \\ 0.5 & 0.5 \end{bmatrix}$ is $\begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$.
2. But if we let $\Omega = \mathbb{Z}$ and consider

$$\begin{aligned} \mathrm{P}\left(X_n = i - 1 \mid X_{n-1} = i\right) &= \frac{1}{4}, \\ \mathrm{P}\left(X_n = i + 1 \mid X_{n-1} = i\right) &= \frac{3}{4}, \end{aligned}$$

then the stationary distribution does not exist.

Irreducible Markov Chain

From now on, we consider the finite-state Markov chain for simplicity.

- The states x and y communicate if there exists an integer $n \geq 0$ such that $\mathrm{P}(X_n = x \mid X_0 = y) > 0$ and $\mathrm{P}(X_n = y \mid X_0 = x) > 0$.
- The transition matrix $\boldsymbol{K}$ is irreducible if x communicates with y whenever $x, y \in \Omega$.

Example

$$\boldsymbol{K} = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 \\ 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0.5 & 0.5 \end{bmatrix}$$ is reducible and $$\boldsymbol{K} = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0.25 & 0.25 & 0 \\ 0 & 0 & 0.5 & 0.5 \\ 0.25 & 0 & 0.25 & 0.5 \end{bmatrix}$$ is irreducible.

Periodic

For a transition matrix, the state w is periodic with period $t > 1$ if

1. $\mathrm{P}(X_n = w \mid X_0 = w) > 0$ only holds for $n = tm$ with integer $m > 0$ and
2. t is the largest number of $\{2, 3, ...\}$ for which 1) holds.

The state w is aperiodic if $\mathrm{P}(X_n = w \mid X_0 = w) > 0$ for all n.

Example

$$\boldsymbol{K} = \begin{bmatrix} 0 & 0 & 0.5 & 0.5 \\ 0 & 0 & 0.5 & 0.5 \\ 0.5 & 0.5 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 \end{bmatrix}$$ has period $t = 2$.

Long-Run Property: Finite-State Markov Chain

Theorem

Consider a finite-state Markov chain. Suppose that the transition matrix $\boldsymbol{K}$ *is irreducible, aperiodic, and has stationary distribution* π. *Then, for all starting state* $w_0 \in \Omega$,

$$\lim_{n\to\infty} P\left(X_n = w \mid X_0 = w_0\right) \quad = \quad \pi\left(w\right).$$

- The stationary distribution is reached after large enough iterations. When we simulate random numbers using a Markov chain, we need to discard the early iterations. The discarded ones are called the burn-in period.
- Samples obtained after the burn-in period can be considered as samples from π.

Choose the Transition Kernel

Our goal is to simulate data from some distribution with density $p(x)$. Now the question becomes how to choose K such that the stationary distribution is $\pi(x) = p(x)$.

Fact

If $p(x)$ and $K(x, y)$ satisfies the detailed balance condition, i.e,

$$K(x, y)\, p(x) \quad = \quad K(y, x)\, p(y),$$

for any $x, y \in \Omega$, then $p(x)$ is the stationary distribution of the Markov chain with the transition kernel K.

Proposal Distribution

When we simulate random numbers from a Markov chain, we need a proposal distribution

$$T(x, y) = p(Y = y \mid X = x).$$

Find a proposal distribution $T(x, y)$ that satisfies the detailed balance condition is difficult.

- So with probability $A(x, y)$ we let $X_{n+1} = y$ (accept), and probability $1 - A(x, y)$ we let $X_{n+1} = x$ (reject).
- For $X_{n+1} \neq x$, the transition is

$$K(x, y) = T(x, y) A(x, y).$$

Hence, we should seek A such that the detailed balance condition is fulfilled.

Detailed Balance: Symmetric Proposal

Start with the case where the proposal distribution is symmetric, i.e., $T(x, y) = T(y, x)$.

Example

$Y \mid X = x \sim N\left(x, \sigma^2\right)$ is symmetric, since

$$T(x, y) = \frac{1}{\sqrt{2\sigma^2}} \exp\left\{-\frac{(x-y)^2}{2\sigma^2}\right\}$$

The transition kernel is

$$K(x, y) = T(x, y) A(x, y) + [1 - A(x, y)] \delta_x(y),$$

where $\delta_x(y) = 1$ if $x = y$, and 0 otherwise. For $y \neq x$, the detailed balance condition reduces to

$$\pi(x) A(x, y) = \pi(y) A(y, x).$$

Deriving $A(x, y)$: Symmetric Proposal

The detailed balance condition is fulfilled, if we choose

$$A(x, y) = \lambda(x, y)\,\pi(y), \quad A(y, x) = \lambda(x, y)\,\pi(x),$$

such that $\lambda(x, y) = \lambda(y, x)$. Since $A(\cdot, \cdot)$ is a probability, we must have

$$\lambda(x, y) \leq \min\left\{\frac{1}{\pi(y)}, \frac{1}{\pi(x)}\right\}.$$

The maximum of $A(\cdot, \cdot)$ with respect to λ is attained at

$$\lambda(x, y) = \min\left\{\frac{1}{\pi(y)}, \frac{1}{\pi(x)}\right\}.$$

Hence,

$$A(x, y) = \lambda(x, y)\,\pi(y) = \min\left\{1, \frac{\pi(y)}{\pi(x)}\right\}.$$

Metropolis Algorithm

Algorithm 5: Metropolis Algorithm

Choose an initial state $x^{(0)}$;

for $t = 1$ *in* $1 : n$ **do**

  Sample a candidate y from $T\left(x^{(t)}, y\right)$;

  Calculate the ratio $R\left(x^{(t)}, y\right) = \frac{\pi(y)}{\pi\left(x^{(t)}\right)}$;

  Draw $U \sim \mathrm{U}[0,1]$;

  Update

$$x^{(t+1)} = \begin{cases} y, & \text{if } U \leq R\left(x^{(t)}, y\right), \\ x^{(t)}, & \text{otherwise.} \end{cases}$$

end

Metropolis Algorithm: Example

Since we have $R\left(x^{(t)}, y\right) = \frac{\pi(y)}{\pi\left(x^{(t)}\right)}$ in the Metropolis algorithm, we only need to know $\pi\left(\right)$ up to a normalizing constant.

Example

Suppose that we want to sample random numbers from a distribution with density

$$p(x) \quad = \quad c \exp\left(-x^2\right)\left[2 + \sin(5x) + \sin(2x)\right],$$

where c is a normalizing constant.

Metropolis-Hastings Algorithm

The Metropolis-Hastings algorithm allows asymmetric proposal distributions, as long as $T(x, y) > 0$ if and only if $T(y, x) > 0$.

Algorithm 6: Metropolis-Hastings Algorithm

Choose an initial state $x^{(0)}$;
for $t = 1$ *in* $1 : n$ **do**
    Sample a candidate y from $T\left(x^{(t)}, y\right)$;
    Calculate the ratio $R\left(x^{(t)}, y\right) = \frac{\pi(y)T\left(y, x^{(t)}\right)}{\pi\left(x^{(t)}\right)T\left(x^{(t)}, y\right)}$;
    Draw $U \sim \mathrm{U}[0, 1]$;
    Update

$$x^{(t+1)} = \begin{cases} y, & \text{if } U \leq R\left(x^{(t)}, y\right), \\ x^{(t)}, & \text{otherwise.} \end{cases}$$

end

Examples of MCMC Algorithms

Many different MCMC algorithms differ mainly in how the candidate y is sampled.

- In the random-walk Metropolis algorithm, $y = x^{(t)} + \epsilon$, where ϵ is sampled from some distribution, e.g., Uniform $[-a, a]$, Normal, etc.
- In multiplicative random walk, $y = x^{(t)}\epsilon$.
- In independence sampler, y is sampled from $g(y)$ that does not depend on $x^{(t)}$.
- In hit and run, we first sample a direction $\rho^{(t)}$ and then a random length $\lambda^{(t)}$. Then $y = x^{(t)} + \lambda^{(t)}\rho^{(t)}$.
- The Langevin Metropolis-Hastings algorithm explores the shape of the target distribution by $y = x^{(t)} + d^{(t)} + \sigma\epsilon$, where $\epsilon \sim N(0, I)$ and

$$d^{(t)} = \frac{\sigma^2}{2}\frac{\partial \log \pi\left(x^{(t)}\right)}{\partial x}.$$

Gibbs Sampler: Conditioning

It can be the case that it is not easy to sample from the joint distribution of $X \in \mathbb{R}^d$. But it is easy to sample from the conditional distributions.

- Suppose that $X = (X_1, ..., X_p)$, where $X_i \in \mathbb{R}^{d_i}$.
- Let $\pi_{i|-i}(X_i \mid X_{-i})$ be the conditional distribution of X_i given X_{-i}, where $X_{-i} = \begin{pmatrix} X_1 & \cdots & X_{i-1} & X_{i+1} & \cdots & X_p \end{pmatrix}$.

Algorithm 7: Gibbs Sampler

Choose an initial state $x^{(0)}$;
for $t = 1$ *in* $1 : n$ **do**
  for $i = 1$ *in* $1 : p$ **do**
    Draw $X_i^{(t+1)} \sim \pi_{i|-i}\left(X_i \mid X_1^{(t+1)}, ..., X_{i-1}^{(t+1)}, X_{i+1}^{(t)}, ..., X_p^{(t)}\right)$;
  end
end

Gibbs Sampler: Example

Example

Suppose that we can generate data from $N\left(\mu, \sigma^2\right)$. For a bivariate normal distribution $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}\right)$, we know that

$$X_2 \mid X_1 = x_1 \quad \sim \quad N\left(\mu_2 + \rho\sigma_1^{-1}\sigma_2\left(x_1 - \mu_1\right), \sigma_2^2\left(1 - \rho^2\right)\right).$$

Sample random numbers from the bivariate normal with Gibbs sampler.

Why Does Gibbs Sampler Work?

The transition kernel $K(x, y)$ is

$$\pi_{1|-1}(y_1 \mid x_2, ..., x_p)\, \pi_{2|-2}(y_2 \mid y_1, x_3, ..., x_p) \cdots \pi_{p|-p}(y_p \mid y_1, ..., y_{p-1}).$$

If we consider the continuous random variables, the transition kernel satisfies

$$\pi(y) = \int K(x, y)\, \pi(x)\, dx,$$

where $\pi(x)$ is the marginal distribution. Here $\pi()$ is not treated as a generic symbol.

Gibbs Sampler: Another Example

Gibbs sampler simulates random number from "correct" conditional distribution. But it can work poorly if the correlations between components are high, especially when the conditional distribution has a low variance so that the step size becomes small. The same issue can occur to Metropolis-Hastings as well.

Example

Suppose that we can generate data from $N\left(\mu, \sigma^2\right)$. For a bivariate normal distribution $\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N\left(\begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}, \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}\right)$, we know that

$$X_2 \mid X_1 = x_1 \quad \sim \quad N\left(\mu_2 + \rho\sigma_1^{-1}\sigma_2\left(x_1 - \mu_1\right), \sigma_2^2\left(1-\rho^2\right)\right).$$

Sample random numbers from the bivariate normal with Gibbs sampler when $\rho = 0.99$.

Burn-In Period

The stationary distribution is reached after large enough iterations.

- If the iterations have not proceeded long enough, the simulated numbers may be unrepresentative of the target distribution.

To diminish the influence of the starting values, we can discard the early simulations, known as the burn-in.

- There is no golden standard on how long the burn-in period should be.

Serial Correlation

It is obvious that X_{n+1} and X_n are not independent draws. Inference from autocorrelated draws is generally less precise than from the same number of independent draws.

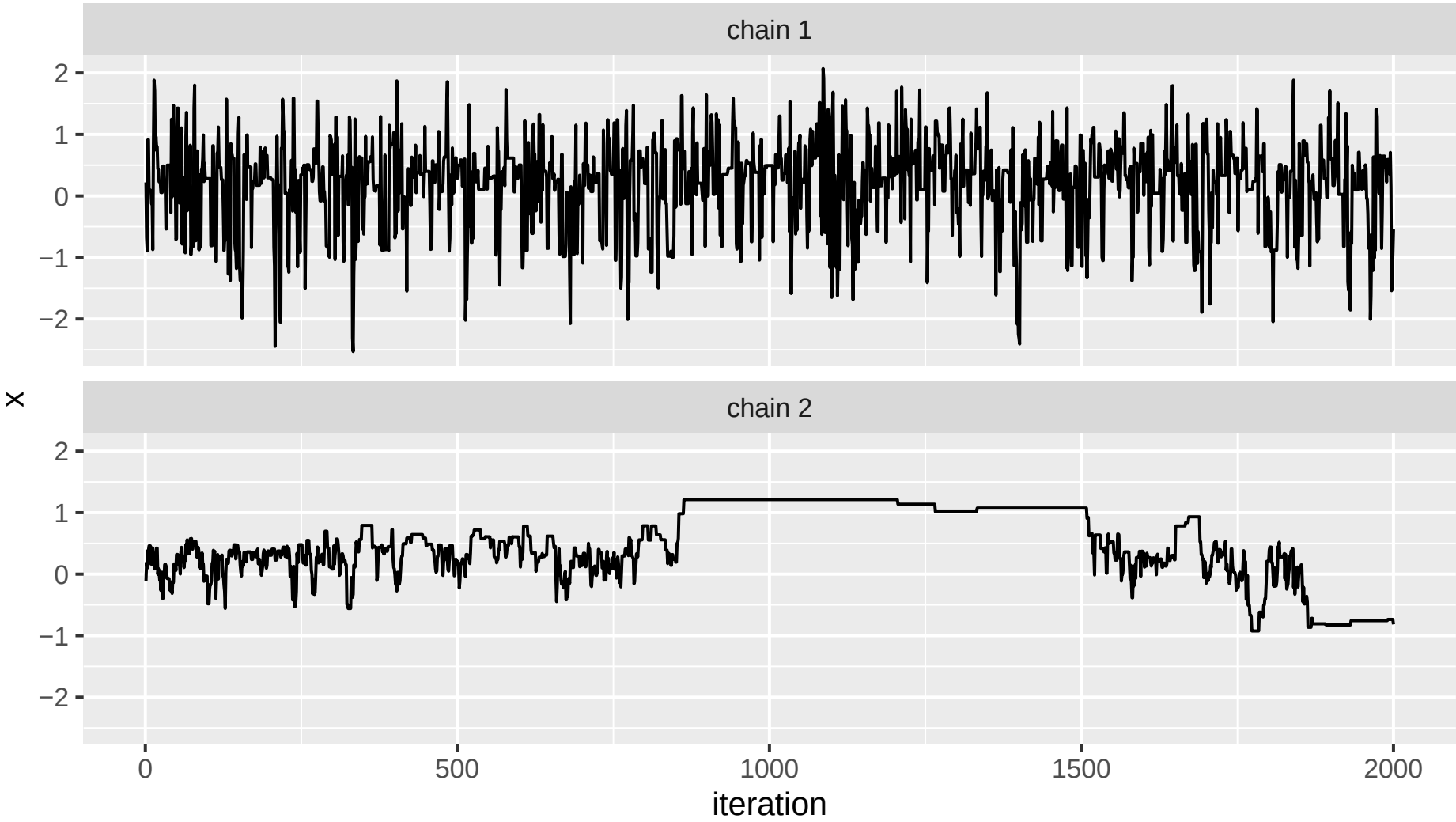
- However, such serial correlation is not necessarily a problem.
- Remember that, at convergence, we reach the stationary distribution.

Some prefer thinning the sequence by only keeping every kth draw from a sequence in order to reduce serial correlation.

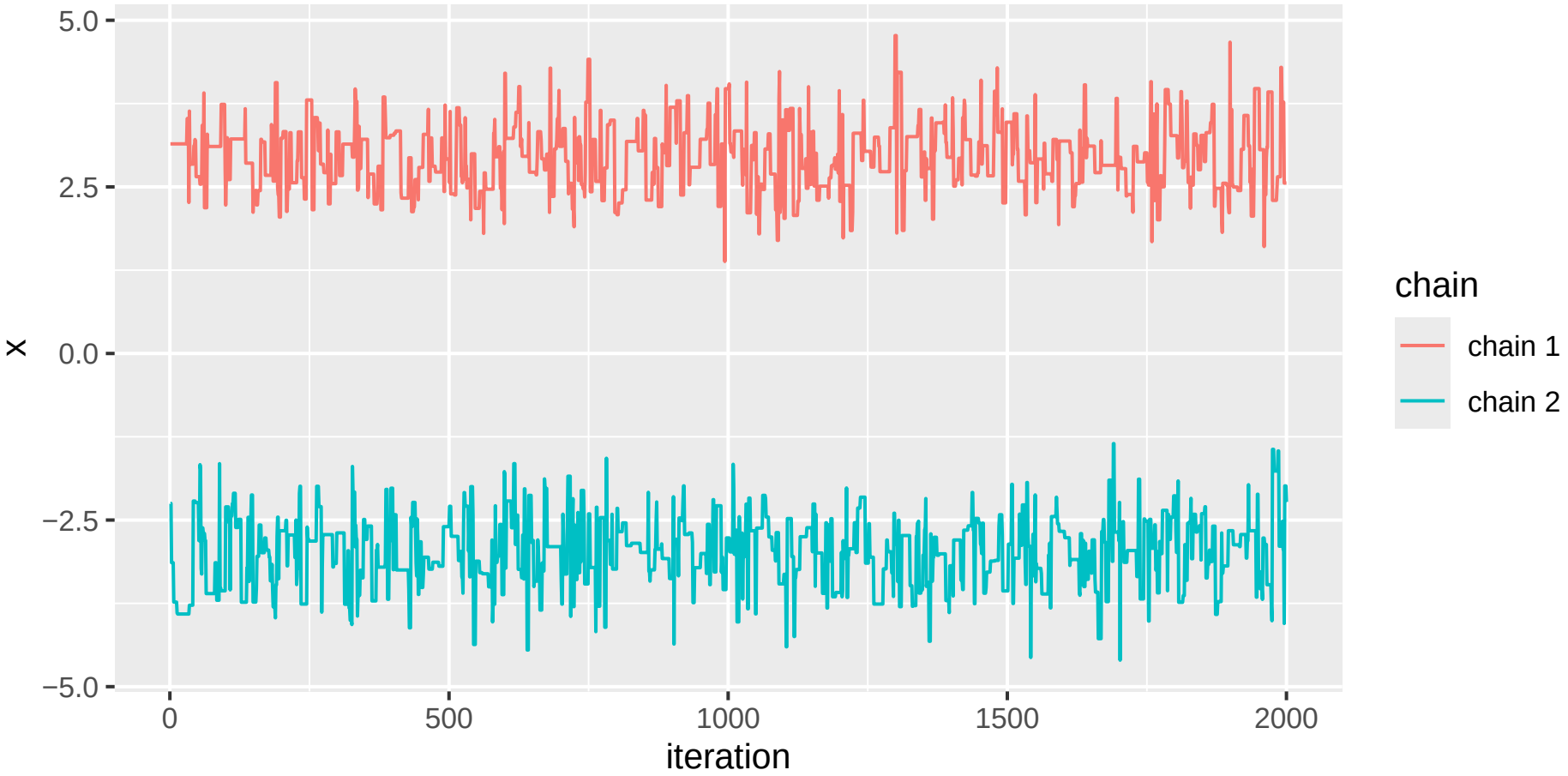
- But whether or not the Markov chain is thinned, it can be used for inferences, provided that it has reached convergence.

Mixing

We want the Markov chain to show good mixing.



Several Markov Chains From Same Model



One suggestion is to generate several independent Markov chains, starting from widely separated places.

Gelman-Rubin $\hat{R}$ Statistic: Variation

One way to assess convergence is the Gelman-Rubin $\hat{R}$ statistic. Suppose that we have simulated m chains each with n iterations after the burn-in period. Say we have a univariate statistic $y_{ij} = f(x_{ij})$, where x_{ij} is the ith value in the jth chain.

- The variation within the chains is measured by

$$W = \frac{1}{m}\sum_{j=1}^{m}\left[\frac{1}{n-1}\sum_{i=1}^{n}(y_{ij}-\bar{y}_{\cdot j})^2\right],$$

  where $\bar{y}_{\cdot j}$ is the average of $\{y_{ij}\}$.
- The variation between the chains is measured by

$$B = \frac{n}{m-1}\sum_{j=1}^{m}(\bar{y}_{\cdot j}-\bar{y}_{\cdot\cdot})^2,$$

  where $\bar{y}_{\cdot\cdot}$ is the average of all $\bar{y}_{\cdot j}$.

Gelman-Rubin $\hat{R}$ Statistic: Expression

Consider

$$\hat{V} = \frac{n-1}{n}W + \frac{1}{n}B.$$

- If the Markov chains have reached stationary as $n \to \infty$, then $\hat{V}$ is unbiased for $\mathrm{Var}(Y)$.
- $\mathrm{E}[W]$ approaches $\mathrm{Var}(Y)$ when $n \to \infty$.

The Gelman-Rubin $\hat{R}$ statistic is then

$$\hat{R} = \sqrt{\frac{\hat{V}}{W}},$$

which declines to 1 as $n \to \infty$.

- It is suggested that we keep simulating the Markov chain until $\hat{R} < 1.1$, or even 1.01.

Variants of Gelman-Rubin $\hat{R}$

Several different versions of $\hat{R}$ have been proposed.

1. One suggestion is to change $\hat{V}$ to

$$\hat{V} = \frac{n-1}{n}W + \frac{1}{n}\left(1+\frac{1}{m}\right)B.$$

   to account for the possibility that $\hat{V}$ is too low.

2. Another suggestion is to split each chain into two parts, yields $2m$ chains of length $n/2$ each. Then compute the $\hat{R}$, pretending that we have simulated $2m$ chains of length $n/2$.
   - This can be useful to detect the case where each chain does not reach stationary but the chains cover a common distribution, e.g, two chains exhibit an X-shape.

Geweke Diagnositic

Suppose that we have simulated a Markov chain $\{x_i\}$. The idea is that, the average of the first $100p_1\%$ observations should be close to the last $100p_2\%$ observations, where $p_1 + p_2 < 1$ to ensure that these two segments do not overlap.

Let $\bar{X}_i$ be the average of the ith segment. Then, the Geweke method computes

$$Z = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\text{Var}\left(\bar{X}_1\right) + \text{Var}\left(\bar{X}_2\right)}},$$

which converges to $N(0,1)$, as $n \to \infty$.

Hamiltonian Monte Carlo

- The Metropolis algorithm and the Gibbs sampler often move too slowly through the target distribution when the dimension of the target distribution is high.
- Hamiltonian Monte Carlo (HMC) moves much quicker through the target distribution.
  - For each component in the target distribution, HMC adds a momentum variable and the proposal distribution largely depends on the momentum variable.
  - Both the component in the target distribution and the momentum are updated in the MCMC algorithm.

Hamiltonian Dynamics

The idea of HMC originates from the Hamiltonian dynamics in physics.

- The state of a system consists of the position $x \in \mathbb{R}^d$ and the momentum $\phi \in \mathbb{R}^d$ of same dimension.
- The Hamiltonian is a function of x and ϕ, denoted by $H(x, \phi)$.
- The position and the momentum can change over time t. The change is described by the Hamilton's equations:

$$\frac{dx_i}{dt} = \frac{\partial H}{\partial \phi_i}, \quad \text{and} \quad \frac{d\phi_i}{dt} = -\frac{\partial H}{\partial x_i},$$

for $i = 1, ..., d$.

Potential and Kinetic Energy

For HMC, the Hamiltonian is usually

$$H(x, \phi) = U(x) + V(\phi),$$

where $U(x) = -\log \pi(x)$ is called the potential energy and $V(\phi)$ is called the kinetic energy.

- We want to sample from $\pi(x)$. Hence, ϕ is artificial.
- We often let $\phi \sim N(0, M)$, independent of X, for a prespecified covariance matrix M, and $V(\phi)$ the negative log density of ϕ.
- The Hamilton's equations become

$$\frac{dx}{dt} = M^{-1}\phi, \quad \text{and} \quad \frac{d\phi}{dt} = \frac{\partial \log \pi(x)}{\partial x},$$

arranged as column vectors.

Augmentation

Since X and ϕ are independent, their joint density is

$$\begin{aligned} p(x,\phi) = \pi(x)\,p(\phi) &= \exp\{-U(x) - V(\phi)\} \\ &= \exp\{-H(x,\phi)\}. \end{aligned}$$

We have augmented the problem from sampling X from $\pi(x)$ to sampling (X,ϕ) form $\exp\{-H(x,\phi)\}$.

1. We first sample ϕ from $N(0,M)$, independent of current x.
   - Since $\phi \sim N(0,M)$, we already sample ϕ from the desired distribution.
2. We then sample X, where the new state is proposed by Hamiltonian dynamics by solving the differential equations.

Solve Differential Equation

To solve the differential equations, we consider an approximation known as the leapfrog method. For some stepsize $\epsilon > 0$, we perform half-step updates as

$$\begin{aligned}
\phi\left(t+\frac{\epsilon}{2}\right) &= \phi(t)+\frac{\epsilon}{2}\frac{\partial\phi(t)}{\partial t}=\phi(t)+\frac{\epsilon}{2}\frac{\partial\log\pi(x(t))}{\partial x},\\
x(t+\epsilon) &= x(t)+\epsilon\frac{\partial x\left(t+\frac{\epsilon}{2}\right)}{\partial t}=x(t)+\epsilon M^{-1}\phi\left(t+\frac{\epsilon}{2}\right),\\
\phi(t+\epsilon) &= \phi\left(t+\frac{\epsilon}{2}\right)+\frac{\epsilon}{2}\frac{\partial\phi(t+\epsilon)}{\partial t}=\phi\left(t+\frac{\epsilon}{2}\right)+\frac{\epsilon}{2}\frac{\partial\log\pi(x(t+\epsilon))}{\partial x}.
\end{aligned}$$

Starting from $t=0$, we can get a trajectory at times $\{\epsilon, 2\epsilon, ..., L\epsilon\}$, and approximate the values for $x(L\epsilon)$ and $\phi(L\epsilon)$.

Leapfrog Method to Sample X

Suppose that the current state is (x, ϕ).

1. Update ϕ with a half-step update by

$$\phi \leftarrow \phi + \frac{\epsilon}{2} \frac{\partial \log \pi(x)}{\partial x}.$$

2. For $\ell = 1, ..., L-1$,
   1. Update the position: $x \leftarrow x + \epsilon M^{-1} \phi$.
   2. Update the momentum:

$$\phi \leftarrow \phi + \epsilon \frac{\partial \log \pi(x)}{\partial x}.$$

3. Make one last update on the position: $x \leftarrow x + \epsilon M^{-1} \phi$.
4. Make one last half-step update of the momentum

$$\phi \leftarrow \phi + \frac{\epsilon}{2} \frac{\partial \log \pi(x)}{\partial x}.$$

Metropolis Step

Suppose that the state after such L updates is (x^*, ϕ^*). We negate the momentum and the new proposal state is $(x^*, -\phi^*)$.

- We determine whether to accept the proposal using the Metropolis algorithm, where the acceptance probability is

$$A\left((x,\phi),(x^*,-\phi^*)\right) = \min\left\{1, \frac{\exp\left\{-H\left(x^*,-\phi^*\right)\right\}}{\exp\left\{-H\left(x,\phi\right)\right\}}\right\}.$$

- If the proposed state is accepted, then we accept x^* as a new state for x, but don't care about ϕ^*.
- No matter we accept or reject the proposal, we will draw a new momentum in the next iteration, independent of previous momentum.

Properties of HMC

Some crucial properties of the Hamiltonian dynamics for MCMC updates include

1. deterministic updates. The Hamiltonian dynamics is deterministic. After running the leapfrog loop L times, we always move the initial state (x_0, ϕ_0) to the same proposal (x^*, ϕ^*).
2. reversible. The mapping from the state at time t, denoted by $(x(t), \phi(t))$, to the state at time $t+s$, denoted by $(x(t+s), \phi(t+s))$, is one-to-one and has an inverse mapping. If we negate the momentum, we will come back from $(x(t+s), -\phi(t+s))$ to $(x(t), -\phi(t))$.

Tuning Parameters

Tuning of HMC can occur in several places such as

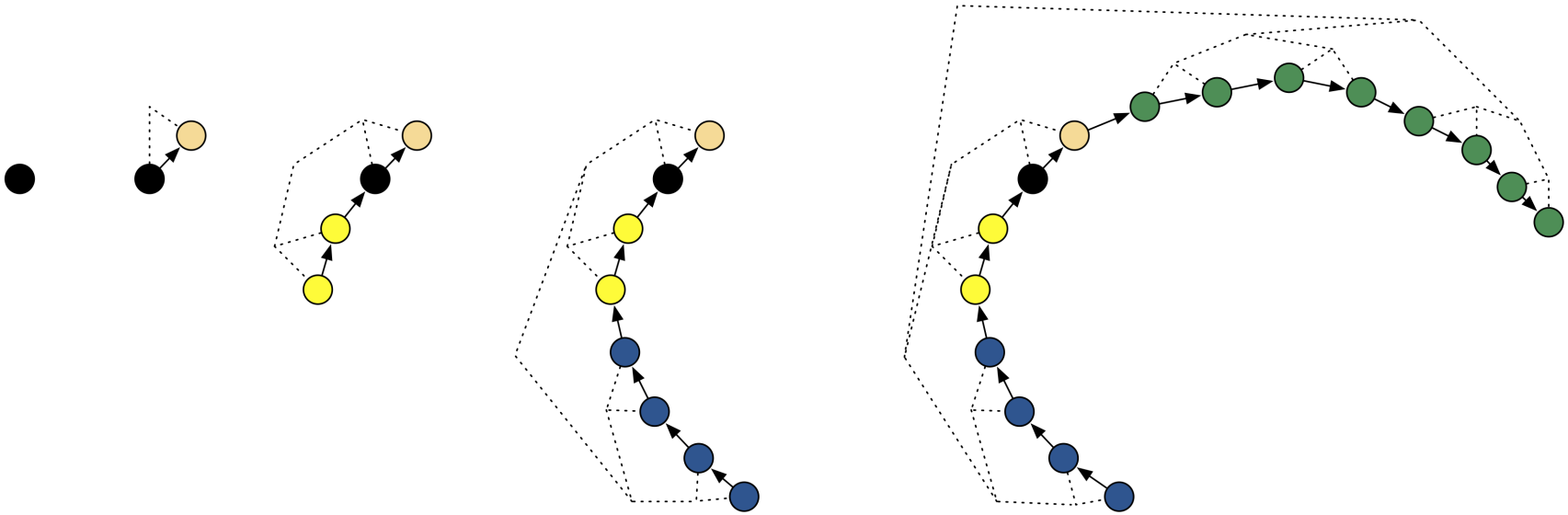
1. the distribution for the momentum (often independent normal),
2. the scaling factor ϵ,
3. the number of leapfrog steps L per iteration.

Some theory suggest that we can tune HMC such that the acceptance probability is around 65%.

No-U-Turn Sampler

The no-U-turn sampler (NUTS) allows us to automatically tune the number of steps L: we increases L until the simulated dynamics is long enough such that the proposed position θ^* starts to move back towards the initial position θ if we run more steps.

- This is measured by the angle between $\theta^* - \theta$ and current momentum ϕ^*.



Basic Algorithm of NUTS

A basic NUTS works as follows. Given the initial status,

1. Sample $u \mid \theta, \phi \sim \text{Uniform}\left[0, \exp\left\{-H\left(\theta, \phi\right)\right\}\right]$.
2. Apply the leapfrog method (with some modification) until a U-turn occurs.
3. Sample from the points in

$$\left\{(\theta, \phi):\ \exp\left\{-H\left(\theta, \phi\right)\right\} \geq u\right\}$$

that the leapfrog step has visited and the detailed balance condition is fulfilled.

Adaptively Tune ϵ

A too small ϵ will waste computation by taking needlessly tiny steps, and a too large will cause high rejection rates.

- In HMC, we tune ϵ in the warm-up stage of MCMC such that the average acceptance probability δ is the user specified value.
- In NUTS, there is no Metropolis accept/reject step. But we can still compute the ratio as if we were using the accept/reject step and set ϵ such that the pseudo acceptance probability is the user specified value.

In stan, the default is $\delta = 0.8$.

Computer Intensive Statistics and Applications
Chapter 2: Monte Carlo Integration

Shaobo Jin

Department of Mathematics

Intractable Integral in Statistics

Many quantities of interest in statistics can be formulated as integrals such as

$$\mu = \int_{x \in D} f(x)\, dx < \infty.$$

But no closed form expressions are available and we want to approximate it. Some easy examples are

1. We want to approximate the probability

$$\mathrm{P}(X \leq x) = \int_{-\infty}^{x} f(x)\, dx.$$

2. We want to approximate the expectation

$$\mathrm{E}[X] = \int x f(x)\, dx.$$

More Intractable Integral in Statistics

1. In engineering, we consider a system such that the strength of the system is a random variable S and the load is a random variable L. The system fails is $S < L$. By the law of total probability, the probability of failure is

$$\mathrm{P}(S < L) = \int \mathrm{P}(s < L) f_S(s) \, ds.$$

2. In Bayesian statistics, we want to obtain the predictive distribution to predict the future value

$$p(y_{\text{new}} \mid y) = \int p(y_{\text{new}} \mid \theta) \pi(\theta \mid y) \, d\theta.$$

Convergence in Distribution

Let $X \in \mathbb{R}^d$ be a $d \times 1$ random vector of random variables. Its distribution function is $F_X(x) = P(X_1 \leq x_1, ..., X_d \leq x_d)$.

Definition (Convergence in Law/Distribution, Weak Convergence)

X_n converges in law (converges in distribution or converges weakly) to X if $F_{X_n}(x) \to F_X(x)$ as $n \to \infty$ for all points x at which $F_X(x)$ is continuous. It is denoted by $X_n \xrightarrow{d} X$ or $X_n \xrightarrow{\mathcal{L}} X$.

Convergence in Probability and Almost Surely

Definition (Convergence in probability)

X_n converges in probability to X if, for every $\epsilon > 0$,

$$\lim_{n\to\infty} \mathrm{P}\left((X_n - X)^T (X_n - X) > \epsilon^2\right) = 0.$$

It is denoted by $X_n \xrightarrow{P} X$. If X is a constant, then we also say X_n is consistent for X.

Definition (Convergence almost surely)

X_n converges almost surely to X if

$$\mathrm{P}\left(\lim_{n\to\infty} X_n = X\right) = 1.$$

It is denoted by $X_n \xrightarrow{a.s.} X$.

Continuous Mapping Theorem

Theorem (Continuous Mapping Theorem)

Let $X_1, X_2, \ldots$ be a sequence of random vectors. Let $g : \mathbb{R}^k \to \mathbb{R}^m$ be continuous at every point of a set C such that $P(X \in C) = 1$.

1. *If $X_n \xrightarrow{d} X$, then $g(X_n) \xrightarrow{d} g(X)$.*
2. *If $X_n \xrightarrow{P} X$, then $g(X_n) \xrightarrow{P} g(X)$.*
3. *If $X_n \xrightarrow{a.s.} X$, then $g(X_n) \xrightarrow{a.s.} g(X)$.*

Big O and Small O Operator

Definition

Let X_1, X_2, ... be a sequence of random vectors.

1. The sequence is bounded in probability, if for any $\epsilon > 0$ there exists $M > 0$, such that

$$\mathrm{P}\left(X_n^T X_n \leq M^2\right) \quad > \quad 1 - \epsilon, \text{ for all } n.$$

   It is denoted by $X_n = O_{\mathrm{P}}(1)$.
2. If the sequence converges in probability to zero, it is denoted by $X_n \xrightarrow{P} 0$ or $X_n = o_{\mathrm{P}}(1)$.
3. For a sequence of positive numbers a_n, we say $X_n = O_{\mathrm{P}}(a_n)$ if $X_n/a_n = O_{\mathrm{P}}(1)$.
4. For a sequence of positive numbers a_n, we say $X_n = o_{\mathrm{P}}(a_n)$ if $X_n/a_n = o_{\mathrm{P}}(1)$.

Big O: Example

Example

Let $X_1, ..., X_n$ be iid from $N\left(\mu, \sigma^2\right)$ with finite variance.

1. Show that $\sum_{i=1}^{n} X_i = O_{\mathrm{P}}(n)$.
2. Show that $\sum_{i=1}^{n} (X_i - \mu) = o_{\mathrm{P}}(n)$.

Law of Large Numbers

Theorem (Law of Large Numbers)

Let $X_1, X_2, \ldots$ be iid random vectors, and $\bar{X}_n = n^{-1}\sum_{i=1}^{n} X_i$.

1. *(weak law) If $E\left[\sqrt{X_i^T X_i}\right] < \infty$, then $\bar{X}_n$ converges in probability to $\mu = E[X_i]$.*
2. *(strong law) $E\left[\sqrt{X_i^T X_i}\right] < \infty$ and $\mu = E[X_i]$ if and only if $\bar{X}_n$ converges almost surely to μ.*

Central Limit Theorem

Theorem (Lindeberg-Lévy Central Limit Theorem (CLT))

Let $X_1, X_2, \ldots$ be iid random vectors, with mean μ and finite covariance matrix Σ. Then, $\sqrt{n}\left(\bar{X}_n - \mu\right)$ converges in distribution to a normal distribution $N(0, \Sigma)$, i.e., $\sqrt{n}\left(\bar{X}_n - \mu\right) \xrightarrow{d} N(0, \Sigma)$.

Delta Method

Theorem (Delta Method)

Let $g : \mathbb{R}^k \to \mathbb{R}$ *be a mapping such that* $\frac{\partial g(x)}{\partial x}$ *is continuous in a neighborhood of* $\mu \in \mathbb{R}$. *Let* X_1, X_2, ... *be a sequence of random vectors, such that* $\sqrt{n}\left(X_n - \mu\right) \xrightarrow{d} N\left(0, \Sigma\right)$. *Then,*

$$\sqrt{n}\left(g\left(X_n\right) - g\left(\mu\right)\right) \quad \xrightarrow{d} \quad N\left(0, \left(\frac{\partial g\left(x\right)}{\partial x^T}\right) \Sigma \left(\frac{\partial g\left(x\right)}{\partial x^T}\right)^T\right).$$

Monte Carlo Approximation

Suppose that we want to evaluate

$$\mu = \int_{x \in D} f(x)\, dx,$$

where $x \in \mathbb{R}^d$, but the closed form expression is hard to obtain. We always assume $\mu < \infty$ without stating it.

Algorithm 1: Independent Monte Carlo

---

1 Find a suitable factorization $f(x) = h(x)\, p(x)$ such that $p(x)$ is a density ;
2 Treat μ as $\mu = \mathrm{E}[h(x)]$;
3 Simulate n iid random numbers from $p(x)$;
4 Approximate μ by

$$\hat{\mu}^{\mathrm{IMC}} = \frac{1}{n} \sum_{i=1}^{n} h(x_i).$$

Why Does Independent MC Work?

Suppose that $0 < \mathrm{Var}\left[h\left(X\right)\right] < \infty$. Then,

1. $\mathrm{E}\left[\hat{\mu}^{\mathrm{IMC}}\right] = \int h\left(x\right) p\left(x\right) dx = \mu$, unbiased.
2. By the Law of Large Numbers, $\hat{\mu}^{\mathrm{IMC}} - \mu = o_{\mathrm{P}}\left(1\right)$, and $\hat{\mu}^{\mathrm{IMC}} \overset{a.s.}{\to} \mu$.
3. By the Central Limit Theorem,

$$\sqrt{n}\left(\hat{\mu}^{\mathrm{IMC}} - \mu\right) \quad \overset{d}{\to} \quad N\left(0, \mathrm{Var}\left[h\left(X\right)\right]\right),$$

and

$$\hat{\mu}^{\mathrm{IMC}} - \mu \quad = \quad O_{\mathrm{P}}\left(n^{-1/2}\right),$$

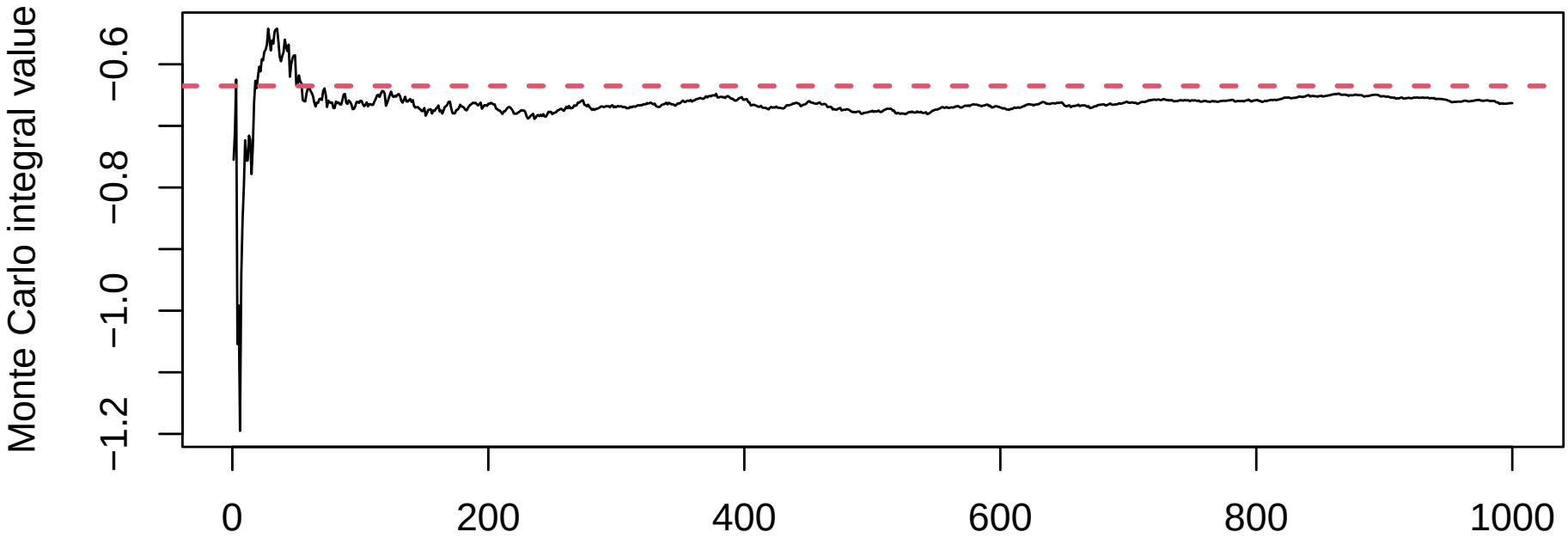
independent of dimension of integral!

Monte Carlo Integral: Example

Example

Approximate the integral

$$\mu = \int_0^\infty \log(x) \exp(-2x)\, dx.$$

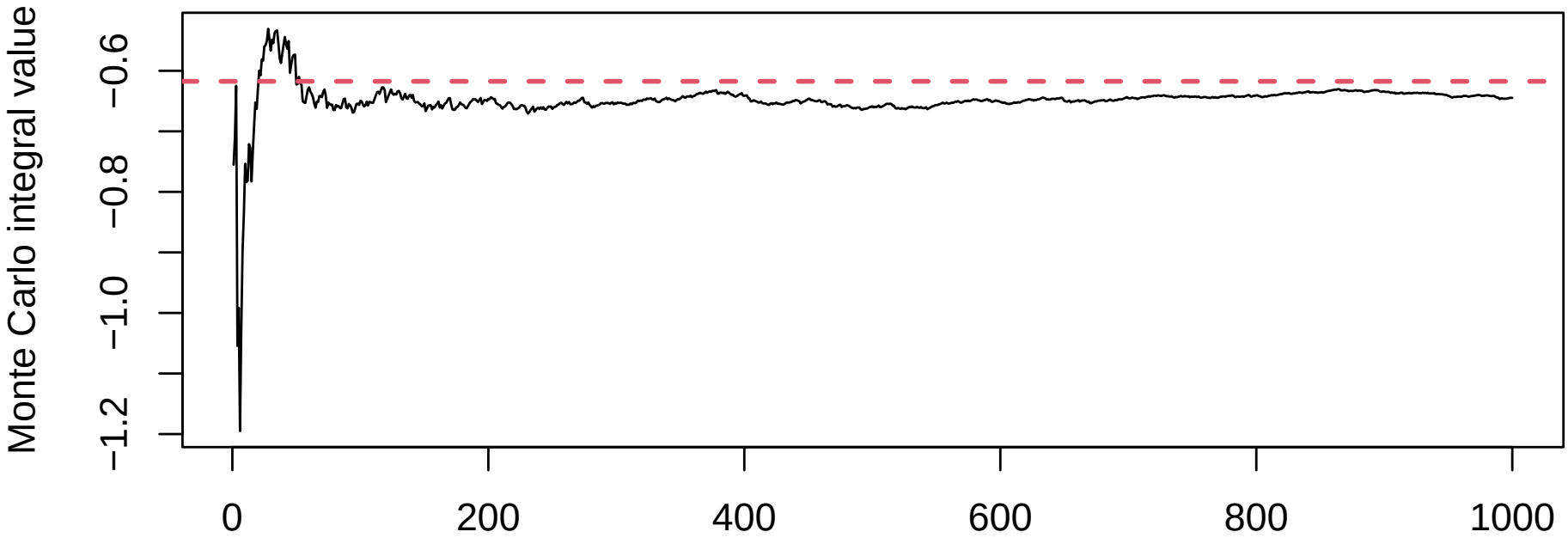


Monte Carlo Integral: Example

Example

Approximate the integral

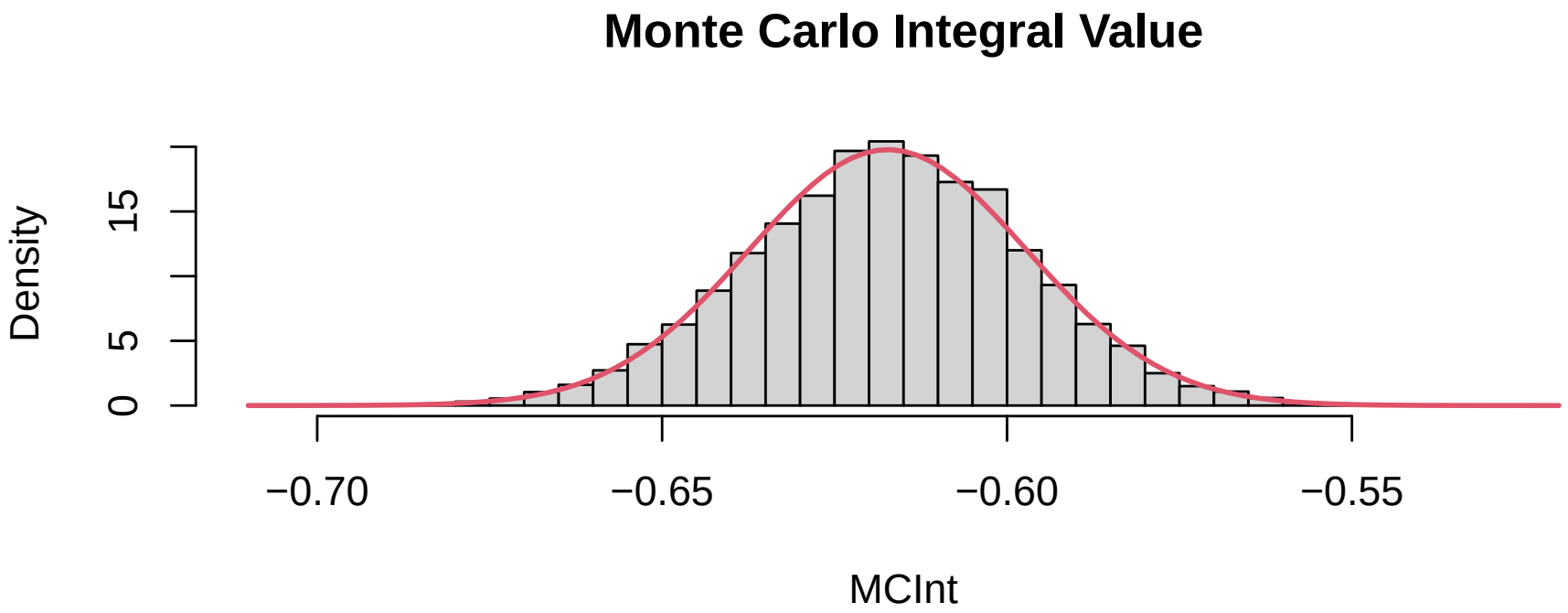
$$\mu = \int_0^{0.5} \log(x) \exp(-2x)\, dx.$$



Randomness

Since we simulate random numbers from a distribution with density $p(x)$, $\hat{\mu}^{\text{IMC}}$ is a random variable.

- For a large enough n, the distribution of $\hat{\mu}^{\text{IMC}}$ can be approximated by a normal distribution.
- This means that we can construct confidence interval for $\hat{\mu}^{\text{IMC}}$.

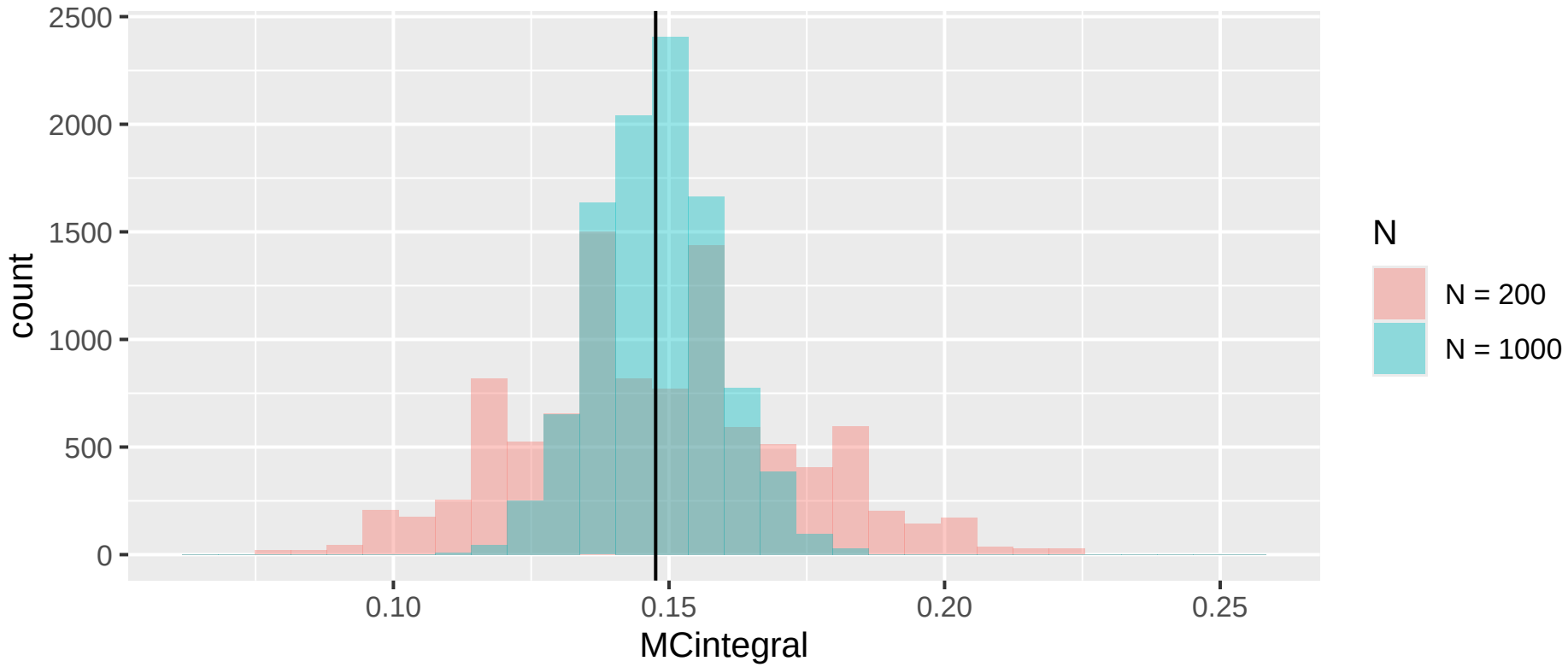


Example: Effect of n

We want to approximate $\mathrm{P}(X > 2)$, where $X \sim \text{Cauchy}(0, 1)$. That is,

$$\mu = \int_{-\infty}^{\infty} 1_{(2,\infty)}(x)\, p(x)\, dx,$$

where $p(x)$ is the density of Cauchy $(0, 1)$.



Remarks: Effects of h and p

$$\mu = \int_D f(x)\,dx < \infty,$$

1. By the Central Limit Theorem,

$$\sqrt{n}\left(\hat{\mu}^{\mathrm{IMC}} - \mu\right) \xrightarrow{d} N\left(0, \sigma^2\right).$$

   Hence, we want σ^2 to be small without the need of choosing a huge n.

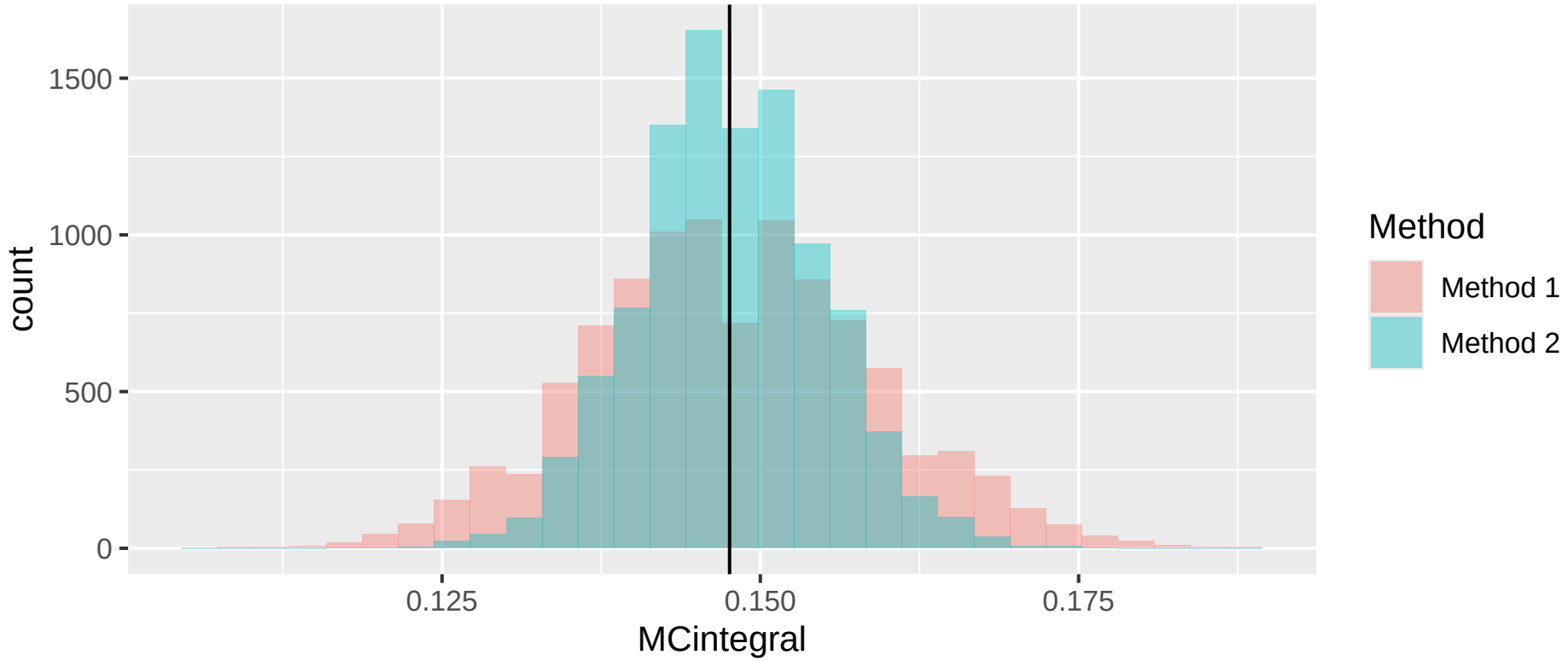
2. Since $f(x) = h(x)\,p(x)$, we want to pick a $p(x)$ such that we can easily simulate random numbers from $p(x)$.

Example: Effect of $h(x)$

To approximate the $\mathrm{P}(X > 2)$, where $X \sim \text{Cauchy}(0, 1)$,

1. Method 1: $\mu = \int_{-\infty}^{\infty} 1_{(2,\infty)}(x)\, p(x)\, dx$,
2. Method 2: $\mu = \int_{-\infty}^{\infty} 2^{-1} 1_{(2,\infty)}(|x|)\, p(x)\, dx$ using symmetry.

They have the same $p(x)$ but different $h(x)$.



Importance Sampling

It is not always the case that we can recognize a $p(x)$ as a density. But we can choose a $g(x)$ such that we can easily sample random numbers from and rewrite

$$\mu = \int f(x)\,dx = \int \frac{f(x)}{g(x)} g(x)\,dx = \mathrm{E}\left[\frac{f(X)}{g(X)}\right],$$

where the density of $X \in \mathbb{R}^d$ is $g(x)$.

Algorithm 2: Importance sampling

1 Simulate n iid random numbers from the trial distribution $g(x)$;
2 Approximate μ by

$$\hat{\mu}^{\mathrm{IS}} = \frac{1}{n}\sum_{i=1}^{n} \frac{f(x_i)}{g(x_i)},$$

provided that the ratio f/g can be computed.

Importance Sampling: Example

Example

1. Suppose that we want to approximate the integral

$$\mu = \int_0^\infty \log(1+x)\, x \exp(-4x)\, dx.$$

We sample X from $\text{Exp}(1)$.

2. We want to approximate $\text{P}(X > 2)$ by importance sampling, where $X \sim \text{Cauchy}(0,1)$. That is,

$$\mu = \int_{-\infty}^\infty 1_{(2,\infty)}(x)\, p(x)\, dx,$$

where $p(x)$ is the density of $\text{Cauchy}(0,1)$.

A Property

Theorem

Let μ be approximated by importance sampling where X is sampled from a distribution with density $g(x)$. Suppose that $g(x) > 0$ whenever $f(x) \neq 0$. Then, $E\left[\hat{\mu}^{IS}\right] = \mu$ (unbiased), $\hat{\mu}^{IS} \xrightarrow{P} \mu$ (consistent), and $Var\left(\hat{\mu}^{IS}\right) = \sigma_g^2/n$, where

$$\sigma_g^2 = \int_{\{x;\, g(x)>0\}} \frac{f^2(x)}{g(x)} dx - \mu^2.$$

Confidence Interval

The central limit theorem implies that the distribution of the importance sampling estimator $\hat{\mu}^{\mathrm{IS}}$ can be approximated by a normal distribution $N\left(\mu, \sigma_g^2/n\right)$. To approximate the distribution and obtain the confidence interval, we need to estimate σ_g^2. One estimator is

$$\hat{\sigma}_g^2 = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{f(x_i)}{g(x_i)} - \hat{\mu}^{\mathrm{IS}}\right]^2.$$

An approximated $1-\alpha$ confidence interval is

$$\hat{\mu}^{\mathrm{IS}} \pm \lambda_{1-\alpha/2}\hat{\sigma}_g/\sqrt{n},$$

where $\lambda_{1-\alpha/2}$ is the $1-\alpha/2$ quantile of $N(0,1)$.

Effect of $g(x)$

$$n\sigma_g^2 = \int_{g(x)>0} \frac{f^2(x)}{g(x)}dx - \mu^2 = \int_{g(x)>0} \frac{[f(x)-\mu g(x)]^2}{g(x)}dx.$$

We need to choose $g(x)$ such that $g(x) > 0$ whenever $f(x) \neq 0$. But even so, the choice of $g(x)$ can still have big impacts.

1. If g is (nearly) proportional to f, then the numerator in the second integral can be small.
2. If g is not proportional to f, then, the ratio in the second integral is magnified by small values of $g(x)$.
3. It is even possible that some choices of $g(x)$ will lead to $\sigma_g^2 = \infty$.

To choose a good $g(x)$ requires some guessing and possibly numerical search.

Importance Weight

Now suppose that we know $f(x) = h(x)\, p(x)$, where $p(x)$ is a density. Then,

$$\hat{\mu}^{\text{IS}} = \frac{1}{n}\sum_{i=1}^{n}\frac{f(x_i)}{g(x_i)} = \frac{1}{n}\sum_{i=1}^{n} w(x_i)\, h(x_i),$$

where $w(x) = p(x)/g(x)$ is the importance weight.

Lemma

Suppose that $\int_{g(x)>0} h^2(x)\, p(x)\, dx < \infty$. *Then a sufficient condition for a finite* $Var\left(\hat{\mu}^{IS}\right)$ *is that* $w(x)$ *is bounded.*

Check Finite Variance

Example

Suppose that we want to approximate the integral

$$\mu = \int_0^\infty \log(1+x) \frac{b^a}{\Gamma(a)} x^{a-1} \exp(-bx)\, dx,$$

where $h(x) = \log(1+x)$ and $p(x) = \frac{b^a}{\Gamma(a)} x^{a-1} \exp(-bx)$. We sample X from $\text{Exp}(c)$.

Optimal $g(x)$

What is the smallest $\text{Var}_p(\hat{\mu})$ we can get?

Theorem

Consider the importance distributions that satisfy

$$\{x;\ g(x) > 0\} \quad = \quad \{x;\ p(x) > 0\}.$$

Then,

$$g(x) \quad = \quad \frac{|h(x)|}{E_p[|h(x)|]} p(x)$$

is a probability density and the resulting $Var\left(\hat{\mu}^{IS}\right)$ *is*

$$Var\left(\hat{\mu}^{IS}\right) \quad = \quad \frac{1}{n}\left(\{E_p[|h(x)|]\}^2 - \mu^2\right).$$

Scaling Creates Problems

So far we have assumed that we can evaluate both $p(x)$ and $g(x)$. In practice, it is often the case that we know $p(x)$ up to a scaling constant

$$p(x) = c\tilde{p}(x),$$

where we can easily compute $\tilde{p}(x)$, but the exact value of c is tedious to obtain.

Example

In Bayesian statistics, the posterior distribution of a parameter θ is

$$\pi(\theta \mid \boldsymbol{x}) = \frac{f(\boldsymbol{x} \mid \theta)\pi(\theta)}{\int f(\boldsymbol{x} \mid \theta)\pi(\theta)\, d\theta},$$

where $c^{-1} = \int f(\boldsymbol{x} \mid \theta)\pi(\theta)\, d\theta$ is often difficult to obtain.

Idea of Normalized Importance Sampling

Suppose that $g(x) > 0$ whenever $h(x)\,p(x) \neq 0$. Then,

$$\begin{aligned}\mu = \int_{h(x)p(x)\neq 0} h(x)\,p(x)\,dx &= \int_{g(x)>0} h(x)\,\frac{p(x)}{g(x)}\,g(x)\,dx \\ &= c\mathrm{E}_g\left[\tilde{w}(X)\,h(X)\right],\end{aligned}$$

where we define the importance weight $\tilde{w}(x) = \tilde{p}(x)/g(x)$.

However, we still need to approximate c. If we assume $g(x) > 0$ whenever $p(x) \neq 0$, then

$$c^{-1} = \int_{p(x)\neq 0} \tilde{p}(x)\,dx = \int_{g(x)>0} \frac{\tilde{p}(x)}{g(x)}\,g(x)\,dx = \mathrm{E}_g\left[\tilde{w}(X)\right].$$

Normalized Importance Sampling

Under the assumption that $g(x) > 0$ whenever $p(x) \neq 0$, we get

$$\mu = \int h(x)\, p(x)\, dx \quad = \quad \frac{\mathrm{E}_g\left[\tilde{w}(X)\, h(X)\right]}{\mathrm{E}_g\left[\tilde{w}(X)\right]}.$$

The importance sampling estimators to the numerator and denominator are

$$\mathrm{E}_g\left[\tilde{w}(X)\, h(X)\right] \approx \frac{1}{n}\sum_{i=1}^{n} \tilde{w}(x_i)\, h(x_i),$$

$$\mathrm{E}_g\left[\tilde{w}(X)\right] \approx \frac{1}{n}\sum_{i=1}^{n} \tilde{w}(x_i).$$

The ratio is the normalized importance sampling estimator

$$\hat{\mu}^{\mathrm{NIS}} \quad = \quad \frac{\sum_{i=1}^{n} \tilde{w}(x_i)\, h(x_i)}{\sum_{i=1}^{n} \tilde{w}(x_i)}.$$

Normalized Importance Sampling: Algorithm

Algorithm 3: Normalized importance sampling

1 Simulate n iid random numbers from the proposal distribution $g(x)$;

2 Compute the weight $\tilde{w}(x) = \frac{\tilde{p}(x)}{g(x)}$;

3 Approximate μ by

$$\hat{\mu}^{\mathrm{NIS}} = \frac{\sum_{i=1}^{n} \tilde{w}(x_i) h(x_i)}{\sum_{i=1}^{n} \tilde{w}(x_i)}.$$

Normalized Importance Sampling: Example

Example

Suppose that we want to approximate the integral

$$\mu = \int_0^\infty \log(1+x) \times cx^{a-1} \exp(-bx)\, dx,$$

where $p(x) = cx^{a-1}\exp(-bx)$ with $a = 2$, and $b = 4$, and c is the unknown normalizing constant. We sample X from $\text{Exp}(1)$.

Why Does Normalized Importance Sampling Work?

Theorem

Suppose that $g(x) > 0$ *whenever* $p(x) \neq 0$. *Let* $p(x) = c\tilde{p}(x)$. *Then,*

1. $\hat{\mu}^{NIS}$ *is a consistent estimator for* μ.
2. $\hat{\mu}^{NIS}$ *is also asymptotically normal.*

Note here that the normalized importance sampler needs $g(x) > 0$ whenever $p(x) \neq 0$, which is stronger than the ordinary importance sampler that requires $g(x) > 0$ whenever $h(x)\,p(x) \neq 0$.

General MCMC Integral

Suppose that we want to approximate

$$\mu = \int h(x)\,\pi(x)\,dx,$$

for $x \in \mathbb{R}^d$. We can also sample directly from $\pi(x)$ using MCMC.

Algorithm 4: General MCMC Integral

1. Sample a Markov chain for a given stationary distribution $\pi(x)$: $x^{(1)}$, ..., $x^{(n)}$ (after burn-in) ;
2. Approximate μ by

$$\hat{\mu}^{\text{MCMC}} = \frac{1}{n}\sum_{i=1}^{n} h(x_i).$$

Long-Run Property

Theorem

Consider a finite-state Markov chain. Suppose that the transition matrix $\boldsymbol{K}$ *is irreducible, aperiodic, and has stationary distribution* π. *Then, for all starting state* $w_0 \in \Omega$,

1. *ergodic theorem: For any initial state,*

$$n^{-1} \sum_{i=1}^{n} h\left(X_i\right) \quad \overset{a.s.}{\to} \quad E\left[h\left(X\right)\right].$$

2. *central limit theorem: Let* $\sigma^2 = Var\left[h\left(X\right)\right]$ *and* $\rho_j = corr\left(h\left(X^{(1)}\right), h\left(X^{(j+1)}\right)\right)$. *Then,*

$$\sqrt{n}\left[\frac{1}{n}\sum_{i=1}^{n} h\left(X_i\right) - E\left[h\left(X\right)\right]\right] \quad \overset{d}{\to} \quad N\left(0, \sigma^2\left(1 + 2\sum_{j=1}^{\infty}\rho_j\right)\right).$$

Effective Sample Size

If we have an iid sample of size n, then

$$\sqrt{n}\left[\frac{1}{n}\sum_{i=1}^{n} h\left(X_i\right) - \mathrm{E}\left[h\left(X\right)\right]\right] \quad \xrightarrow{d} \quad N\left(0, \sigma^2\right).$$

If we have a converged Markov chain of length n,

$$\sqrt{n}\left[\frac{1}{n}\sum_{i=1}^{n} h\left(X_i\right) - \mathrm{E}\left[h\left(X\right)\right]\right] \quad \xrightarrow{d} \quad N\left(0, \sigma^2\left(1 + 2\sum_{j=1}^{\infty}\rho_j\right)\right).$$

The variance of $\hat{\mu}^{\mathrm{MCMC}}$ is larger than the variance of $\hat{\mu}^{\mathrm{IMC}}$. We define

$$n_{\mathrm{eff}} \quad = \quad \frac{n}{1 + 2\sum_{j=1}^{\infty}\rho_j}$$

as the effective sample size.

Estimate Effective Sample Size

We can also estimate the effective sample size, if we have m Markov chains of length n.

- Following the Gelman-Rubin $\hat{R}$ statistic, we can estimate σ^2 by

$$\hat{V} = \frac{n-1}{n} W + \frac{1}{n} B.$$

- The autocorrelations can be estimated by

$$\hat{\rho}_t = 1 - \frac{\sum_{j=1}^{m} \sum_{i=t+1}^{n} \left(y_{i,j} - y_{i-t,j}\right)^2}{2m\left(n-t\right)\hat{V}}.$$

- The effective sample size is estimated by

$$\hat{n}_{\text{eff}} = \frac{mn}{1 + 2\sum_{t=1}^{T} \hat{\rho}_t},$$

where T is the first odd positive integer such that $\hat{\rho}_{T+1} + \hat{\rho}_{T+2}$ is negative.

Alternative Confidence Interval

Suppose that we can divide the Markov chain of length n into b batches (e.g., 20 or proportional to $n^{1/3}$) of m consecutive observations each.

- Let $\bar{y}_j$ be the average of batch j.
- We will treat $\{\bar{y}_j\}$ as iid normal random variables.

An approximate confidence interval is

$$\frac{1}{b}\sum_{j=1}^{b}\bar{y}_j \pm t_{1-\alpha/2}\,(b-1)\sqrt{\frac{1}{b\,(b-1)}\sum_{j=1}^{b}(\bar{y}_j-\bar{y})^2},$$

where $\bar{y}$ is the average of $\{\bar{y}_j\}$. This is just the usual t-confidence interval.

Rao-Blackwell Theorem in Statistical Inference

Theorem (Rao-Blackwell Theorem)

Let $\hat{\theta}$ be an unbiased estimator of θ. Suppose that $T = T(X)$ is a sufficient statistic for θ. Then, $\theta^* = E\left[\hat{\theta} \mid T\right]$ is a uniformly minimum variance unbiased estimator of θ, i.e.,

$$Var\left(\hat{\theta}\right) \quad \geq \quad Var\left(\theta^*\right).$$

A weaker version is of the theorem is based on the low of total variance:

$$\mathrm{Var}\left(X\right) \quad = \quad \mathrm{Var}\left(\mathrm{E}\left[X \mid Y\right]\right) + \mathrm{E}\left(\mathrm{Var}\left[X \mid Y\right]\right) \geq \mathrm{Var}\left(\mathrm{E}\left[X \mid Y\right]\right).$$

Rao-Blackwellization in Monte Carlo

If we are interested in $\mathrm{E}\left[f\left(X,Y\right)\right]$, then

$$\mathrm{Var}\left(f\left(X,Y\right)\right) \quad \geq \quad \mathrm{Var}\left(\mathrm{E}\left[f\left(X,Y\right)\mid Y\right]\right).$$

That is, instead of simulating (X_i, Y_i) to compute $n^{-1}\sum_{i=1}^{n} f\left(X_i, Y_i\right)$, we can simulate only Y_i and compute

$$\frac{1}{n}\sum_{i=1}^{n}\mathrm{E}\left[f\left(X_i,Y_i\right)\mid Y_i\right].$$

This also suggests that we should compute as many analytical steps as possible before Monte Carlo approximation.

Rao-Blackwellization: Example

Example

Consider a Bayesian model, where $X_i \mid \mu, \lambda \sim N\left(\mu, \lambda^{-1}\right)$, $\mu \sim N\left(\mu_0, \lambda_0^{-1}\right)$, and $\lambda \sim \text{Gamma}\left(a_0, b_0\right)$. Then,

$$\mu \mid \lambda, \text{data} \sim N\left(\frac{\lambda_0\mu_0 + \lambda\sum_{i=1}^n x_i}{\lambda_0 + n\lambda}, \frac{1}{\lambda_0 + n\lambda}\right),$$

$$\lambda \mid \mu, \text{data} \sim \text{Gamma}\left(a_0 + \frac{n}{2}, b_0 + \frac{1}{2}\sum_{i=1}^n x_i^2 - \sum_{i=1}^n x_i\mu + \frac{n}{2}\mu^2\right).$$

We want to approximate $\text{E}\left[\lambda \mid \text{data}\right]$.

Rao-Blackwellization: Another Example

Example

Suppose that we want to estimate $\mathrm{E}\left[X1\left(X>0\right)\right]$, where $X \sim N\left(0,1\right)$.

1. Approach 1: Independent Monte Carlo

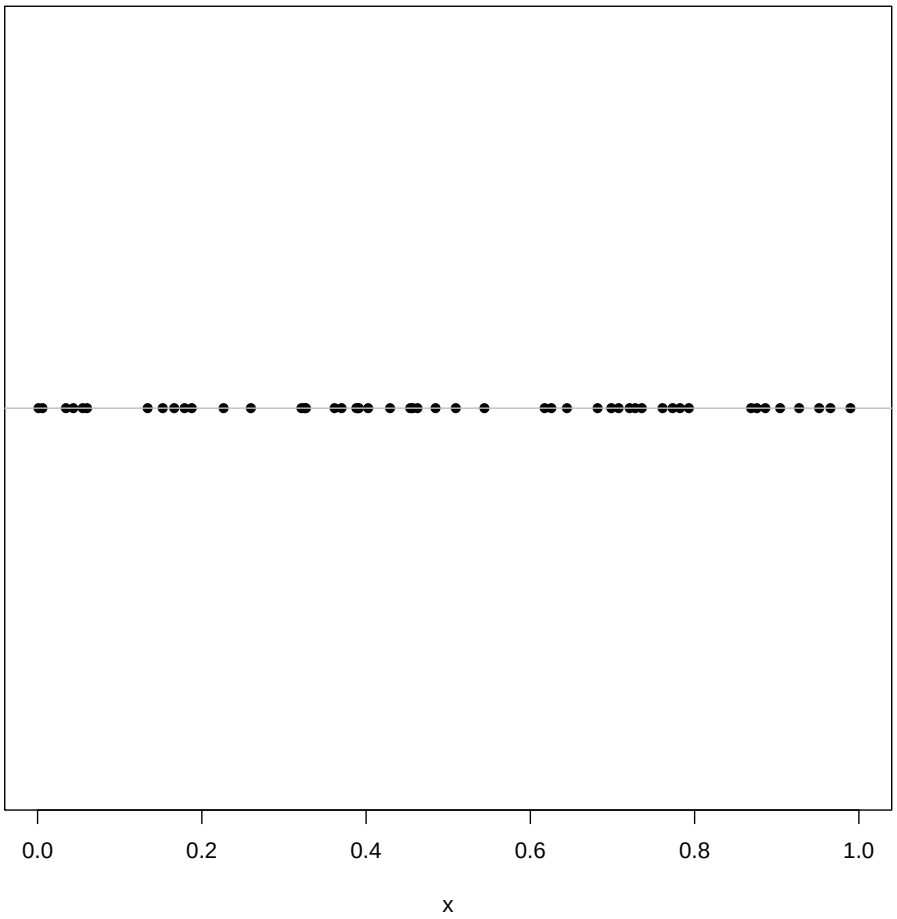
$$\mathrm{E}\left[X1\left(X>0\right)\right] = \int x1\left(x>0\right)\phi\left(x\right)dx \approx \frac{1}{n}\sum_{i=1}^{n} x_i 1\left(x_i>0\right).$$

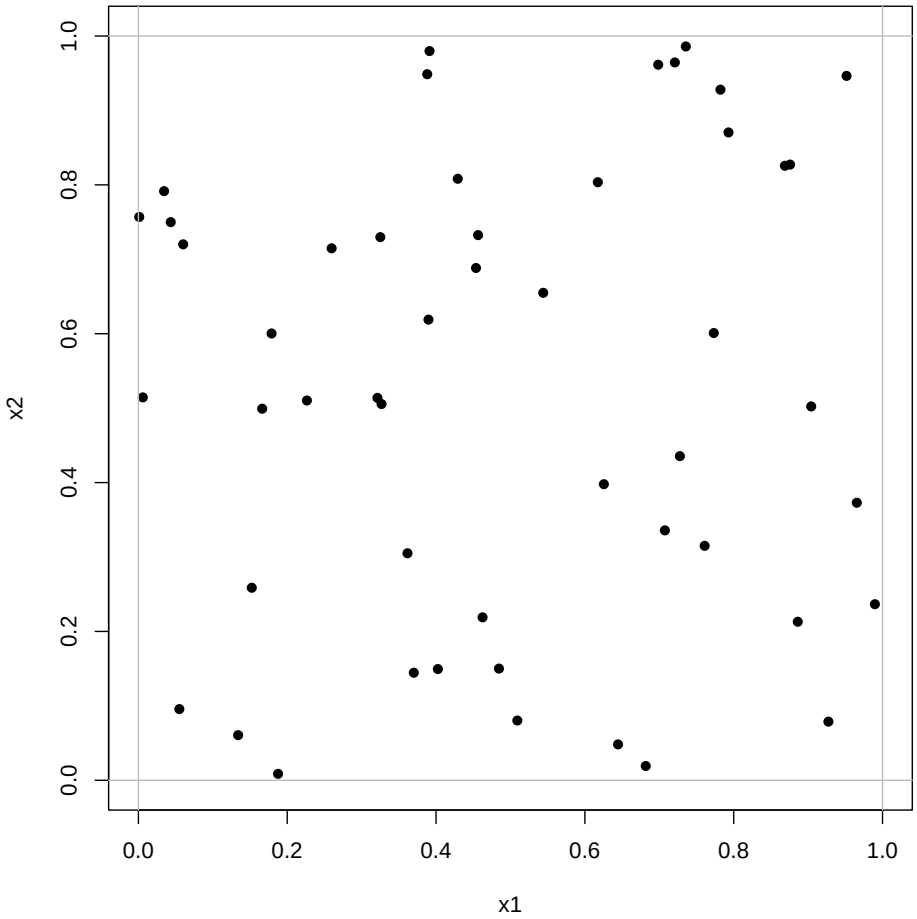
2. Approach 2: Conditioning

$$\mathrm{E}\left[X1\left(X>0\right)\right] = \mathrm{P}\left(X>0\right)\mathrm{E}\left[X \mid 1\left(X>0\right)\right].$$

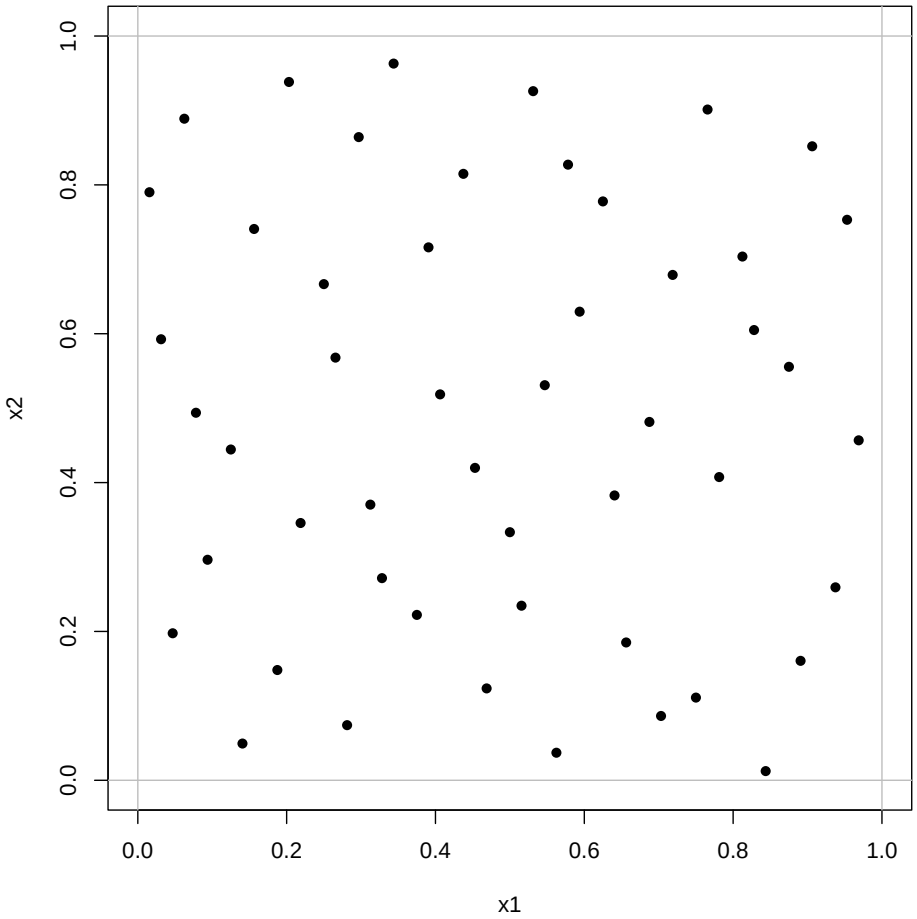
We approximate $\mathrm{E}\left[X \mid 1\left(X>0\right)\right]$.

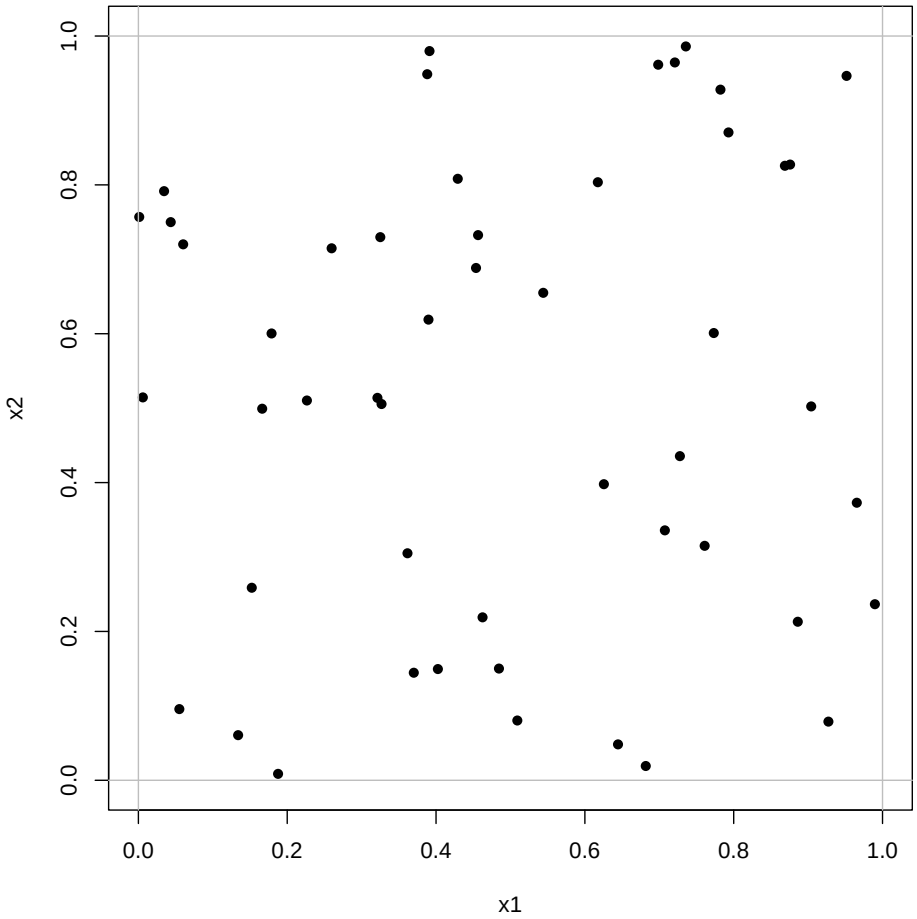
Curse of Dimensionality: Clumsy Allocation





Quasi-Random Numbers





Quasi-Monte Carlo Integral

In quasi-Monte Carlo methods, it is customary to focus on numerical integration over the unit hypercube

$$\mu = \int_{x \in [0,1]^d} f(x)\, dx.$$

- In case the domain is not a hypercube, we make transformations to obtain a hypercube, e.g., through the cumulative distribution function, we can transform $\mathbb{R}$ to $[0, 1]$.

Let X be a uniform distributed random variable/vector on $[0, 1]$. Then, $\mu = \mathrm{E}[f(X)]$. The quasi-Monte Carlo approximation is

$$\hat{\mu}^{\mathrm{QMC}} = \frac{1}{n} \sum_{i=1}^{n} f(x_i),$$

for some cleverly chosen $\{x_i\}$.

Discrepancy

Our starting point is that we want the discrepancy of the points $\{x_i\}$ to be low.

- Given a collection $\mathcal{A}$ of subsets in $[0,1]^d$, we define the discrepancy as

$$D_n\left(\{x_i\}_{i=1}^n, \mathcal{A}\right) = \sup_{A\in\mathcal{A}} \left| \frac{\#\{x_i \in A\}}{n} - \text{volume}(A) \right|.$$

- For example, we can take $\mathcal{A}$ to be the collections of all rectangles of the form

$$[0, v_1] \times [0, v_2] \times \cdots \times [0, v_d].$$

The resulting discrepancy is called the star discrepancy D_n^*.

Uniform Allocation

A natural attempt to achieve low discrepancy is to use a grid, in the spirit of Riemann sum. For example,

- consider $[0,1]$ and divide it into K equally spaced intervals, i.e., $x_i = \frac{2i-1}{2n}$ for $i = 1, ..., K$.
- consider $[0,1]^2$ and divide it into K^2 equally spaced squares, i.e., take $x_{ij} = \left(\frac{2i-1}{2n}, \frac{2j-1}{2n}\right)$ for $(i,j) \in \{1, ..., K\}$.

If we choose K points per dimension, the total number of points in the grid is $n = K^d$.

- Grows too quickly with either K or d.

Another limitation is that, if we find out n is not large enough want to increase it, we need to construct the whole grid again.

Halton Sequence

It is computationally more efficient if we have an infinite sequence and just take the first n points.

- If we want to include more points, just start with merging $(n+1)$th point, etc.

The Halton sequence is such an example.

- The Halton sequence starts with a set of coprime integers that are greater than 1, called the bases.
- For example, if the dimension is $d = 2$, then we choose $(2, 3)$.
- Another example, if the dimension is $d = 5$, then we choose $(2, 3, 5, 7, 11)$.

Create Halton Sequence

Take the dimension $d = 2$ as an example. We choose the coprime integers $(2, 3)$. The sequence is

For the first dimension

$$
\begin{array}{cccccc}
\text{initial} & 0 & & & & \\
\left(x + 2^{-1}\right): & \frac{1}{2} & & & & \\
\left(x + 2^{-2}\right): & \frac{1}{4} & \frac{3}{4} & & & \\
\left(x + 2^{-3}\right): & \frac{1}{8} & \frac{5}{8} & \frac{3}{8} & \frac{7}{8} & \\
\left(x + 2^{-4}\right): & \frac{1}{16} & \frac{9}{16} & \frac{5}{16} & \cdots & \\
\vdots & & & & &
\end{array}
$$

For the second dimension

$$
\begin{array}{cccccc}
\text{initial} & 0 & & & & \\
\left(x + 3^{-1}\right): & \frac{1}{3} & \frac{2}{3} & & & \\
\left(x + 3^{-2}\right): & \frac{1}{9} & \frac{4}{9} & \frac{7}{9} & & \\
\left(x + 2 \cdot 3^{-2}\right): & \frac{2}{9} & \frac{5}{9} & \frac{8}{9} & & \\
\left(x + 3^{-3}\right): & \frac{1}{27} & \frac{10}{27} & \frac{19}{27} & \cdots & \\
\vdots & & & & &
\end{array}
$$

We often ignore the zero point $(0, 0)$ when we use the Halton sequence in practice.

Halton as Low Discrepancy Sequence

The infinite sequence $x_1, x_2, ..., \in [0,1]^d$ is a low discrepancy sequence if

$$D_n^*(x_1, ..., x_n) = O\left(\frac{(\log n)^d}{n}\right), \quad n \to \infty.$$

- Any finite positive power of $\log n$ is asymptotically negligible compared to n.
- Hence, a low discrepancy sequence satisfies $D_n^* = O\left(n^{-1+\epsilon}\right)$ for any $\epsilon > 0$.

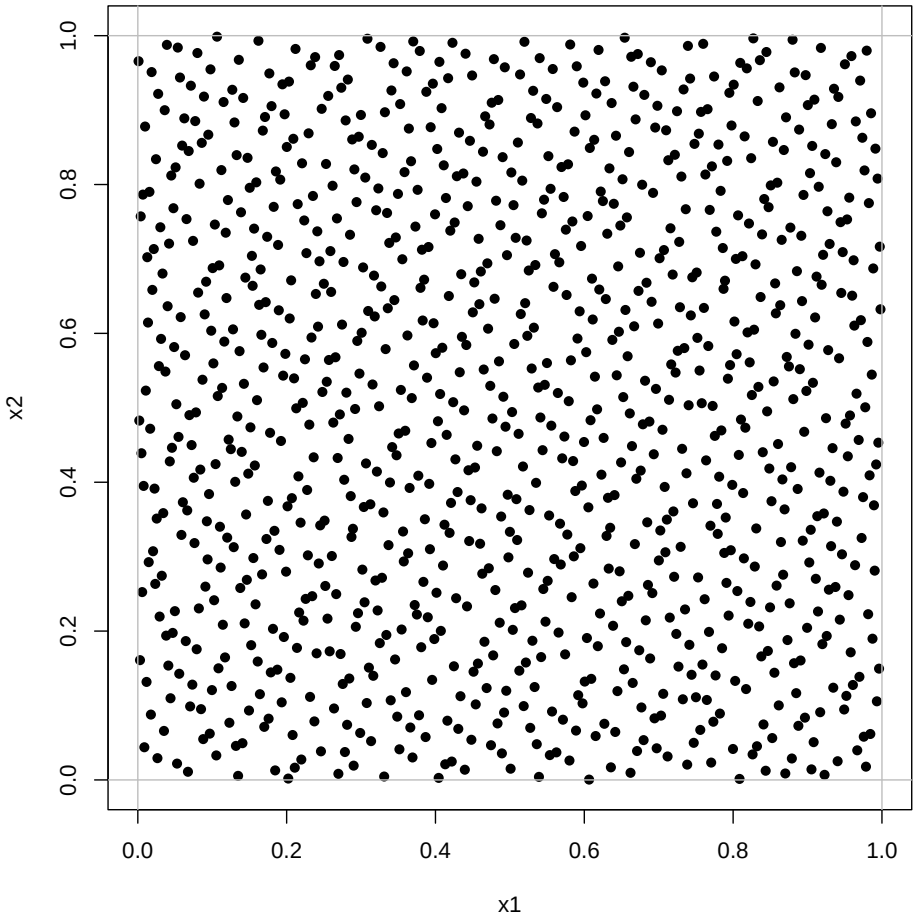
The star discrepancy of the first n Halton points has the upper bound

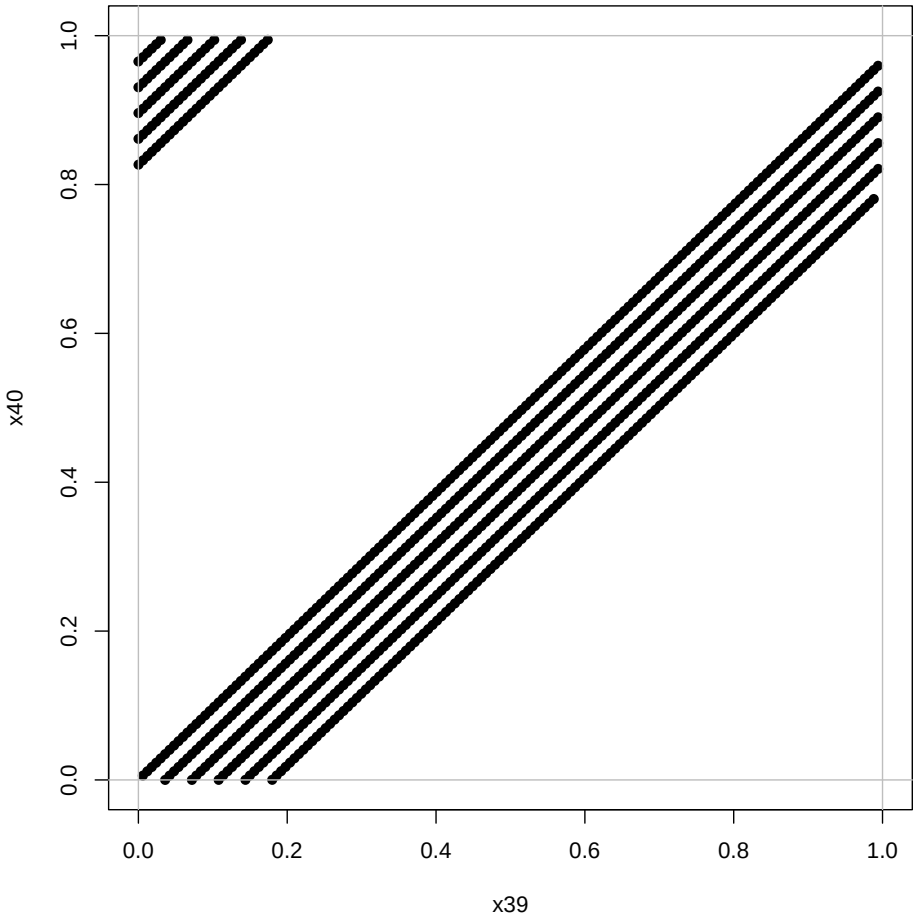
$$D_n^* \leq C_d \frac{(\log n)^d}{n} + O\left(\frac{(\log n)^{d-1}}{n}\right),$$

where C_d does not depend on n. Hence, it is a low-discrepancy sequence.

An Illustration: $n = 1000$, $d = 40$

But C_d increases in the same rate as d^d, meaning that discrepancy can be large if d is very large. This is caused by the fact that we need a large base if d is large.





Digital Nets

Let $d \geq 1$ and $b \geq 2$ be integers. An elementary interval in base b is a rectangle of the form

$$\left[\frac{c_1}{b^{k_1}}, \frac{c_1+1}{b^{k_j}}\right) \times \cdots \times \left[\frac{c_d}{b^{k_d}}, \frac{c_d+1}{b^{k_d}}\right),$$

for integers $k_j \geq 0$ and $0 \leq c_j < b^{k_j}$.

Definition

Let $m \geq t \geq 0$, $b \geq 2$, and $d \geq 1$ be integers. The sequence x_1, ..., $x_{b^m} \in [0,1)^d$ is a (t, m, d)-net in base b, if every elementary interval in base b of volume b^{t-m} contains exactly b^t points of the sequence.

- Since $m \geq t$, a smaller t indicates that the volume b^{t-m} is smaller and the points are more "equidistribution".

Sobol Sequence

- The (t, m, d)-net in base b has exactly b^m points. The infinite sequence x_1, x_2, ..., $\in [0,1)^d$ is a (t, d)-sequence in base b if for all $k \geq 0$ and $m \geq t$, the sequence x_{kb^m+1}, ..., $x_{(k+1)b^m} \in [0,1)^d$ is a (t, m, d)-net in base b, i.e., a series of (t, m, d)-nets, one after another.
- A Sobol sequence is a (t, d)-sequence in base $b = 2$, where the value of t depends on d.
- Because of the structure of a (t, m, d)-net in base b, if we use $n = b^m$ points from a (t, d)-sequence, then the next sample size that retains "equidistribution" is $n = 2b^m$.

Koksma-Hlawka Inequality

When we replace randomly sampled points by deterministic ones, we can no longer use the law of large numbers or the central limit theorem.

Theorem (Koksma-Hlawka Inequality)

For $d \geq 1$ and $x_1, ..., x_n \in [0,1]^d$,

$$\left| \frac{1}{n} \sum_{i=1}^{n} f(x_i) - \int_{[0,1]^d} f(x)\, dx \right| \leq D_n^*(x_1, ..., x_n)\, V_{HK}(f),$$

where $V_{HK}(f)$ denotes the total variation of f in the sense of Hardy and Krause.

- $V_{HK}(f)$ depends on the property of $f(x)$. If f' is continuous, then

$$V_{HK}(f) = \int_0^1 |f'(x)|\, dx.$$

QMC Versus MC

$$\left| \frac{1}{n} \sum_{i=1}^{n} f(x_i) - \int_{[0,1]^d} f(x)\, dx \right| \leq D_n^*(x_1, ..., x_n)\, V_{HK}(f)$$

- This result shows that as long as we have a low frequency sequence and $V_{HK} < \infty$, then we will achieve

$$\left| \hat{\mu}^{\mathrm{QMC}} - \mu \right| = O\left(n^{-1+\epsilon}\right)$$

  for any $\epsilon > 0$.
  - For large enough n, QMC is expected to be more accurate than MC with rate $O\left(n^{-1/2}\right)$.
- However, error analysis based on the central limit theorem is easy for MC.
  - For QMC, the error bound involves unknown quantities that are often even harder to obtain than μ.

Randomized QMC

It is hard to estimate of QMC integral error since QMC is deterministic.

- As a remedy, we inject some randomness into $\{x_i\}$, but still keep low discrepancy.

Definition

Random variables $X_i \in [0,1]^d$ for $i \geq 1$ comprise a randomized quasi-Monte Carlo rule, if there exist $B < \infty$ and $N > 0$ such that

$$\mathrm{P}\left(D_n^*(X_1, ..., X_n) < B(\log n)^d / n\right) = 1, \text{ for all } n \geq N,$$

and $X_i \sim \text{Uniform}\,[0,1]^d$ for all $i \geq 1$.

Properties of Randomized QMC

The randomized QMC approximation is still

$$\hat{\mu}^{\mathrm{RQMC}} = \frac{1}{n}\sum_{i=1}^{n} f(x_i).$$

- $\mathrm{E}\left[\hat{\mu}^{\mathrm{RQMC}}\right] = \mu$, i.e, $\hat{\mu}^{\mathrm{RQMC}}$ is unbiased.
- If $V_{HK} < \infty$, then

$$\mathrm{Var}\left(\hat{\mu}^{\mathrm{RQMC}}\right) < B^2 \left[V_{NK}(f)\right]^2 \frac{(\log n)^{2d}}{n^2} = O\left(n^{-2+2\epsilon}\right),$$

for large enough n.

- Randomized QMC is asymptotically more stable than Monte Carlo, since $\mathrm{Var}\left(\hat{\mu}^{\mathrm{MC}}\right) = O\left(n^{-1}\right)$.

Confidence Interval

Suppose that we can repeat randomized QMC independently R times. Then,

$$\hat{\mu}^{\mathrm{RQMC}} = \frac{1}{R}\sum_{r=1}^{R}\hat{\mu}_r^{\mathrm{RQMC}},$$

where $\hat{\mu}_r^{\mathrm{RQMC}}$ is the estimate at rth replication.

- We can estimate the variance by

$$\widehat{\mathrm{Var}}(\hat{\mu}^{\mathrm{RQMC}}) = \frac{1}{R(R-1)}\sum_{r=1}^{R}\left(\hat{\mu}_r^{\mathrm{RQMC}} - \hat{\mu}^{\mathrm{RQMC}}\right)^2.$$

- A t-confidence interval is then

$$\hat{\mu}^{\mathrm{RQMC}} \pm t_{1-\alpha/2}(R-1)\sqrt{\widehat{\mathrm{Var}}(\hat{\mu}^{\mathrm{RQMC}})}.$$

Cranley-Patterson Rotation

A simple way to obtain randomized QMC is the Cranley-Patterson rotation by taking

$$X_i \quad = \quad a_i + U \quad \mod 1,$$

where $U \sim \text{Uniform}\,[0,1]^d$.

- The Cranley-Patterson rotation of low discrepancy points retains low discrepancy.
- However, it is not commonly applied to a (t, m, d)-net. It is more commonly used for a lattice rule.

Scrambling

The general idea of scrambling for a (t, m, d)-net in base b is to

1. chop $[0,1]^d$ into b slices, and shuffle those slices in random order,
2. chop each of b slices into b thinner slices, and shuffle the thinner slices in random order within their respective original slices,
3. chop each of b^2 thinner slices into b even thinner slices, and shuffle the even thinner slices in random order within their respective thinner slices,
4. proceed the algorithm.

State Space Model

A state space model is a type of probabilistic models that consists of latent variables X and observed variables Y.

- For example, for $t \geq 1$,

$$\begin{aligned} Y_t \mid X_t &\sim p\left(y_t \mid X_t\right), \\ X_{t+1} \mid X_t &\sim p\left(x_{t+1} \mid X_t\right), \end{aligned}$$

  independently, where $p\left(\right)$ is used as a generic symbol for densities.

We are often interested in the posterior densities $p\left(x_{1:t} \mid y_{1:t}\right)$ and $p\left(x_t \mid y_{1:t}\right)$, or

$$\mu_t = \mathrm{E}\left[h_t\left(x_{1:t}\right) \mid y_{1:t}\right] = \int h_t\left(x_{1:t}\right) p\left(x_{1:t} \mid y_{1:t}\right) dx_{1:t},$$

for some function $h_t\left(x_{1:t}\right)$.

State Space Model: One Example

A typical example of a state space model is the location problem.

- Let X_t be the location of an object at time t. The dynamic is

$$X_{t+1} = X_t + u_t + V_t,$$

  where u_t (velocity) is known, and V_t is the unknown disturbance.

- The measurement model is

$$Y_t = h(X_t) + E_t,$$

  where E_t is the unknown disturbance, and the function $h(x)$ denotes the height of the position x.

- We want to know where we are at time t (x_t), i.e., compute the density $p(x_t \mid y_{1:t})$.

Evaluate Expectation

Suppose that we want to evaluate

$$\mu_t = \mathrm{E}\left[h_t\left(x_{1:t}\right) \mid y_{1:t}\right] = \int h_t\left(x_{1:t}\right) p\left(x_{1:t} \mid y_{1:t}\right) dx_{1:t}.$$

At any time $t \geq 1$, the posterior density satisfies

$$p\left(x_{1:t} \mid y_{1:t}\right) = \frac{p\left(y_{1:t} \mid x_{1:t}\right) p\left(x_{1:t}\right)}{p\left(y_{1:t}\right)}.$$

1. The normalizing constant $p\left(y_{1:t}\right)$ can be difficult to obtain.
2. The dimension of the integral is high as t increases, i.e., new data are collected sequentially.

Approximate Expectation

$$\mu_t = \int h_t(x_{1:t})\, p(x_{1:t} \mid y_{1:t})\, dx_{1:t}.$$

- If we can simulate $\left\{x_{1:t}^{(i)}\right\}_{i=1}^{N}$ from $p(x_{1:t} \mid y_{1:t})$, then we approximate μ_t by independent Monte Carlo.
- If simulating $\left\{x_{1:t}^{(i)}\right\}_{i=1}^{N}$ from $p(x_{1:t} \mid y_{1:t})$ is not an trivial task, we can approximate it by normalized importance sampling because

$$\mu_t = \int h_t(x_{1:t}) \frac{p(y_{1:t} \mid x_{1:t})\, p(x_{1:t})}{\int p(y_{1:t} \mid x_{1:t})\, p(x_{1:t})\, dx_{1:t}} dx_{1:t},$$

where $Z = \int p(y_{1:t} \mid x_{1:t})\, p(x_{1:t})\, dx_{1:t}$ is the intractable normalizing constant for $p(x_{1:t} \mid y_{1:t})$.

Normalized Importance Sampling

Let $g()$ be the importance distribution. Then,

$$\begin{aligned}
\mu_t &= \int h_t(x_{1:t}) \frac{p(y_{1:t} \mid x_{1:t}) p(x_{1:t})}{\int p(y_{1:t} \mid x_{1:t}) p(x_{1:t}) dx_{1:t}} dx_{1:t} \\
&= \frac{\int h_t(x_{1:t}) p(y_{1:t} \mid x_{1:t}) p(x_{1:t}) / g(x_{1:t}) \cdot g(x_{1:t}) dx_{1:t}}{\int p(y_{1:t} \mid x_{1:t}) p(x_{1:t}) / g(x_{1:t}) \cdot g(x_{1:t}) dx_{1:t}} \\
&= \frac{\int h_t(x_{1:t}) w_t g(x_{1:t}) dx_{1:t}}{\int w_t g(x_{1:t}) dx_{1:t}},
\end{aligned}$$

where the importance weight is

$$w_t(x_{1:t}) = \frac{p(y_{1:t} \mid x_{1:t}) p(x_{1:t})}{g(x_{1:t})}.$$

Here $g()$ can also depend on $y_{1:t}$, but we drop it for simplicity.

Normalized Importance Sampling: Approximation

The normalized importance sampling approximation is

$$\hat{\mu}_t = \sum_{i=1}^{N} \frac{w_t\left(x_{1:t}^{(i)}\right)}{\sum_{j=1}^{N} w_t\left(x_{1:t}^{(j)}\right)} h_t\left(x_{1:t}^{(i)}\right),$$

where $\left\{x_{1:t}^{(i)}\right\}_{i=1}^{N}$ is simulated from $g\left(x_{1:t}\right)$. Each $x_{1:t}^{(i)}$ is called a particle.

Equivalently, normalized importance sampling approximates the distribution of $x_{0:t} \mid y_{1:t}$ by an empirical distribution

$$\hat{\mathrm{P}}\left(x_{1:t} = x_{1:t}^{(i)} \mid y_{1:t}\right) = \frac{w_t\left(x_{1:t}^{(i)} \mid y_{1:t}\right)}{\sum_{j=1}^{N} w_t\left(x_{1:t}^{(j)} \mid y_{1:t}\right)}.$$

Sequential Update of Posterior $p\left(x_{1:t} \mid y_{1:t}\right)$

However, generating the whole trajectory $\left\{x_{1:t}^{(i)}\right\}$ for every t is demanding, especially when t becomes large. Luckily, many elements can be updated sequentially.

Using Bayes formula, we can show that the posterior density of our state space model becomes

$$\begin{aligned} p\left(x_{1:t} \mid y_{1:t}\right) &= \frac{p\left(y_t \mid x_{1:t}, y_{1:t-1}\right) p\left(x_{1:t} \mid y_{1:t-1}\right)}{p\left(y_t \mid y_{1:t}\right)} \\ &= \frac{p\left(y_t \mid x_t\right) p\left(x_t \mid x_{t-1}\right)}{p\left(y_t \mid y_{1:t-1}\right)} p\left(x_{1:t-1} \mid y_{1:t-1}\right), \end{aligned}$$

Hence, we can update $p\left(x_{1:t} \mid y_{1:t}\right)$ recursively.

Sequential Update of Posterior $p(x_t \mid y_{1:t})$

Likewise, the Bayes formula implies

$$p(x_t \mid y_{1:t}) = \frac{p(y_t \mid x_t, y_{1:t-1})\, p(x_t \mid y_{1:t-1})}{p(y_t \mid y_{1:t-1})} = \frac{p(y_t \mid x_t)\, p(x_t \mid y_{1:t-1})}{p(y_t \mid y_{1:t-1})}.$$

Here, $p(x_t \mid y_{1:t-1})$ can be viewed as the density used for forecasting:

$$\begin{aligned} p(x_t \mid y_{1:t-1}) &= \int p(x_t \mid x_{t-1}, y_{1:t-1})\, p(x_{t-1} \mid y_{1:t-1})\, dx_{t-1} \\ &= \int p(x_t \mid x_{t-1})\, p(x_{t-1} \mid y_{1:t-1})\, dx_{t-1}. \end{aligned}$$

However, the integrals we need (e.g., $p(y_t \mid y_{1:t-1})$ and $p(x_t \mid y_{1:t-1})$) often do not closed form expressions.

Sequential Importance Sampling (SIS): Idea

The normalized importance sampling can be used to handle integrals with unknown normalizing constant. We can further modify it so that we don't need to resimulate $\left\{x_{1:t-1}^{(i)}\right\}_{i=1}^{N}$ when we need to incorporate data at time t.

This means that the marginal $x_{1:t-1}$ from $g(x_{1:t})$ should be the same as the distribution $g(x_{1:t-1})$ we used to simulate $\left\{x_{1:t-1}^{(i)}\right\}_{i=1}^{N}$.

- The trick is to use the conditional distribution

$$g(x_{1:t}) = g(x_{1:t-1})\, g(x_t \mid x_{1:t-1}).$$

- This means that the importance distribution satisfies

$$g(x_{1:t}) = g(x_1) \prod_{k=2}^{t} g(x_k \mid x_{1:k-1}).$$

Back to State Space Model

For our state space model, we can show that

$$\begin{aligned} w_t\left(x_{1:t}\right) &= \frac{p\left(y_{1:t} \mid x_{1:t}\right) p\left(x_{1:t}\right)}{g\left(x_{1:t}\right)} \\ &= w_{t-1}\left(x_{1:t-1}\right) \times \frac{p\left(y_{1:t} \mid x_{1:t}\right) p\left(x_t \mid x_{1:t-1}\right)}{p\left(y_{1:t-1} \mid x_{1:t-1}\right) g\left(x_t \mid x_{1:t-1}\right)} \\ &= w_{t-1}\left(x_{1:t-1}\right) \times \frac{p\left(y_t \mid x_t\right) p\left(x_t \mid x_{1:t-1}\right)}{g\left(x_t \mid x_{1:t-1}\right)}. \end{aligned}$$

A special example of $g()$ that further simplifies the importance weights is

$$g\left(x_{1:t}\right) = p\left(x_{1:t}\right) = p\left(x_1\right) \prod_{k=2}^{t} p\left(x_k \mid x_{1:k-1}\right).$$

Approximate μ by Sequential Importance Sampling

Hence, the importance weights can be updated recursively and the sequential importance sampling (SIS) approximation of μ_t is

$$\hat{\mu}_t = \sum_{i=1}^{N} \frac{w_t\left(x_{1:t}^{(i)}\right)}{\sum_{j=1}^{N} w_t\left(x_{1:t}^{(j)}\right)} h_t\left(x_{1:t}^{(i)}\right).$$

Equivalently, we still let

$$\hat{\mathrm{P}}\left(x_{1:t} = x_{1:t}^{(i)} \mid y_{1:t}\right) = \frac{w_t\left(x_{1:t}^{(i)}\right)}{\sum_{j=1}^{N} w_t\left(x_{1:t}^{(j)}\right)},$$

but we update $w_t(x_{1:t})$ sequentially. This is often called a particle filter.

State Space Model: Linear Gaussian State Space Model

Let

$$\begin{aligned} X_1 &\sim N\left(0, \sigma^2\right), \quad \text{initial density} \\ Y_t &= \beta X_t + W_t, \quad \text{observation density} \\ X_{t+1} &= \gamma X_t + V_{t+1}, \quad \text{transition density} \end{aligned}$$

where $W_t \sim N\left(0, \sigma_y^2\right)$, $V_t \sim N\left(0, \sigma_x^2\right)$, and errors are all mutually independent. We are interested in $\mu_t = \mathrm{E}\left[x_t \mid Y_{1:t} = y_{1:t}\right]$.

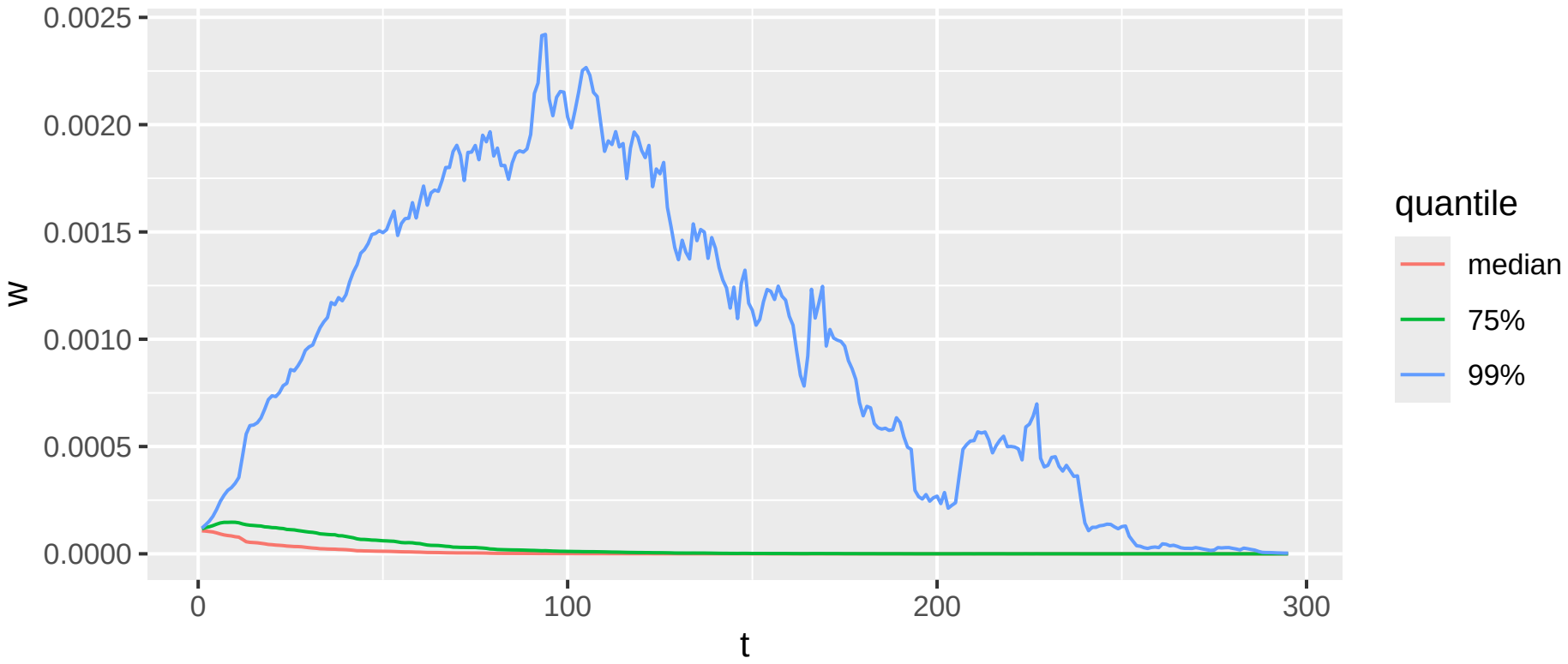
This model means that, for $t \geq 1$,

$$\begin{aligned} Y_t \mid X_t &\sim N\left(\beta X_t, \sigma_y^2\right), \\ X_{t+1} \mid X_t &\sim N\left(\gamma X_t, \sigma_x^2\right). \end{aligned}$$

Example: pH Values from Lake Washington, $N = 10000$



However, Something is Wrong



In fact, it has been proved that, as the dimension of integral increases, the maximum of weights converge in probability to 1!

Bootstrap Particle Filter

To avoid the degenerate weights, a simple but revolutionary idea is to resample the simulated trajectories (i.e., kill trajectories with low weights). This is known as SIS with resampling or bootstrap particle filter.

- We consider a multinomial distribution with possible outcomes $\left\{x_{1:t}^{(i)}\right\}_{i=1}^{N}$, with probabilities

$$\hat{\mathrm{P}}\left(x_{1:t} = x_{1:t}^{(i)} \mid y_{1:t}\right) = \frac{w_t\left(x_{1:t}^{(i)}\right)}{\sum_{j=1}^{N} w_t\left(x_{1:t}^{(j)}\right)}, \quad i = 1, \ldots N.$$

- We resample with replacement from such multinomial distribution to get N new trajectories.
- The weight of each resampled trajectory is reset to $1/N$.

After Resampling

Before resampling, we have the trajectories $\left\{x_{1:t}^{(i)}\right\}_{i=1}^{N}$, each with probability

$$\hat{\mathrm{P}}\left(x_{1:t} = x_{1:t}^{(i)} \mid y_{1:t}\right) = \frac{w_t\left(x_{1:t}^{(i)} \mid y_{1:t}\right)}{\sum_{j=1}^{N} w_t\left(x_{1:t}^{(j)} \mid y_{1:t}\right)}, \quad i = 1, \ldots N.$$

After resampling, we obtain new trajectories $\left\{\tilde{x}_{1:t}^{(i)}\right\}_{i=1}^{N}$, each has probability

$$\hat{\mathrm{P}}\left(x_{1:t} = \tilde{x}_{1:t}^{(i)} \mid y_{1:t}\right) = \frac{1}{N}, \quad i = 1, \ldots N.$$

Thus,

$$\hat{\mu}_t = \frac{1}{N}\sum_{i=1}^{N} h_t\left(\tilde{x}_{1:t}^{(i)}\right).$$

Effects of Resampling

Without resampling, only one particle receives a nonzero importance weight. Hence, it fails to represent the posterior distribution.

With resampling, the weights are more spread out.

1. Resampling does not create bias in the sense that

$$\mathrm{E}\left[\frac{1}{N}\sum_{i=1}^{N} h_t\left(\tilde{x}_{1:t}^{(i)}\right)\right] = \mathrm{E}\left[\sum_{i=1}^{N}\frac{w_t\left(x_{1:t}^{(i)}\right)}{\sum_{j=1}^{N} w_t\left(x_{1:t}^{(j)}\right)} h_t\left(x_{1:t}^{(i)}\right)\right].$$

2. The price is that resampling increases the variance:

$$\mathrm{Var}\left[\frac{1}{N}\sum_{i=1}^{N} h_t\left(\tilde{x}_{1:t}^{(i)}\right)\right] \geq \mathrm{Var}\left\{\sum_{i=1}^{N}\frac{w_t\left(x_{1:t}^{(i)}\right)}{\sum_{j=1}^{N} w_t\left(x_{1:t}^{(j)}\right)} h_t\left(x_{1:t}^{(i)}\right)\right\}.$$

More General Model

Beyond our state space models, we can have more complicated models such as

$$\begin{aligned} X_t \mid X_{t-1} &\sim p\left(x_t \mid X_{1:t-1}, \theta\right), \\ Y_t \mid X_t &\sim p\left(y_t \mid X_{1;t}, \theta\right), \end{aligned}$$

where θ is the vector of unknown parameters.

In the context of Bayesian statistics, we are interested in the posterior

$$p\left(x_{1:t}, \theta \mid y_{1:t}\right)$$

and $\mathrm{E}\left[h_t\left(X_{1:t}, \theta\right) \mid y_{1:t}\right]$.

Computer Intensive Statistics and Applications
Chapter 3: Bootstrap

Shaobo Jin

Department of Mathematics

Bootstrap Principle

The main principle of the bootstrap is that our data can be used to approximate the population. A large number of new observations drawn from this approximated population can be used to estimate distribution properties.

1. $\boldsymbol{X} = (X_1, ..., X_n) \sim \boldsymbol{P}$ (not necessarily iid sample) with distribution function $\boldsymbol{F}$ and the statistical model $\boldsymbol{P} \in \{\boldsymbol{P}_\theta; \theta \in \Theta\}$. X_i can be unidimensional or multidimensional.
2. We are interested in the parameter $\theta = \theta(\boldsymbol{F})$, the distribution properties of a given estimator $\hat{\theta} = \hat{\theta}(\boldsymbol{X})$, and the distribution properties of a given function $T(\boldsymbol{X}, \boldsymbol{F})$.
3. Our data is $\boldsymbol{x} = (x_1, ..., x_n)$ from $\boldsymbol{X} = (X_1, ..., X_n)$.
4. Estimate $\boldsymbol{F}$ or $\boldsymbol{P}$ by some estimator $\hat{\boldsymbol{F}}_n$ or $\boldsymbol{P}^*$. $\boldsymbol{P}^*$ is called the bootstrap distribution.

Bootstrap Principle

5 Draw independently B random samples $\boldsymbol{x}^*_{(j)}$ of size n from $\boldsymbol{P}^*$, $j = 1, ..., B$, that is $\boldsymbol{X}^*_{(j)} \sim \boldsymbol{P}^*$. The samples $\boldsymbol{x}^*_{(j)}$ are called the bootstrap samples.

6 For each j, estimate θ using the bootstrap sample $\boldsymbol{x}^*_{(j)}$ as $\hat{\theta}^*_j = \hat{\theta}\left(\boldsymbol{x}^*_{(j)}\right)$. The bootstrap replications of $\hat{\theta}$ are $\hat{\theta}^*_1$, ..., $\hat{\theta}^*_B$. The bootstrap replications of the function $T(\boldsymbol{X}, \boldsymbol{F})$ are $T^*_{(j)} = T\left(\boldsymbol{X}^*_{(j)}, \hat{\boldsymbol{F}}_n\right)$, $j = 1, ..., B$.

7 Pretend that the bootstrap replications are an iid sample from the distribution of $\hat{\theta}$ or $T(\boldsymbol{X}, \boldsymbol{F})$.

The main idea is that when considering a function $Q\left(\hat{\theta}, \theta\right)$ using the observed data, the bootstrap treats $\hat{\theta}$ as the truth and $\hat{\theta}^*_j$ as the estimator, i.e., $Q\left(\hat{\theta}^*_j, \hat{\theta}\right)$.

Bootstrap: An Example

Suppose that we have observed iid data $x_1, ..., x_n$ from a statistical model $X \sim P$ with distribution function F. Our focus is the median of this distribution, denoted by $m(F)$.

- In classical inference, we can use the sample median $m_{(n)}$ to estimate the population median $m(F)$.
- The "bias" can be computed by $T(\boldsymbol{X}, \boldsymbol{F}) = m_{(n)} - m(F)$.
- However, we do not know the true value $m(F)$.

In bootstrap, the first steps are the same as classical inference:

2. The parameter is $m(F)$. We are interested in $T = m_{(n)} - m(F)$.
3. Our data is $\boldsymbol{x} = (x_1, ..., x_n)$ from $\boldsymbol{X} = (X_1, ..., X_n)$.

Bootstrap: An Example

4 If we choose the nonparametric approach, we estimate $\boldsymbol{F}$ by some estimator $\hat{\boldsymbol{F}}_n$ using $\boldsymbol{x}$ as

$$\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} 1(x_i \leq x).$$

The estimated distribution P^* is $\mathrm{P}(X = x_i) = n^{-1}$ for any i. The estimator of $m(F)$ is $m_{(n)} = m\left(\hat{F}_n\right)$.

Bootstrap: An Example

From step 5, we need Monte Carlo simulation.

5 Given step 4 in our example, draw independently B bootstrap samples $\boldsymbol{x}^*_{(j)} = \left(x^*_{1j}, ..., x^*_{nj}\right)$ of size n from P^*, $j = 1, ..., B$.

6 For each j, estimate $m\left(\hat{F}_n\right)$ using the bootstrap sample $\boldsymbol{x}^*_{(j)}$ as $m_{(n),j}$, where $m_{(n),j}$ is the sample median of $x^*_{1j}, ..., x^*_{nj}$. Compute the difference

$$T_j \quad = \quad m_{(n),j} - m_{(n)},$$

since $m_{(n)}$ is the median of the distribution P^*.

7 View $m_{(n),j} - m_{(n)}$, $j = 1, ..., B$, as an iid sample from the distribution of $m_{(n)} - m(F)$. For example, the bias of the estimator $m_{(n)}$ can be approximated by

$$\frac{1}{B}\sum_{j=1}^{B} m_{(n),j} - m_{(n)}.$$

Parametric and Nonparametric Bootstrap

4 Estimate $\boldsymbol{F}$ by some estimator $\hat{\boldsymbol{F}}_n$ (e.g., parametric or nonparametric). Denote the estimated distribution by $\boldsymbol{P}^*$. $\boldsymbol{P}^*$ is called the bootstrap distribution.

5 Draw independently B random samples $\boldsymbol{x}^*_{(j)}$ of size n from $\boldsymbol{P}^*$, $j = 1, ..., B$, that is $\boldsymbol{X}^*_{(j)} \sim \boldsymbol{P}^*$. The samples $\boldsymbol{x}^*_{(j)}$ are called the bootstrap samples.

Parametric and nonparametric approaches can make some difference.

1. The parametric bootstrap assumes a parametric form of the statistical model $\boldsymbol{P}$ index by some parameter θ.
   - For example, we assume $X \sim N\left(\mu, \sigma^2\right)$. The bootstrap samples are drawn from $N\left(\hat{\mu}, \hat{\sigma}^2\right)$.
2. The nonparametric bootstrap estimates $\boldsymbol{F}$ by the empirical distribution function.
   - The estimated distribution is $\mathrm{P}\left(X = x_i\right) = n^{-1}$ for any i.

Bootstrap: Median Example Again

If we want to use parametric bootstrap, we need to assume some parametric form.

4 If we assume our data are iid draws from an exponential distribution with mean λ, the maximum likelihood estimator of λ is $\hat{\lambda} = \bar{X}$. The estimated distribution P^* is an exponential distribution with mean $\bar{X}$, denoted by $\text{Exp}\left(\bar{X}\right)$. Our estimator of the median is $\bar{X} \log 2$.

5 Given step 4 in our example, draw independently B bootstrap samples $\boldsymbol{x}^*_{(j)} = \left(x^*_{1j}, ..., x^*_{nj}\right)$ of size n from $\text{Exp}\left(\bar{X}\right)$, $j = 1, ..., B$.

Empirical Distribution Function

For notation simplicity, we consider $X \in \mathbb{R}$.

Definition

Let x_1, ..., x_n be a sequence of measurements, then the fraction

$$\hat{F}_n(x) = \frac{1}{n}\sum_{i=1}^{n} 1(x_i \leq x)$$

is called the empirical cumulative distribution function (ecdf) or the empirical distribution function.

Example

Find the ecdf of the data 1, 1, 2, 3.

Plug-In Principle in Nonparametric Bootstrap

Real Word	Bootstrap World
cdf F	ecdf $\hat{F}_n$
$X \sim F$	$X^* \sim \hat{F}_n$
$\theta(F)$	$\theta\left(\hat{F}_n\right)$
$\hat{\theta}(\boldsymbol{X})$	$\hat{\theta}^* = \hat{\theta}(\boldsymbol{X}^*)$
$T(\boldsymbol{X}, F)$	$T\left(\boldsymbol{X}^*, \hat{F}_n\right)$

For example, if $\theta(F) = \mathrm{E}(X^r)$, then

$$\theta\left(\hat{F}_n\right) = \int x^r d\hat{F}_n = n^{-1} \sum_{i=1}^{n} x_i^r,$$

the sample moments.

Why Does Bootstrap Work? An Example

For simplicity, suppose that $X_1, ..., X_n$ is an iid sample from the distribution P, that the distribution function F is continuous, and that $\sigma^2 = \mathrm{Var}(X) < \infty$ is known. Our parameter of interest is $\mu = \mathrm{E}[X]$. We estimate it by $\mu\left(\hat{F}_n\right) = \bar{x}$.

- By the central limit theorem,

$$Z = \frac{\sqrt{n}\left(\bar{X} - \mu\right)}{\sigma} \quad \xrightarrow{d} \quad N(0,1).$$

- In the bootstrap world, given $\hat{F}_n$,

$$Z^* = \frac{\sqrt{n}\left(\bar{X}^* - \bar{x}\right)}{\sigma} \quad \xrightarrow{d} \quad N(0,1).$$

- Hence,

$$\mathrm{P}(Z < z) - \mathrm{P}\left(Z^* < z \mid \hat{F}_n\right) \quad = \quad o_P(1).$$

More Generally: Consistency

Bootstrap aims to approximate the exact distribution of $T(\boldsymbol{X}, \boldsymbol{F})$ using the distribution of $T\left(\boldsymbol{X}^*, \hat{\boldsymbol{F}}_n\right)$.

- For example, we are interested in the distribution of $T(\boldsymbol{X}, \boldsymbol{F}) = \left(\hat{\theta} - \theta\right)/\hat{\sigma}$.
- The bootstrap counterpart is $\left(\hat{\theta}^* - \hat{\theta}\right)/\hat{\sigma}^*$.
- We basically use the distribution of $\left(\hat{\theta}^* - \hat{\theta}\right)/\hat{\sigma}^*$ given $\hat{\boldsymbol{F}}_n$ to approximate the distribution of $\left(\hat{\theta} - \theta\right)/\hat{\sigma}$.

The bootstrap is consistent if

$$\sup_t \left| \mathrm{P}\left(T(\boldsymbol{X}, \boldsymbol{F}) \le t\right) - \mathrm{P}\left(T\left(\boldsymbol{X}^*, \hat{\boldsymbol{F}}_n\right) \le t \mid \hat{\boldsymbol{F}}_n\right) \right| \xrightarrow{P} 0,$$

where the second probability is still a random variable.

Why Does Nonparametric Bootstrap Work?

By the Glivenko-Cantelli theorem, we know that

$$\mathrm{P}\left(\sup_x \left|\hat{F}_n(x) - F(x)\right| > \epsilon\right) \quad \to \quad 0.$$

Hence for sufficiently large n, sampling from F is similar to sampling from $\hat{F}_n$.

However, we still need more assumptions in order for bootstrap to work. Suppose that $T(\boldsymbol{X}, \boldsymbol{F}) \xrightarrow{d} T_A$ with distribution function $F_A(t)$. We need to assume

1. For any sequence of distributions $\{G_n\}$ such that $\lim_{n\to\infty} G_n(t) = F(t)$, we have $\lim_{n\to\infty} P\left(T\left(\tilde{\boldsymbol{X}}, G_n\right) \le t\right) = F_A(t)$.
2. $F_A(t)$ is a continuous function.

Why Does Nonparametric Bootstrap Work?

To simplify the assumptions a little, the bootstrap is consistent if, for every x,

$$\begin{aligned} \mathrm{P}\left(T\left(\boldsymbol{X}, \boldsymbol{F}\right) \leq x\right) &\rightarrow F_A\left(x\right), \\ \mathrm{P}\left(T\left(\boldsymbol{X}^*, \hat{\boldsymbol{F}}_n\right) \leq x \mid \hat{\boldsymbol{F}}_n\right) &\xrightarrow{P} F_A\left(x\right), \end{aligned}$$

and F_A is a continuous function.

Example

Consider again bootstrapping the mean. The central limit theorem implies that the distribution functions of

$$Z = \frac{\sqrt{n}\left(\bar{X} - \mu\right)}{\sigma} \quad \text{and} \quad Z^* = \frac{\sqrt{n}\left(\bar{X}^* - \bar{X}\right)}{\hat{\sigma}}$$

converge to $\Phi\left(\right)$, the distribution function of $N\left(0, 1\right)$.

Bootstrap Can Fail

Consider an iid sample X_1, ..., X_n from Uniform $[0, \theta]$. The maximum likelihood estimator of θ is

$$\hat{\theta} \quad = \quad \max_i X_i = X_{[n]}.$$

- We can show that $\sqrt{n}\left(X_{[n]} - \theta\right)$ does not converge to a normal distribution. In fact,

$$Q = \frac{n\left(\theta - X_{[n]}\right)}{\theta} \quad \xrightarrow{d} \quad \text{Exp}\left(1\right).$$

- If nonparametric bootstrap works, the distribution of $Q^* = n\left(x_{[n]} - X^*_{[n]}\right)/x_{[n]}$ should converge to distribution of Q.
- But

$$P\left(Q^* = 0\right) \quad \to \quad 1 - e^{-1}.$$

How Do We Know Bootstrap Works in Practice?

We can

- show that the conditions in the previous slides hold;
- or, run some simulations to get some ideas about whether it works.

In case, bootstrap does not work, we can try to modify the bootstrap algorithm.

Example

In the uniform distribution example, we taking a bootstrap sample of m instead of n, such as $m = n^{1/3}$ and $m = n^{1/2}$ and check whether it works.

Parametric Bootstrap

The main difference between parametric bootstrap and nonparametric bootstrap is that we simulate X^* from $F\left(x;\hat{\theta}\right)$, instead of $\hat{F}_n$, where $F\left(x;\theta\right)$ is the distribution function with parameter θ.

In order for the parametric bootstrap to work, we can establish similar conditions as in the nonparametric bootstrap, but with $F\left(x;\hat{\theta}\right)$, instead of $\hat{F}_n$.

An Extreme Example

One important property of bootstrap is that it can often achieve more accurate approximations than the usual asymptotic approaches.

Example

Consider again the uniform distribution but we apply parametric bootstrap. It is easy to show that

$$\mathrm{P}\left(\frac{n\left(x_{[n]}-X^*_{[n]}\right)}{x_{[n]}} > t \mid X_{[n]}\right) = \left(1-\frac{t}{n}\right)^n,$$

$$\mathrm{P}\left(\frac{n\left(\theta-X_{[n]}\right)}{\theta} > t\right) = \left(1-\frac{t}{n}\right)^n.$$

But the asymptotic method relies on

$$\mathrm{P}\left(\frac{n\left(\theta-X_{[n]}\right)}{\theta} > t\right) \to e^{-t}.$$

Error of Approximation

In general, we are interested in the exact distribution of a random quantity $T(\boldsymbol{X}, \boldsymbol{F})$.

- Its exact distribution is difficult to derive, so we use its asymptotic distribution to approximate the exact distribution.

It is often the case that $T(\boldsymbol{X}, \boldsymbol{F}) = Z + O_P\left(n^{-1/2}\right)$, where $Z \sim N(0,1)$. Hence the error of approximation is $O_P\left(n^{-1/2}\right)$.

- For example, $\sqrt{n}\left(\hat{\theta} - \theta\right)$ often converges in distribution to a normal distribution.

The Quantity $T(\boldsymbol{X}, \boldsymbol{F})$

The choice of $T(\boldsymbol{X}, \boldsymbol{F})$ can have effects on the accuracy of approximation.

Example

Consider again the uniform distribution but we apply parametric bootstrap and consider $T = n\left(\theta - X_{[n]}\right)$. It is easy to show that

$$\begin{aligned} \mathrm{P}\left(n\left(x_{[n]} - X^*_{[n]}\right) > t \mid X_{[n]}\right) &= \left(1 - \frac{t}{nx_{[n]}}\right)^n, \\ \mathrm{P}\left(n\left(\theta - X_{[n]}\right) > t\right) &= \left[1 - \frac{t}{n\theta}\right]^n. \end{aligned}$$

We achieve an approximation error of order $O_P\left(n^{-1}\right)$, same order as the asymptotic approximation.

Pivot

The "magic" behind some "good" choices of $T(\boldsymbol{X}, \boldsymbol{F})$ is that it is a pivot.

Definition

A quantity is a pivot if it is a function of $\boldsymbol{X}$ and the parameter but its distribution does not depend on the parameter. A quantity is an asymptotic pivot if its asymptotic distribution does not depend on the parameter, despite that its distribution depends on the parameter.

Example

In the uniform distribution example, $n\left(\theta - X_{[n]}\right)/\theta$ is a pivot but $n\left(\theta - X_{[n]}\right)$ is not.

Edgeworth Expansion

Suppose that S_n is a statistic with a limiting distribution $N(0,1)$. Then, the Edgeworth expansion means that

$$\mathrm{P}(S_n \leq s) = \Phi(s) + n^{-1/2} p_1(s)\,\phi(s) + O\left(n^{-1}\right),$$

where $p_1(x)$ is a polynomial of degree 2.

- In the special case where $S_n = \sqrt{n}\left(\bar{X} - \mu\right)/\sigma$ for iid data with $\sigma^2 = \mathrm{Var}(X)$,

$$p_1(s) = \frac{\mathrm{E}\left[(X-\mu)^3\right]}{6\sigma^3}(1-s)^2.$$

In the bootstrap world, the Edgeworth expansion yields

$$\mathrm{P}\left(S_n \leq s \mid \hat{F}\right) = \Phi(s) + n^{-1/2} p_1^*(s)\,\phi(s) + O\left(n^{-1}\right).$$

Second-Order Accuracy

Hence,

$$\mathrm{P}\left(S_n \leq s\right) - \mathrm{P}\left(S_n \leq s \mid \hat{F}\right) \quad = \quad n^{-1/2}\left[p_1\left(s\right) - p_1^*\left(s\right)\right]\phi\left(s\right) + O\left(n^{-1}\right).$$

If we are lucky such that $p_1\left(s\right) - p_1^*\left(s\right) = O_P\left(n^{-1/2}\right)$, we obtain the second-order accuracy as

$$\mathrm{P}\left(S_n \leq s\right) - \mathrm{P}\left(S_n \leq s \mid \hat{F}\right) \quad = \quad O_P\left(n^{-1}\right).$$

- In the special case where $S_n = \sqrt{n}\left(\bar{X} - \mu\right)/\sigma$ for iid data with $\sigma^2 = \mathrm{Var}\left(X\right)$, $\mathrm{E}\left[\left(X - \mu\right)^3\right]/\sigma^3$ is the skewness and it is asymptotically normal.

If S_n is not a pivot, then the leading $\Phi\left(\right)$ terms will not cancel each other, leading to $O_P\left(n^{-1/2}\right)$.

Regression

Consider a regression model

$$Y \quad = \quad X\beta + \epsilon,$$

where $\mathrm{E}[\epsilon] = 0$, and $\epsilon = \begin{bmatrix} \epsilon_1 & \cdots & \epsilon_n \end{bmatrix}^T$ are iid from some distribution P. But $Y = \begin{bmatrix} Y_1 & \cdots & Y_n \end{bmatrix}^T$ are not iid. We only assume that they are independent.

- The estimator $\hat{\beta}$ of β can be estimated using X and Y. For example, the OLS estimator of β is $\hat{\beta} = \left(X^T X\right)^{-1} X^T y$.
- The residual vector is $\hat{\epsilon} = Y - X\hat{\beta}$.

Residual Bootstrap

Algorithm 1: Residual bootstrap

Fit the regression model using observed data ;

Obtain the residuals $\hat{\epsilon} = Y - X\hat{\beta}$;

Normalize the residuals, if needed, and obtain $\tilde{\epsilon}$ such that $n^{-1}\sum_{i=1}^{n}\tilde{\epsilon}_i = 0$;

for *each integer* j *from 1 to* B **do**

  Draw a random sample ϵ_j^* of size n from the empirical distribution of $\tilde{\epsilon}$, i.e., sample with replacement from $\tilde{\epsilon}$;

  Calculate the bootstrap response $Y_j^* = X\hat{\beta} + \epsilon_j^*$;

  Obtain the bootstrap OLS estimator $\hat{\beta}_j^* = \left(X^T X\right)^{-1} X^T y_j^*$;

end

Casewise Bootstrap

Algorithm 2: Casewise bootstrap

for *each integer* j *from 1 to* B **do**
  Sample with replacement of size n from (y_i, x_i) ;
  Fit the model using the bootstrap sample (y_i^*, x_i^*), $i = 1, ..., n$;
  Obtain the bootstrap estimate $\hat{\beta}_j^*$;
end

Assumptions for Regression

When we perform regression analysis, we often assume that ϵ_1, ..., ϵ_n are iid from $N\left(0, \sigma^2\right)$. This assumption is ciritical for the t-test and F-test in the regression. This assumption actually contains two parts:

1. error terms are normally distributed,
2. the variances of the error terms are the same, i.e., homoscedasticity.

Once the assumption is violated, the OLS estimator often remains valid, but the t-test and the F-test are invalid.

- We can use bootstrap to overcome the violated assumptions.

Residual Bootstrap or Casewise Bootstrap?

In residual bootstrap, we sample the residuals from the empirical distribution (resample with replacement). Hence,

- it still requires homoscedasticity.
- it also assumes Y is random, but X is not random
  - For example, we want to compare the caffeine content of Lindvalls Mörkrost och Lindvalls Brygg.

In casewise bootstrap, we resample the pair (y_i, x_i). Hence,

- it does not require homoscedasticity,
- it requires both Y and X are random.

method = "case" eller method = “residual”?

Boot {car} R Documentation

Bootstrapping for regression models

Description

This function provides a simple front-end to the `boot` function in the **boot** package that is tailored to bootstrapping based on regression models. Whereas `boot` is very general and therefore has many arguments, the `Boot` function has very few arguments.

Usage

```
Boot(object, f=coef, labels=names(f(object)), R=999,
  method=c("case", "residual"), ncores=1, ...)
```

Autocorrelation

A common assumption in regression analysis is that our data are independent.

- The independence assumption is often violated in time series data due to autocorrelation.

When we resample time series data, the bootstrapped time series should keep their dependence as much as possible.

- When we deal with iid data previously, the bootstrap sample is iid conditional on the observed data.

Blockwise Bootstrap

One idea is that we can divide the time series into several blocks. If the dependence is weak and the blocks are long, the dependence is preserved.

Algorithm 3: Blockwise Bootstrap

Specify a fixed block length L such that $n = KL$;
Divide the time series into K non-overlapping blocks of length L. Denote the blocks by B_1, ..., B_K ;
for *each integer* j *from 1 to* B **do**
    Sample with replacement k_1^*, ..., k_K^* from 1, ..., K ;
    The bootstrapped time series is $B_{k_1^*}$, ..., $B_{k_K^*}$;
end

Blockwise Bootstrap: Some Details

- It is common to choose a length of order n^γ for $\gamma \in (0, 1)$ such that the length tends to infinity and the number of blocks also tends to infinity as $n \to \infty$.
- A limitation is that the bootstrapped series is not stationary, because the the joint distribution of resampled observations close to a join between blocks differs from that in the centre of a block.

Stationary Bootstrap

The stationary bootstrap takes blocks whose lengths L are geometrically distributed with density

$$\mathrm{P}(L = j) = (1-p)^{j-1} p, \quad j = 1, 2, ...$$

The resampled series are stationary with mean block length p^{-1}.

Algorithm 4: Stationary Bootstrap

1 Specify a tuning parameter $0 < p < 1$;
2 **for** *each integer* j *from 1 to* B **do**
3 Generate independently L_1, L_2 ... from a geometric distribution with parameter p ;
4 Draw independently I_1, I_2, ... uniformly distributed on 1, ..., n ;
5 The bootstrapped time series is $X_j^* = (B_{I_1,L_1}, B_{I_2,L_2}, ...)$, where $B_{a,b} = (X_a, ..., X_{a+b-1})$;
6 **end**

Stationary Bootstrap: Some Details

- If $j > n$, then we let $X_j = X_i$, where $i = j \bmod n$.
- How many L_k and I_k are simulated depends on when X_j^* reaches length n. We stop once n observations in the pseudo-time series are generated.
- Conditional on $X = (X_1, ..., X_n)$, X^* is stationary.

Confidence Set

One of the most important tasks of bootstrap is to construct confidence sets/intervals. Consider an iid sample $X_1, ..., X_n$ from P_θ. We want a confidence set $C(\boldsymbol{X})$ for $g(\theta)$ such that

$$\mathrm{P}(g(\theta) \in C(\boldsymbol{X})) = 1 - \alpha.$$

We will introduce

1. percentile bootstrap confidence interval,
2. bootstrap pivotal method,
   1. basic bootstrap confidence interval,
   2. normal bootstrap confidence interval,
   3. studentized bootstrap confidence interval,
3. BC_a-Confidence Interval.

Approach 1: Percentile Bootstrap Interval

From the bootstrap sample we can obtain the bootstrap replications $\hat{\theta}_1^*$, ..., $\hat{\theta}_B^*$. Let the $100\alpha/2$th and $100\,(1-\alpha/2)$th percentiles of the bootstrap replications be L^* and U^*, respectively, such that

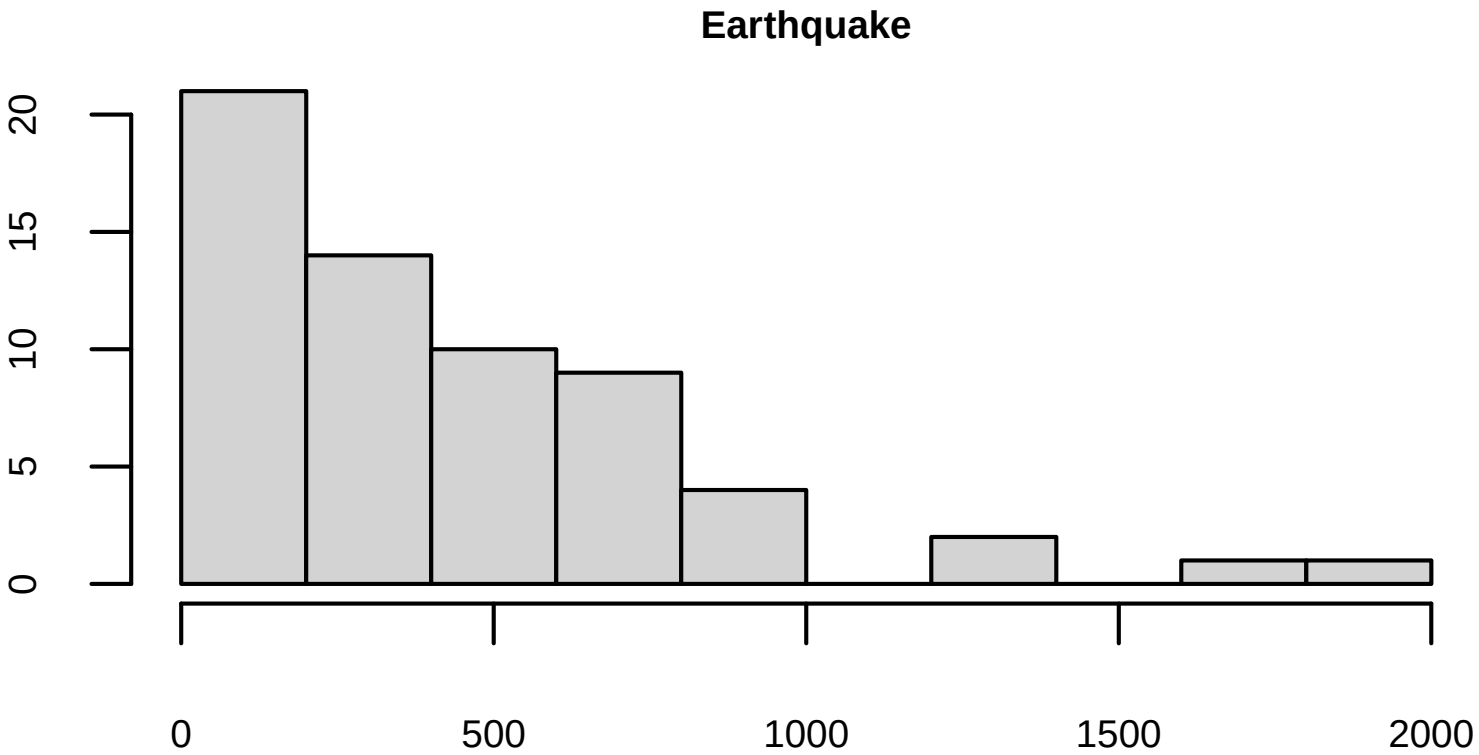
$$\mathrm{P}^*\left(L^* \le \hat{\theta}^* \le U^*\right) \;=\; 1-\alpha.$$

Hereafter, we use P^* to denote that the probability is evaluated for the bootstrap sample, sampled from some $\hat{F}$ (e.g., ecdf or $\mathrm{P}_{\hat{\theta}}$).
The percentile bootstrap confidence interval is

$$\{\theta:\; L^* \le \theta \le U^*\}.$$

Example: Earthquake Data

The time intervals in days between 63 successive serious earthquakes world-wide have been recorded.



We assume that our data are iid from $\text{Exp}(\theta)$ with mean θ. We want to estimate $p = P(X > 1500) = \exp\left(-\frac{1500}{\theta}\right)$.

Approach 2: Bootstrap Pivotal Method

Let the random variable $Q(\boldsymbol{X}, \theta)$ be a pivot. Deriving the exact distribution of $Q(\boldsymbol{X}, \theta)$ may not be easy, but we can estimate the quantiles of the distribution of $Q(\boldsymbol{X}, \theta)$ by boostrapping.

- We can find L^* and U^* from the bootstrap sample such that

$$\mathrm{P}^*\left(L^* \leq Q\left(\boldsymbol{X}^*, \hat{\theta}\right) \leq U^*\right) \quad = \quad 1-\alpha.$$

- Then we approximate

$$\mathrm{P}_\theta\left(L^* \leq Q(\boldsymbol{X}, \theta) \leq U^*\right) \quad = \quad 1-\alpha.$$

The bootstrap confidence set is

$$\{\theta: \ L^* \leq Q(\boldsymbol{X}, \theta) \leq U^*\}$$

Basic Bootstrap Confidence Interval

Let $Q(\boldsymbol{X}, \theta) = \hat{\theta} - \theta$, although it may not be a pivot. The bootstrap replicates are $Q^*_{(j)} = \hat{\theta}^*_{(j)} - \hat{\theta}$. Then,

$$1 - \alpha \quad = \quad \mathrm{P}^* \left(Q^*_L \le \hat{\theta}^* - \hat{\theta} \le Q^*_U \right).$$

Hence,

$$\begin{aligned} 1 - \alpha &= \mathrm{P} \left(Q^*_L \le \hat{\theta} - \theta \le Q^*_U \right) \\ &= \mathrm{P} \left(\hat{\theta} - Q^*_U \le \theta \le \hat{\theta} - Q^*_L \right). \end{aligned}$$

The basic bootstrap confidence interval is

$$\left\{ \theta : \ \hat{\theta} - Q^*_U \le \theta \le \hat{\theta} - Q^*_L \right\}.$$

Basic Bootstrap Interval

Suppose that θ_L^* and θ_U^* are the quantiles such that

$$\mathrm{P}^*\left(\theta_L^* \le \hat{\theta}^* \le \theta_U^*\right) \;=\; 1-\alpha.$$

Then,

$$\mathrm{P}^*\left(Q_L^* \le \hat{\theta}^* - \hat{\theta} \le Q_U^*\right) \;=\; 1-\alpha$$

means that

$$Q_L^* = \hat{\theta}_L^* - \hat{\theta}, \qquad Q_U^* = \hat{\theta}_U^* - \hat{\theta}.$$

Hence, the basic bootstrap confidence interval can also be expressed as

$$\left\{\theta:\; 2\hat{\theta} - \hat{\theta}_U^* \le \theta \le 2\hat{\theta} - \hat{\theta}_L^*\right\}.$$

Normal Bootstrap Confidence Interval

Let

$$Q(\boldsymbol{X}, \theta) = \frac{\hat{\theta} - \theta}{\sqrt{\hat{V}}},$$

where $\hat{V}$ is an estimator of $V = \text{Var}\left(\hat{\theta}\right)$, i.e., $\sqrt{\hat{V}}$ is the standard error. If the distribution of Q is $N(0,1)$, the usual asymptotic confidence interval is

$$1 - \alpha = \text{P}\left(\hat{\theta} - \lambda_{1-\alpha/2}\sqrt{\hat{V}} \leq \theta \leq \hat{\theta} + \lambda_{1-\alpha/2}\sqrt{\hat{V}}\right),$$

where λ_α is the quantile of $N(0,1)$. The bootstrap counterpart is

$$1 - \alpha = \text{P}^*\left(\hat{\theta}^* - \lambda_{1-\alpha/2}\sqrt{\hat{V}^*} \leq \hat{\theta} \leq \hat{\theta}^* + \lambda_{1-\alpha/2}\sqrt{\hat{V}^*}\right),$$

Normal Bootstrap Confidence Interval

1. We can estimate $V = \text{Var}\left(\hat{\theta}\right)$ using the variance of the bootstrap sample.
2. We can also estimate the bias of $\hat{\theta}$ by

$$\text{Bias} \quad = \quad \frac{1}{B}\sum_{j=1}^{B}\hat{\theta}^*_{(j)} - \hat{\theta}.$$

Hence, we can approximate

$$1-\alpha \quad = \quad \text{P}\left(\hat{\theta} - \text{Bias} - \lambda_{1-\alpha/2}\sqrt{\hat{V}^*} \le \theta \le \hat{\theta} - \text{Bias} + \lambda_{1-\alpha/2}\sqrt{\hat{V}^*}\right).$$

The normal bootstrap confidence interval is

$$\left\{\theta:\ \hat{\theta} - \text{Bias} - \lambda_{1-\alpha/2}\sqrt{\hat{V}^*} \le \theta \le \hat{\theta} - \text{Bias} + \lambda_{1-\alpha/2}\sqrt{\hat{V}^*}\right\}.$$

Studentized Bootstrap Confidence Interval

Let

$$Q\left(\boldsymbol{X}, \theta\right) \quad = \quad \frac{\hat{\theta} - \theta}{\sqrt{\hat{V}}},$$

where $\hat{V}$ is an estimator of Var $\left(\hat{\theta}\right)$, i.e., $\sqrt{\hat{V}}$ is the standard error. Then,

$$1 - \alpha \quad = \quad \mathrm{P}^* \left(Q_L^* \leq \frac{\hat{\theta}^* - \hat{\theta}}{\sqrt{\hat{V}^*}} \leq Q_U^* \right).$$

The studentized bootstrap confidence interval is

$$\left\{ \theta : \ \hat{\theta} - Q_U^* \sqrt{\hat{V}} \leq \theta \leq \hat{\theta} - Q_L^* \sqrt{\hat{V}} \right\}.$$

Variance Estimator

In order to use the normal bootstrap CI and the studentized bootstrap CI, we need to estimate $\text{Var}\left(\hat{\theta}\right)$.

- We have used the sample variance of the bootstrap replicates in the normal boostrap CI.

Alternative approaches include the

- nonparametric delta method, a model-by-model approach,
- bootstrap the bootstrap sample.

Bootstrap Variances For Studentized Intervals

We can apply a nested bootstrap to approximate the variance.

1. Draw a bootstrap sample.
2. Estimate the statistic as usual. Denote the estimate by $\hat{\theta}^*_{(j)}$.
3. Bootstrap the bootstrap sample, and estimate the variance by using the variance of estimates across the bootstrapped estimates. Denote the estimate of the variance by $\hat{V}^*_{(j)}$.
4. Calculate $\hat{t}^*_{(j)} = \left(\hat{\theta}^*_{(j)} - \hat{\theta}\right) / \sqrt{\hat{V}^*_{(j)}}$.

Order of Accuracy

The above confidence intervals are often first-order accurate, i.e.,

$$\mathrm{P}\left(L \leq \theta \leq U\right) \quad = \quad 1 - \alpha + O\left(n^{-1/2}\right).$$

The bias-corrected and accelerated (BC_a) CI is second-order accurate, i.e.,

$$\mathrm{P}\left(L \leq \theta \leq U\right) \quad = \quad 1 - \alpha + O\left(n^{-1}\right).$$

Second-Order Accurate

Suppose that there exists a monotone transformation $\eta = h(\theta)$ such that $\hat{\eta} = h\left(\hat{\theta}\right)$ satisfies

$$\frac{\hat{\eta} - \eta}{\sqrt{(1 + a\eta)^2}} + w \quad = \quad N(0,1) + o_{\mathrm{P}}(1),$$

for bias w and acceleration coefficient a.

To achieve second-order accuracy, we choose w and a such that

$$\mathrm{P}\left(\frac{\hat{\eta} - \eta}{\sqrt{(1 + a\hat{\eta})^2}} + w < z\right) \quad = \quad \Phi(z) + O\left(n^{-1}\right).$$

This is often possible because of the pivot and Edgeworth expansion trick that we discussed above.

BC_a-Confidence Interval

For known a and w, define

$$\hat{\eta}_{\alpha/2} = \hat{\eta} + (1 + a\hat{\eta}) \frac{\lambda_{\alpha/2} + w}{1 - a\left(\lambda_{\alpha/2} + w\right)}.$$

Then, for a monotone h, we have

$$\mathrm{P}^*\left(\hat{\eta}^* < \hat{\eta}_{\alpha/2}\right) = \Phi\left(\frac{\lambda_{\alpha/2} + w}{1 - a\left(\lambda_{\alpha/2} + w\right)} + w\right) + O\left(n^{-1}\right).$$

The limits of the BC_a interval are the the $\Phi\left(\frac{\lambda_{\alpha/2}+w}{1-a\left(\lambda_{\alpha/2}+w\right)} + w\right)$ and $\Phi\left(\frac{\lambda_{1-\alpha/2}+w}{1-a\left(\lambda_{1-\alpha/2}+w\right)} + w\right)$ quantiles of the bootstrap replicates $\left\{\hat{\theta}_{(j)}\right\}$.

Estimating a and w

We do not need to find the exact form of the transformation h. We only need its existence. But we need to estimate a and w. For example,

1. We can estimate w by

$$\hat{w} = \Phi^{-1}\left(\frac{\#\left\{\hat{\theta}^*_{(j)} < \hat{\theta}\right\}}{B}\right).$$

2. We can estimate a by

$$\hat{a} = \frac{\sum_{i=1}^n \left(\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}\right)^3}{6\left[\sum_{i=1}^n \left(\hat{\theta}_{(\cdot)} - \hat{\theta}_{(i)}\right)^2\right]^{3/2}},$$

where $\hat{\theta}_{(i)}$ is the jackknife estimate of θ and $\hat{\theta}_{(\cdot)}$ is the average of $\hat{\theta}_{(i)}$.

Transformed Confidence Interval

Suppose that we have obtained a CI for θ as

$$\{\theta : \ L \leq \theta \leq U\}.$$

Let $\eta = h(\theta)$ be a monotone smooth transformation. The confidence interval for η is then

$$\{\eta : \ h(L) \leq \eta \leq h(U)\}.$$

Confidence Interval and Hypothesis Test

1. Using bootstrap confidence interval: suppose that the bootstrap confidence interval is $L \leq \theta \leq U$. Consider H_0: $\theta = \theta_0$. If θ_0 lies in the bootstrap CI, then we cannot reject H_0.
2. We can also bootstrap to approximate a p-value.

Suppose that we observe an iid sample $X_1, ..., X_n$ from $X \sim P_\theta$ and we are interested in $\gamma = g(\theta)$. In hypothesis testing,

1. Construct a test statistic $T = T(X_1, ..., X_n)$ and the observed value is $t_{\text{obs}} = T(x_1, ..., x_n)$.
2. Find the distribution of T: $T \sim P_\theta^{(T)}$.

Hypothesis Testing

3 Compute the p-value

$$p = \int I_{A_{\text{obs}}}(t)\, dP_0^{(T)},$$

where A_{obs} is the region in the direction of the extreme values, $I_A(t)$ is the indicator function, and $P_0^{(T)}$ is the null distribution of T, i.e., the distribution of T under H_0.

4 Reject H_0 if $p < \alpha$ for a given significance level α.

However, it is not always easy to obtain the closed form of the null distribution $P_0^{(T)}$. But we can use bootstrap to approximate the distribution and the p-value.

Parametric Bootstrap Hypothesis Test

Suppose that $X \sim P_{\theta,\lambda}$, where θ is the parameter of interest and λ is the nuisance parameter. The distribution of test statistic T depends on θ and λ.

- Consider H_0: $\theta = \theta_0$.
- The null distribution $P^{(T)}_{\theta_0,\lambda}$ is not completely known due to unknown λ.

Algorithm 5: Parametric bootstrap hypothesis test

1 Estimate $\hat{\lambda}$;
2 **for** $j = 1$ *in* $1 : B$ **do**
3 | Draw a random sample $\{x^*_{i,j}\}$ of size n from $P_{\theta_0,\hat{\lambda}}$;
4 | Calculate the test statistic $T^*_j = T\left(x^*_{1,j}, ..., x^*_{n,j}\right)$;
5 **end**
6 Approximate the p-value by $p_{boot} = \#\left\{j : T^*_j \in A_{\text{obs}}\right\}/B$

Parametric Bootstrap Hypothesis Test: Example

Example

Suppose that we have a sample from

$$X = U_1 Z_1 + (1 - U_1) Z_2,$$

where $U_1 \sim \text{Bernoulli}\left(\frac{1}{2}\right)$, $Z_1 \sim N(0, 1)$, and $Z_2 \sim N(\mu, 1)$. We also have another sample from

$$Y = U_2 V_1 + (1 - U_2) V_2,$$

where $U_2 \sim \text{Bernoulli}\left(\frac{3}{4}\right)$, $V_1 \sim N(0, 1)$, and $V_1 \sim N(\mu + \delta, 1)$. U_1, U_2, Z_1, Z_2, V_1, and V_2 are mutually independent. We want to test

$$H_0 : \delta = 0 \qquad H_1 : \delta > 0,$$

where μ is the nuisance parameter. From our data, $n = 20$, $\bar{X} = 2.576$, and $\bar{Y} = 2.211$.

Non-Parametric Bootstrap Hypothesis Test

If we prefer non-parametric bootstrap, the algorithm is similar to parametric bootstrap test. But we must have a resampling procedure that incorporates H_0.

Algorithm 6: Parametric bootstrap hypothesis test

1 Estimate $\hat{\lambda}$;
2 **for** $j = 1$ *in* $1 : B$ **do**
3 | Draw a random sample $\left\{x^*_{i,j}\right\}$ of size n under H_0 ;
4 | Calculate the test statistic $T^*_j = T\left(x^*_{1,j}, ..., x^*_{n,j}\right)$;
5 **end**
6 Approximate the p-value by $p_{boot} = \#\left\{j : T^*_j \in A_{\text{obs}}\right\}/B$

Non-Parametric Bootstrap Hypothesis Test: Example

Example

Suppose that we have an iid sample X_1, ..., X_n from some distribution and another iid sample Y_1, ..., Y_n from the same distribution but shifted by δ. These two samples are independent. We want to test

$$H_0 : \delta = 0 \qquad H_1 : \delta \neq 0.$$

Computer Intensive Statistics and Applications
Chapter 4: Simulation-Based Methods

Shaobo Jin

Department of Mathematics

General Setup

The expectation and maximization (EM) algorithm is one of the most used methods in statistics. The general setup is

- we can specify a statistical model for a random vector X, say, its density is $f(x \mid \theta)$,
- we observe some data y, corresponding to the random variable Y,
- the observed y is a function of x, as $y = y(x)$, a many-to-one mapping.

In this setting, X is our complete data, Y is our incomplete data, and the density of y satisfies

$$g(y \mid \theta) = \int_{\{x;\, y(x)=y\}} f(x \mid \theta)\, dx.$$

Typical Applications: Missing Value

A typical example of such setup is the missing value problem.

- In the ideal world, we could have observed x, e.g., all responses to a questionnaire,
- In reality, we only observe y, e.g., some questions are not answered or some people choose not to participate.

Typical Applications: Clustering

Clustering is also a common problem.

- We know our data come from several populations as a mixture of distributions, but we don't know which group each observation comes from.
- Let Z be a multinomial random variable that indicates which group the observations comes from:

$$P(Z = k) = p_k \geq 0.$$

- Within group k $(Z = k)$, our data follow some distribution with density function $f_k(y)$.
- Then, the observation vector for a single object is modeled as a finite mixing distribution $\sum_{k=1}^{K} p_k f_k(y)$.

Complete Likelihood and Observed Likelihood

If x were observed, then the likelihood function is

$$L_C(\theta) = f(x \mid \theta).$$

However, we only observe y in practice, the likelihood is

$$L(\theta) = g(y \mid \theta) = \int_{\{x;\, y(x)=y\}} f(x \mid \theta)\, dx.$$

L_C is known as the complete likelihood and L is known as the observed likelihood. The maximum likelihood estimators of unknown parameters maximize the observed likelihood.

EM Algorithm

However, directly maximizing the observed likelihood is computationally not easy. The EM algorithm is often used instead.

Algorithm 1: EM Algorithm

Suppose that the parameter vector is θ. Obtain initial guess $\theta^{(0)}$;

while *at step* t **do**

  E step: Find conditional expectation $\mathrm{E}\left[\log f\left(x \mid \theta\right) \mid y, \hat{\theta}^{(t-1)}\right]$ where the expectation is done to the conditional distribution of $x \mid y$ and the parameter value is $\hat{\theta}^{(t-1)}$;

  M step: Maximize the conditional expectation with respect to θ and obtain the updated value $\boldsymbol{\theta}^{(t)}$;

end

Kullback-Leilber Divergence

Theorem (Theorem 4.1, Kullback-Leibler)

Let p and g be two densities, then

$$E_p\left[\log p\left(X\right)\right] \quad \geq \quad E_p\left[\log g\left(X\right)\right],$$

where the subscript indicates the true distribution of X.

We often call

$$\mathrm{KL}\left(p, g\right) \quad = \quad \mathrm{E}_p\left[\log\left(\frac{p\left(X\right)}{g\left(X\right)}\right)\right] \geq 0$$

the Kullback-Leibler divergence. A large $\mathrm{KL}\left(p, g\right)$ means that g is far away from p.

Conditional Expectation

In the literature, the conditional expectation

$$Q\left(\theta \mid \theta'\right) \quad = \quad \mathrm{E}\left[\log f\left(x \mid \theta\right) \mid y, \theta'\right]$$

is often called the surrogate function.

Theorem (Theorem 4.2)

The surrogate function satisfies

$$Q\left(\theta' \mid \theta'\right) - \log g\left(y \mid \theta'\right) \quad \geq \quad Q\left(\theta \mid \theta'\right) - \log g\left(y \mid \theta\right).$$

Why Does EM Algorithm Work?

Theorem 4.2 implies that

$$Q\left(\theta^{(t-1)} \mid \theta^{(t-1)}\right) - \log g\left(y \mid \theta^{(t-1)}\right) \geq Q\left(\theta \mid \theta^{(t-1)}\right) - \log g\left(y \mid \theta\right).$$

The M-step in the EM algorithm maximizes $Q\left(\theta \mid \theta^{(t-1)}\right)$ as a function of θ. Hence,

$$\begin{aligned} \log g\left(y \mid \theta^{(t)}\right) &\geq Q\left(\theta^{(t)} \mid \theta^{(t-1)}\right) - Q\left(\theta^{(t-1)} \mid \theta^{(t-1)}\right) + \log g\left(y \mid \theta^{(t-1)}\right) \\ &\geq \log g\left(y \mid \theta^{(t-1)}\right). \end{aligned}$$

That is, the observed likelihood increases as we maximize the conditional expectation.

EM Algorithm: Example

The most common model is the Gaussian mixture.

EM Algorithm For Gaussian Mixture

We know that

$$\begin{aligned} Y \mid Z = k &\sim N\left(\mu_k, \sigma_k^2\right), \\ p_k &= P\left(Z = k\right), \quad k = 1, ..., K. \end{aligned}$$

We observed a random sample y_1, ..., y_n, but we do not observe Z. Estimate p_k, μ_k and σ_k^2.

EM Algorithm: Missing Example

EM Algorithm For Missing Value

Suppose that we have n observations from a bivariate normal distribution $N\left(0, \begin{bmatrix} \sigma & \rho\sigma \\ \rho\sigma & \sigma \end{bmatrix}\right)$, where ρ and σ are the unknown parameters. We observe

$$Y = \begin{bmatrix} Y_{11} & Y_{12} \\ - & Y_{22} \end{bmatrix},$$

where Y_{21} is missing. For a bivariate normal distribution

$$\begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N\left(0, \begin{bmatrix} \sigma & \rho\sigma \\ \rho\sigma & \sigma \end{bmatrix}\right),$$

we know that $X_1 \mid X_2 = x_2 \sim N\left[\rho x_2,\ \left(1-\rho^2\right)\sigma\right]$.

Other Details

- We need to determine K. One way is to use the information criterion

$$\begin{aligned} \text{AIC} &= -2 \log g\left(y \mid \hat{\theta}\right) + 2 \times \text{number of parameters}, \\ \text{BIC} &= -2 \log g\left(y \mid \hat{\theta}\right) + \log n \times \text{number of parameters}. \end{aligned}$$

- It is often the case that some errors (e.g., non-convergence) are encountered for complicated models.
- EM algorithm is prone to local maxima. Hence, it is common to use multiple starting values.
- The EM algorithm can become extremely slow after a few steps. Various acceleration techniques have been proposed.

Self-Consistency of EM Algorithm

If $\theta^{(t)}$ maximizes $Q\left(\theta \mid \theta^{(t-1)}\right)$ as a function of θ, then the observed likelihood increases, i.e.,

$$\log g\left(y \mid \theta^{(t)}\right) \geq \log g\left(y \mid \theta^{(t-1)}\right).$$

- Suppose that $\hat{\theta}$ maximizes $\log g\left(y \mid \theta\right)$, i.e., $\hat{\theta}$ is the MLE.
- The MLE must satisfy $Q\left(\hat{\theta} \mid \hat{\theta}\right) \geq Q\left(\theta \mid \hat{\theta}\right)$ for all θ.
  - Otherwise $\log g\left(y \mid \theta\right) \geq \log g\left(y \mid \hat{\theta}\right)$ for some θ.
- Hence, MLE is one of the local maxima.

MCEM Algorithm

It is not always easy to obtain a closed form expression when we evaluate the expectation

$$Q\left(\theta \mid \hat{\theta}^{(t-1)}\right) = \mathrm{E}\left[\log f\left(x \mid \theta\right) \mid y, \hat{\theta}^{(t-1)}\right].$$

In the case where $x = (y, z)$, where z is unobserved, we can often approximate $Q\left(\theta \mid \theta'\right)$ using Monte Carlo integration. The resulting EM algorithm is known as the MCEM algorithm.

In the E step,

- Generate independently z_1, ..., z_M from $p\left(z \mid y, \hat{\theta}^{(t-1)}\right)$.
- Approximate $Q\left(\theta \mid \hat{\theta}^{(t-1)}\right)$ by

$$Q\left(\theta \mid \hat{\theta}^{(t-1)}\right) = \frac{1}{M}\sum_{m=1}^{M} \log f\left(y, z_j \mid \theta\right).$$

Stochastic Nature

Since a Monte Carlo approximation error is introduced at the E step, the monotonicity property of the EM algorithm is lost.

- $\hat{\theta}^{(t)}$ will fluctuate around some value as the algorithm moves closer to convergence.
- Specifying a suitable M and monitoring convergence become very important.
- It is often recommended to use small values of M in the initial stages and increase M as the algorithm moves forward.
- To monitor convergence, one easy tool is to plot $\hat{\theta}^{(t)}$ against the steps. Convergence is indicated by stable and random fluctuation about some value.
- A more sophisticated approach is to derive a confidence interval for $Q\left(\hat{\theta}^{(t-1)} \mid \hat{\theta}^{(t-1)}\right) - Q\left(\hat{\theta}^{(t)} \mid \hat{\theta}^{(t-1)}\right)$ and terminate the algorithm if the upper limit of the confidence interval is below some tolerance level.

Score Function

Let

$$S_C(\theta) = \frac{\partial \log f(x \mid \theta)}{\partial \theta}, \qquad S(\theta) = \frac{\partial \log g(y \mid \theta)}{\partial \theta}$$

be the complete score function and the observed score function, respectively. Then, if we can change the order of integration and differentiation, we have

$$S(\theta) = \mathrm{E}\left[S_C(\theta) \mid Y\right].$$

We can also define the Fisher information:

$$\text{observed}: \quad \mathcal{I}_{\text{obs}} = -\frac{\partial^2 \log g(y \mid \theta)}{\partial \theta \partial \theta^T},$$

$$\text{expected}: \quad \mathcal{I}_{\text{exp}} = \mathrm{var}\left[S(\theta)\right] = -\mathrm{E}\left[\frac{\partial^2 \log g(y \mid \theta)}{\partial \theta \partial \theta^T}\right].$$

Quantify Uncertainty

Since the solution of the EM algorithm converges to the MLE, it is also a random variable and we need to quantity its uncertainty.

- For MLE, the Fisher information is often used to compute the standard errors of the MLE.
- Under some conditions the MLE $\hat{\theta}$ satisfies

$$\mathcal{I}^{1/2}\left(\hat{\theta}-\theta\right) \quad \xrightarrow{d} \quad N\left(0, I\right).$$

- Hence, the standard errors can be obtained as the square root of the diagonal entries of $\mathcal{I}_{\text{obs}}\left(\hat{\theta}\right)$ or $\mathcal{I}_{\text{exp}}\left(\hat{\theta}\right)$.
- Confidence intervals can be constructed accordingly.

Approximate Information Matrix

1. Sometimes we can compute $S_i(\theta) = \frac{\partial \log g(y_i|\theta)}{\partial \theta}$. Then, we can approximate the Fisher information by

$$\hat{\mathcal{I}} = \frac{1}{n}\sum_{i=1}^{n} \frac{\partial \log g(y_i \mid \theta)}{\partial \theta} \frac{\partial \log g(y_i \mid \theta)}{\partial \theta^T}.$$

2. Sometimes it is not straightforward to compute $S_i(\theta)$. The Louis method means that

$$\mathcal{I}_{\text{obs}} = -\mathrm{E}\left[\frac{\partial^2 \log f(x \mid \theta)}{\partial \theta \partial \theta^T} \mid Y\right] - \mathrm{E}\left[S_C(\theta) S_C^T(\theta) \mid Y\right] + S(\theta) S^T(\theta).$$

If we take $\hat{\theta}$ as the MLE, then $S\left(\hat{\theta}\right) = 0$ and we skip the evaluation of observed gradient.

Measurement Error

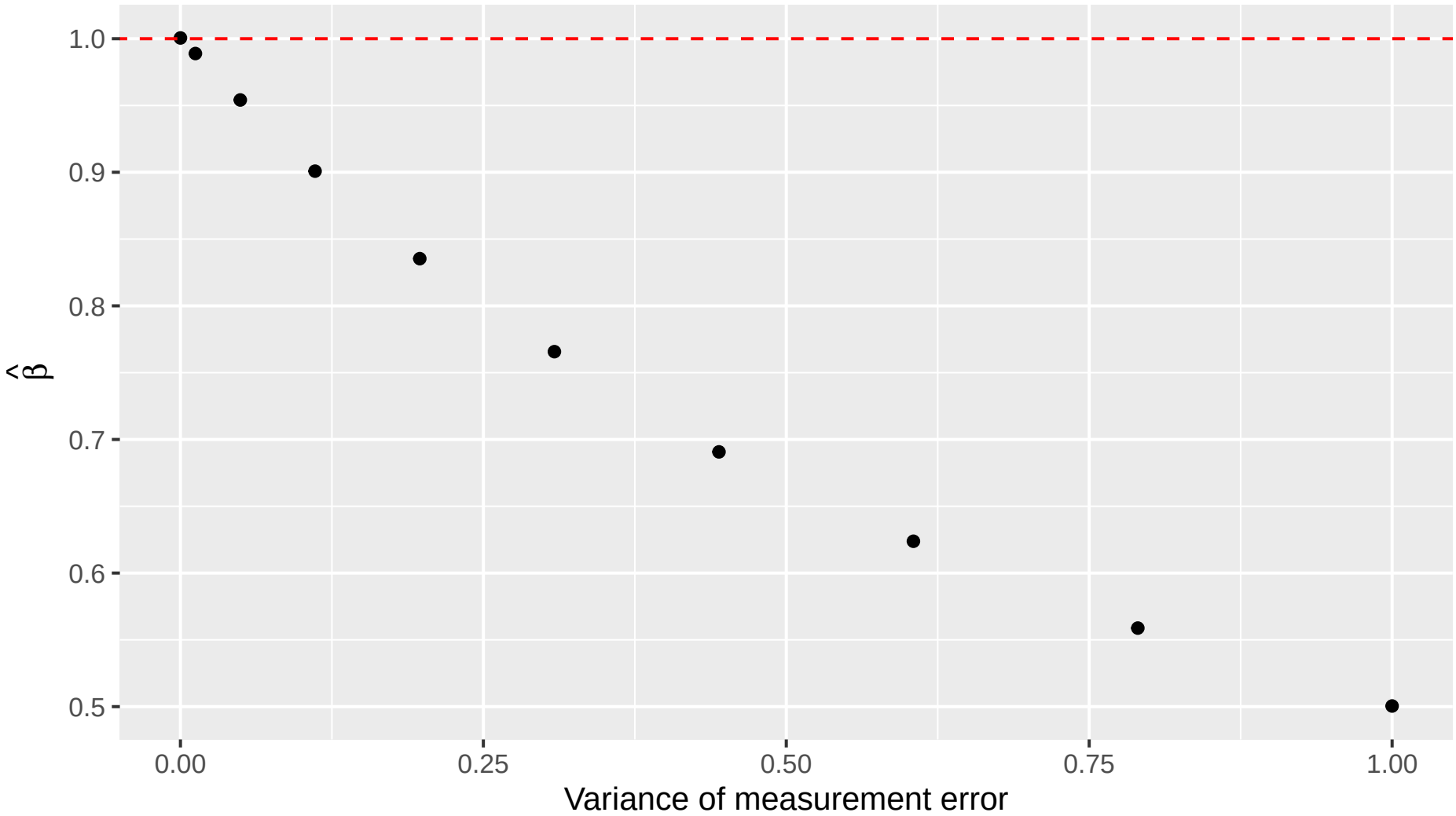
It is often the case that our data are incomplete.

- We want to model $\mathrm{E}[Y \mid U, V]$. But we cannot observe U directly.
- U is called a latent variable.

For example, in the measurement error model,

- instead of observing U, we observe $X = U + \sigma Z$, where $\mathrm{Var}(Z) = 1$ and σZ is the measurement error.
- our observed values are (Y, X, V), and U is the latent variable.
- if $T(Y, U, V)$ is a valid estimator, then $T(Y, X, V)$ is often not.

Example: Measurement Error in Regression



Simulation-Extrapolation Estimation

The simulation-extrapolation (SIMEX) estimation for known σ^2 works as follows.

Algorithm 2: SIMEX Algorithm

Choose a sequence $\{\lambda_1, ..., \lambda_K\}$, where $\lambda_k > 0$ for all k ;
for *k from 1 to K* **do**
  for *b from 1 to B* **do**
    Generate new errors $Z^*_{b,i}$ (zero mean, variance 1) ;
    Obtain new samples $X_{b,i}(\lambda) = X_i + \sqrt{\lambda_k}\sigma Z^*_i$, $\forall i$;
    Calculate $\hat{\theta}_b(\lambda) = T(\boldsymbol{Y}, \boldsymbol{V}, \boldsymbol{X}_b(\lambda_k))$;
  end
  Calculate the average $\bar{\theta}(\lambda_k) = b^{-1}\sum_{b=1}^{B}\hat{\theta}_b(\lambda_k)$;
end
Fit a curve $\hat{f}(\lambda)$ such that $\sum_{k=1}^{K}\left[\hat{f}(\lambda_k) - \bar{\theta}(\lambda_k)\right]^2$ is minimal ;
The SIMEX estimator is $\hat{\theta}_{\text{SIMEX}} = \hat{f}(-1)$

SIMEX for Linear Regression

Example

Consider the linear relationship

$$y_i = \beta_0 + \beta_1 v_i + \beta_2 z_i + \epsilon_i.$$

Instead of observing z, we observe

$$x_i = z_i + \delta_i,$$

where $\text{Var}(\delta_i) = 0.5$.

Introduction

Suppose that we have the response variable Y, and p variables (features) X_1, ..., X_p.

- We want to select the minimal set $\{X_{j_1}, ..., X_{j_m}\}$ from $\{X_1, ..., X_p\}$ to model

$$\mathrm{E}\left[Y \mid X_1, ..., X_p\right] \quad = \quad \mathrm{E}\left[Y \mid X_{j_1}, ..., X_{j_m}\right].$$

- The set of indices $\mathcal{A} = \{j_1, ..., j_m\}$ is called an active set.

We will introduce

1. backward, forward, and stepwise procedures,
2. information criterion,
3. cross validation.

Backward and Forward Selection in Linear Regression

Suppose that

$$\boldsymbol{Y} = \boldsymbol{X}\beta + \epsilon, \quad \epsilon \sim N_n\left(0\sigma^2 \boldsymbol{I}_n\right),$$

where the rank of the $n \times p$ matrix$\boldsymbol{X}$ is p. Define

$$\text{RSS}\left(\mathcal{A}\right) = \sum_{i=1}^{n}\left(y_i - \sum_{j\in\mathcal{A}} x_j\hat{\beta}_j\right)^2,$$

where our hypothesized model is $\sum_{j\in\mathcal{A}} x_j\beta_j$, and $\beta_j = 0$ for $j \notin \mathcal{A}$.

Backward and Forward Selection in Linear Regression

Suppose that $\mathcal{A}_0 \subset \mathcal{A}_1$. Then,

$$F(\mathcal{A}_0, \mathcal{A}_1) = \frac{(\mathrm{RSS}(\mathcal{A}_0) - \mathrm{RSS}(\mathcal{A}_1)) / (p - p_0)}{\mathrm{RSS}(\mathcal{A}_1) / (n - p)} \quad \sim \quad F(p - p_0, n - p),$$

where $\mathcal{A}_1$ has p_0 more features than $\mathcal{A}_0$.

Using F values,

- Forward selection starts with an intercept model and sequentially adds terms.
- Backward elimination starts with a model with all features and sequentially removes terms.
- Stepwise regression is a combination of both by checking whether adding one term or deleting one term.

Backward Elimination in Linear Regression

Algorithm 3: Backward Elimination Using F Test

Specify a significance level α ;
Specify the active set $\mathcal{A}$ that contains all features ;
while *Proceed* **do**
    For each $j \in \mathcal{A}$, calculuate $F\left(\mathcal{A} \setminus \{j\}, \mathcal{A}\right)$;
    Select l such that $F\left(\mathcal{A} \setminus \{l\}, \mathcal{A}\right) = \min_{j \in \mathcal{A}} F\left(\mathcal{A} \setminus \{j\}, \mathcal{A}\right)$;
    Calculate c, where c is the $1-\alpha$ quantile of $F\left(p-1, n-p\right)$ with p being the size of the active set $\mathcal{A}$;
    if $F\left(\mathcal{A} \setminus \{l\}, \mathcal{A}\right) < c$ **then**
        $\mathcal{A} \leftarrow \mathcal{A} \setminus \{l\}$;
    else
        Select the model with active set $\mathcal{A}$ and terminate the procedure ;
    end
end

Forward Selection in Linear Regression

Algorithm 4: Forward Selection Using F Test

Specify a significance level α ;
Specify the active set $\mathcal{A} = \emptyset$ that contains no feature ;
while *Proceed* **do**
    For each $j \in \{1, ..., p\} \setminus \mathcal{A}$, Select l such that $F(\mathcal{A}, \mathcal{A} \cup \{l\}) = \max_{j \in \{1,...,p\} \setminus \mathcal{A}} F(\mathcal{A}, \mathcal{A} \cup \{j\})$;
    Calculate c, where c is the $1 - \alpha$ quantile of $F(1, n - p)$ with p being the size of $\mathcal{A} \cup \{j\}$;
    if $F(\mathcal{A}, \mathcal{A} \cup \{l\}) > c$ **then**
        $\mathcal{A} \leftarrow \mathcal{A} \cup \{l\}$;
    else
        Select the model with active set $\mathcal{A}$ and terminate the procedure ;
    end
end

Stepwise Regression in Linear Regression

Algorithm 5: Stepwise Regression Using F Test

Specify a significance level α ;
Specify the active set $\mathcal{A} = \emptyset$ that contains no feature ;
while *Proceed* **do**
    For each $j \in \{1, ..., p\} \setminus \mathcal{A}$, calculuate $F(\mathcal{A}, \mathcal{A} \cup \{j\})$;
    Select l such that $F(\mathcal{A}, \mathcal{A} \cup \{l\}) = \max_{j \in \{1,...,p\} \setminus \mathcal{A}} F(\mathcal{A}, \mathcal{A} \cup \{j\})$;
    Calculate c, where c is the $1 - \alpha$ quantile of $F(1, n - p)$ with p being the size of $\mathcal{A} \cup \{j\}$;
    if $F(\mathcal{A}, \mathcal{A} \cup \{l\}) > c$ **then**
        For each $j \in \mathcal{A}$, test whether it can be deleted from $\mathcal{A} \cup \{l\}$ in a similar manner to backward elimination. Update $\mathcal{A}$ to the resulting set ;
    end
    if *No feature has been added nor deleted* **then**
        Terminate the procedure ;
    end
end

Multiple Testing

A major limitation of the above procedures is that we will perform multiple F tests.

- Each F test is subject to error with probability α, where α is small.
- The aggregated error of the whole procedure is often very large or even close to 1.

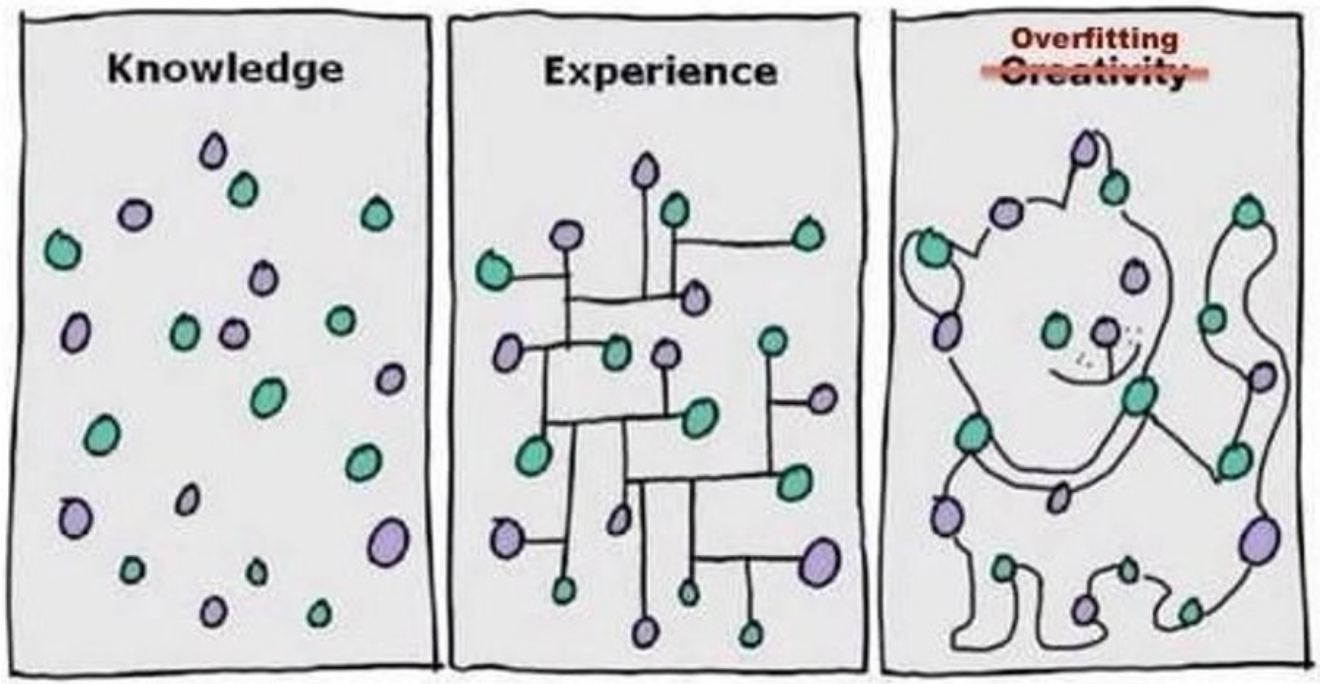
In R, backward elimination, forward selection, and stepwise regression are performed not based on the F values.

- Forward selection starts with an intercept model and sequentially adds the term that gives the lowest AIC/BIC.
- Backward elimination starts with a model with all explanatory variables and sequentially removes the term that has the greatest decrease in AIC/BIC.
- Stepwise regression is a combination of both by checking whether adding one term or deleting one term will yield the smallest AIC/BIC.

Overfitting

Suppose that $\boldsymbol{X}$ follows some distribution with density $p(\boldsymbol{X})$.

- We want our model to be complex enough to fit the data well, such as maximizing $\log p(\boldsymbol{X})$.
- On the other hand, we don't want the model to be too complex to avoid overfitting.



Information Criterion

We cannot simply choose a complex model in order to have a better fit to the current data.

- Suppose that the true data generating process is $p(x)$, but we assume $g(x \mid \theta)$, where p and g may not be the same.
- An information criterion is often of the form

$$-c \log g(\boldsymbol{X}) + \text{penality of model complexity}$$

for some constant c.

AIC: Minimizing Distance of the Fit from the Truth

We can express the KL divergence between the fitted model and the truth is

$$\mathrm{KL}(p, g) = \mathrm{E}[\log p(x^*)] - \underbrace{\int \log\left[g\left(x^* \mid \hat{\theta}\right)\right] p(x^*)\, dx^*}_{\text{define to be } R_n},$$

where we can view x^* as a new observation independent of $\boldsymbol{X}$, and $\hat{\theta} = \hat{\theta}(\boldsymbol{X})$.

- The Akaike information criterion (AIC) aims to estimate the expected value of $-R_n$. Hence, we often say AIC is a nearly unbiased estimator of KL divergence (bar some constant).
- We prefer the model with a small AIC, since a small AIC means that the divergence is small.

Derivation of AIC

Take iid data as an example, it can be shown that

$$\mathrm{E}\left[R_n\right] = \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \hat{\theta}\right) - \frac{1}{n}\mathrm{tr}\left(\mathcal{I}H^{-1}\right) + o_P\left(n^{-1}\right),$$

where

$$\mathcal{I} = \mathrm{Var}\left(\frac{\partial \log g\left(X^* \mid \theta^*\right)}{\partial\theta}\right), \text{ Fisher information}$$

$$H = -\mathrm{E}\left[\frac{\partial^2 \log g\left(x^*, \theta^*\right)}{\partial\theta\partial\theta^T}\right]. \text{ expected Hessian}$$

- If we estimate $\mathrm{tr}\left(\mathcal{I}H^{-1}\right)$ by q, the dimension of θ, we obtain the Akaike information criterion (AIC) is defined as

$$\mathrm{AIC} = -2\log\left[g\left(\boldsymbol{X}\right)\right] + 2q.$$

- Other ways of estimating $\mathrm{tr}\left(\mathcal{I}H^{-1}\right)$ is also available, yielding other information criterion such as Takeuchi information criterion (TIC).

OBS! AIC Expression

Many resources define AIC to be

$$n \log\left(\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2\right) + 2q,$$

where q is the number of parameters in $\mathrm{E}[Y]$. Keep in mind that this expression requires the normality assumption!

Under the assumption $Y_i \sim N\left(\mu_i(\theta), \sigma^2\right)$, the AIC becomes

$$\mathrm{AIC} = n \log\left(\frac{\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2}{n}\right) + 2(q+1) + n\log(2\pi) + n.$$

Bayesian Information Criterion

- Bayesian information criterion penalizes a complex model much more than AIC.

$$\text{BIC} \quad = \quad -2\log\left[g\left(\boldsymbol{X}\right)\right] + \log\left(n\right)q.$$

- We prefer the model with the smallest BIC or a parsimonious model that has BIC near the minimum.
- BIC is an approximation to the so called Bayes factor in Bayesian statistics.

BIC: Consistent Model Selection

We call a model selection procedure is consistent if, with probability approaching 1, it picks the most parsimonious model among the set of models that minimize the Kullback-Leibler divergence to the truth.

- BIC is consistent.
- In contrast, AIC is not consistent and tends to be conservative.
  - The price is that, for BIC, the square risk satisfies

$$\sup_f \sum_{i=1}^{n} \mathrm{E}\left[\left(f(x_i) - \hat{f}(x_i)\right)^2\right] \to \infty, \qquad n \to \infty.$$

  - It is bounded for AIC.
- AIC is only weakly consistent.

Cross Validation (CV)

CV focuses on the prediction property.

Algorithm 6: One version of cross validation

Randomly split the data set into K nonoverlapping groups (K-fold CV) or split the data set into n groups (leave-one-out CV, aka jackknife);
for $k = 1$ *in* $1 : K$ **do**
    Take the kth group as test set and the remaining groups as training set ;
    while *for each model* **do**
        Fit it on the training set and evaluate it on the test set ;
        Retain the model performance (e.g., MSE, misclassification error, log-likelihood) ;
    end
end
Summarize the model performance (e.g., average across K groups) ;
Choose the model that performs the best ;
Refit the chosen model using the entire data set ;

Maybe a Less Discussed Point

Take regression as an example. Suppose that we have fitted a model $\hat{\mu}(x)$ and want to evaluate its prediction property.

- The test error between a new y^* and the prediction $\hat{\mu}(x^*)$ is $L(y^*, \hat{\mu}(x^*))$.
- The expected test error is $\mathrm{E}[L(y^*, \hat{\mu}(x^*)) \mid \hat{\mu}]$.
- The K-fold CV estimate of the expected test error is

$$\mathrm{CV}_K = \frac{1}{n} \sum_{k=1}^{K} \sum_{i \in C_k} L(y_i, \hat{\mu}_{-k}(x_i)),$$

  where C_k collects the indices of observations in the kth fold, and $\hat{\mu}_{-k}$ is the estimated regression function excluding the kth fold.

Such CV is a biased estimator of the expected test error!

Biased CV

In general cases,

$$\mathrm{E}\left[\mathrm{CV}_K - \mathrm{E}\left[L\left(y^*, \hat{\mu}\left(x^*\right)\right) \mid \hat{\mu}\right]\right] \approx \frac{c_0}{n\left(K-1\right)},$$

where c_0 depends only on $L\left(\right)$ and the data generating process.

- For simplicity, we consider a simple linear regression

$$Y = \beta_0 + \beta_1 X + \epsilon$$

  where $\epsilon \sim N\left(0,1\right)$ and $X \sim N\left(0, \sigma^2\right)$. We also assume $n = Km$.
- Then,

$$\mathrm{E}\left[\mathrm{CV}_K - \mathrm{E}\left[L\left(y^*, \hat{\mu}\left(x^*\right)\right) \mid \hat{\mu}\right]\right] \approx \frac{2\sigma^2}{n\left(K-1\right)},$$

$$\mathrm{Var}\left[\mathrm{CV}_K - \mathrm{E}\left[L\left(y^*, \hat{\mu}\left(x^*\right)\right) \mid \hat{\mu}\right]\right] \approx 2\sigma^4\left[\frac{1}{n} + \frac{8}{n^2} + \frac{8}{n^2\left(K-1\right)} + \frac{4}{n^2\left(K-1\right)^2} + \frac{2}{n^2\left(K-1\right)^3}\right].$$

More on Cross Validation

Suppose that the true model is one of our candidate models.

- It has been proved that, for leave-one-out CV,

$$\begin{aligned} \lim_{n\to\infty} \mathrm{P}\left(\text{Choose a model where a nonzero } \beta \text{ is missing}\right) &= 0, \\ \lim_{n\to\infty} \mathrm{P}\left(\text{Choose the most parsimonious true model}\right) &\neq 1. \end{aligned}$$

  Hence, the leave-one-out CV is conservative that selects a model of excessive size.

- It has been proved that you need leave-n_v-out CV if you want

$$\lim_{n\to\infty} \mathrm{P}\left(\text{Choose the most parsimonious true model}\right) = 1,$$

  where $n_v/n \to 1$ as $n \to \infty$.

More on Leave-One-Out Cross Validation

Suppose now that all candidate models are wrong. Consider the squared loss in linear regression

$$L_n \quad = \quad n^{-1} \left(\mathrm{E} \left(\boldsymbol{y} \mid \boldsymbol{X} \right) - \hat{\boldsymbol{y}} \right)^T \left(\mathrm{E} \left(\boldsymbol{y} \mid \boldsymbol{X} \right) - \hat{\boldsymbol{y}} \right).$$

It has been proved that, for any $\epsilon > 0$, we have

$$\lim_{n \to \infty} \mathrm{P} \left(\left| \frac{L_n \text{ of the model selected by CV}}{\text{Minimum } L_n \text{ among all candidate models}} - 1 \right| < \epsilon \right) \quad = \quad 1.$$

Cross Validation and AIC

In fact, we can show that leave-one-out CV is asymptotically equivalent to AIC if the model performance is measured by

$$\sum_{i=1}^{n} \log g\left(x_i \mid \hat{\theta}_{-i}\right),$$

where $\hat{\theta}_{-i}$ is the estimator of θ after removing the ith observation.

Computer Intensive Statistics and Applications
Chapter 5: Density Estimation

Shaobo Jin

Department of Mathematics

Problem

Suppose that we have observed an iid sample X_1, ..., X_n. We want to the density of X, denoted by $f(x)$, using the observed data.

1. Parametric approach: the distribution belongs to a known distribution family with unknown parameters.
2. Nonparametric approach: the unknown density satisfies

$$f \in \mathcal{F} = \left\{ f : \mathbb{R} \to \mathbb{R}, f(x) \geq 0, \int f(x)\, dx = 1, f \text{ is continuous} \right\}.$$

Estimating the density allows us to visualize the data and also provides a starting point for regression analysis.

Nonexistence of Unbiased Estimator

Theorem (Theorem 5.1)

Suppose that $(X_1, \cdots, X_n)$ *is an iid sample of the random variable* X *with unknown density* f. *Consider* $\hat{f}(x) = \hat{f}(x, X_1, ..., X_n)$. *Then, there exists no* $\hat{f} \in \mathcal{G}$ *such that* $E\left[\hat{f}(x)\right] = f(x)$, *for all* $x \in \mathbb{R}$ *and for all* $f \in \mathcal{F}$, *where*

$$\mathcal{G} = \left\{\hat{f} : \mathbb{R} \to \mathbb{R}, \hat{f}(x) \geq 0, \int \hat{f}(x)\, dx = 1, \hat{f}(x) \text{ is continuous in } x \text{ and measurable in } X_1, ..., X_n\right\}.$$

Mean Squared Error

For a fixed x, the mean squared error between our estimator $\hat{f}(x)$ and the true density $f(x)$ is

$$\begin{aligned} \mathrm{MSE}\left(\hat{f}(x)\right) &= \mathrm{E}\left[\left(\hat{f}(x) - f(x)\right)^2\right] \\ &= \mathrm{Var}\left[\hat{f}(x)\right] + \left\{\mathrm{E}\left[\hat{f}(x)\right] - f(x)\right\}^2. \end{aligned}$$

Theorem 5.1 implies that the bias term cannot be zero. Hence, we want to find a balance between the variance and the bias.

1. When the estimator has a large variance, it is called undersmoothing.
2. When the bias becomes too large, it is called oversmoothing.

Under- and Oversmoothing: Example

Example

Suppose that, for a fixed x, we approximate the density by

$$f(x) \approx \frac{\mathrm{P}(x-\epsilon \leq X \leq x+\epsilon)}{2\epsilon} \equiv \frac{p}{2\epsilon}.$$

We can estimate p by $n^{-1}\sum_{i=1}^{n} 1(x-\epsilon \leq x_i \leq x+\epsilon)$. Then,

$$\begin{aligned}
\mathrm{E}\left[\hat{f}(x)\right] &= \mathrm{E}\left[\frac{\sum_{i=1}^{n} 1(x-\epsilon \leq x_i \leq x+\epsilon)}{2\epsilon n}\right] = \frac{p}{2\epsilon}, \\
\mathrm{Var}\left[\hat{f}(x)\right] &= \mathrm{Var}\left[\frac{\sum_{i=1}^{n} 1(x-\epsilon \leq x_i \leq x+\epsilon)}{2\epsilon n}\right] = \frac{p(1-p)}{4\epsilon^2 n}.
\end{aligned}$$

- If ϵ is small, the bias is small and variance is large.
- If ϵ is large, the bias is large and variance is small.

Histogram

Histogram is a standard method to approximate/estimate density.

- Consider a bin with bandwidth $h > 0$ as

$$B_j = [x_0 + (j-1)\,h,\ x_0 + jh],$$

where j is an integer (positive or negative), and x_0 is the origin of the histogram.

- For $x \in B_j$, we approximate $f(x)$ by $h^{-1}\mathrm{P}(X \in B_j)$, and estimate $\mathrm{P}(X \in B_j)$ by $n^{-1}\sum_{i=1}^{n} 1(X_i \in B_j)$.
- The histogram is given by

$$\begin{aligned}\hat{f}_h(x) &= \sum_j 1(x \in B_j)\,\frac{1}{h}\left[\sum_{i=1}^{n}\frac{1}{n}1(X_i \in B_j)\right]. \\ &= \frac{1}{nh}\sum_{i=1}^{n}\sum_j 1(X_i \in B_j)\,1(x \in B_j).\end{aligned}$$

Histogram: Bias and Variance

It is easy to show $\hat{f}_h(x) \geq 0$ and

$$\int \hat{f}_h(x)\, dx = 1.$$

Consider a fixed x. Then, if $x \in B_j$,

$$\begin{aligned} \mathrm{E}\left[\hat{f}(x)\right] &= \frac{\mathrm{P}(X \in B_j)}{h}, \\ \mathrm{Var}\left[\hat{f}(x)\right] &= \frac{\mathrm{P}(X \in B_j)\left[1 - \mathrm{P}(X \in B_j)\right]}{nh^2}. \end{aligned}$$

Hence, histogram is a biased estimator unless $f(x)$ is constant over B_j. The bias depends on h.

Histogram: Taylor Expansion

Suppose that the true density $f(x)$ is smooth enough. Let b_j be the midpoint of B_j. Taylor expansion for $x \in B_j$ yields

$$f(x) = f(b_j) + f'(b_j)(x - b_j) + o(h), \quad \text{as } h \to 0.$$

Hence, skipping proof,

$$\begin{aligned} \mathrm{E}\left[\hat{f}(x)\right] - f(x) &= f'(b_j)(b_j - x) + o(h), \\ \mathrm{Var}\left[\hat{f}(x)\right] &= \frac{1}{nh} f(x) + o\left(\frac{1}{nh}\right), \end{aligned}$$

as $h \to 0$ and $nh \to \infty$.

Mean Squared Error

For a fixed x, the mean squared error satisfies

$$\begin{aligned} \text{MSE}\left(\hat{f}(x)\right) &= \text{Var}\left[\hat{f}(x)\right] + \left\{\text{E}\left[\hat{f}(x)\right] - f(x)\right\}^2 \\ &= \frac{1}{nh} f(x) + \left[f'(b_j)\right]^2 (b_j - x)^2 + o\left(\frac{1}{nh}\right) + o\left(h^2\right). \end{aligned}$$

As $h \to 0$ and $nh \to \infty$, $\text{MSE}\left(\hat{f}(x)\right) \to 0$. Hence, $\hat{f}(x)$ is a consistent estimator of $f(x)$.

Mean Integrated Squared Error

The mean integrated squared error (MISE) for the goodness of estimation:

$$\text{MISE}\left(\hat{f}\right) = \text{E}\left[\int\left[\hat{f}(x) - f(x)\right]^2 dx\right].$$

The MISE of the histogram satisfies

$$\text{MISE}\left(\hat{f}\right) = \frac{1}{nh} + \frac{h^2}{12}\left\|f'(x)\right\|_2^2 + o\left(\frac{1}{nh}\right) + o\left(h^2\right),$$

where

$$\left\|f'(x)\right\|_2 = \sqrt{\int\left[f'(x)\right]^2 dx}.$$

Optimal Binwidth

The leading term of MISE is called the asymptotic MISE:

$$\text{AMISE}\left(\hat{f}\right) = \frac{1}{nh} + \frac{h^2}{12}\int \left[f'(x)\right]^2 dx.$$

We can minimize AMISE to obtain the optimal binwidth.

- Minimizing $\text{AMISE}\left(\hat{f}\right)$ as a function in h yields

$$h_0 = \left[\frac{6}{n\int \left[f'(x)\right]^2 dx}\right]^{1/3},$$

  the same order as $n^{-1/3}$.
- With the optimal h_0,

$$\text{AMISE}\left(\hat{f}\right) = \frac{1}{nh} + \frac{h^2}{12}\int \left[f'(x)\right]^2 dx.$$

- However, a dilemma is that we do not know $f'(x)$.

Specify Break Points

Suppose that the jth bin is B_j with width h_j. The bandwidth can vary across bins.

- For $x \in B_j$, the density can be estimated by

$$\frac{\sum_{i=1}^{n} 1\left(X_i \in B_j\right)}{n h_j}.$$

- Hence,

$$\hat{f}\left(x\right) = \sum_{j} 1\left(x \in B_j\right) \frac{\sum_{i=1}^{n} 1\left(X_i \in B_j\right)}{n h_j}.$$

Modifying Histogram

The bins in the histogram

$$\hat{f}_h(x) = \frac{1}{nh}\sum_{i=1}^{n}\sum_{j} 1(X_i \in B_j)\, 1(x \in B_j)$$

are not adapted to x.

- Choose bins first and then check which bin x belongs to.

One alternative is to define bins according to the x of interest.

Alternative

Let X be a random variable with density f. For a fixed x, define $B_h(x) = \left[x - 2^{-1}h, x + 2^{-1}h\right]$. We can approximate the density by

$$\begin{aligned} f(x) &\approx \frac{P(X \in B_h(x))}{h} = \frac{1}{h} \int_{x-2^{-1}h}^{x+2^{-1}h} f(u)\, du \\ &= \int \frac{1}{h} f(u) K\left(\frac{x-u}{h}\right) du = \mathrm{E}\left[\frac{1}{h} K\left(\frac{x-X}{h}\right)\right], \end{aligned}$$

where $K(\cdot)$ is the density of Uniform $[-0.5, 0.5]$.

- The corresponding density estimator is

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right).$$

Kernel Density Estimation

In general, we can follow the above idea and estimate the density by

$$\hat{f}_h(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right),$$

where $K()$ is a suitably chosen function.

Definition

A non-negative function $K(x)$ is a kernel function if $\int K(x)\,dx = 1$ and $K(x) = K(-x)$. Such density estimator using a kernel function is called a kernel density estimator and such h is called bandwidth.

Kernel Function: Examples

$$\text{Uniform}: \quad K(u) = \frac{1}{2} I(|u| \leq 1).$$

$$\text{Triangle}: \quad K(u) = (1 - |u|) I(|u| \leq 1).$$

$$\text{Epanechnikov}: \quad K(u) = \frac{3}{4}\left(1 - u^2\right) I(|u| \leq 1).$$

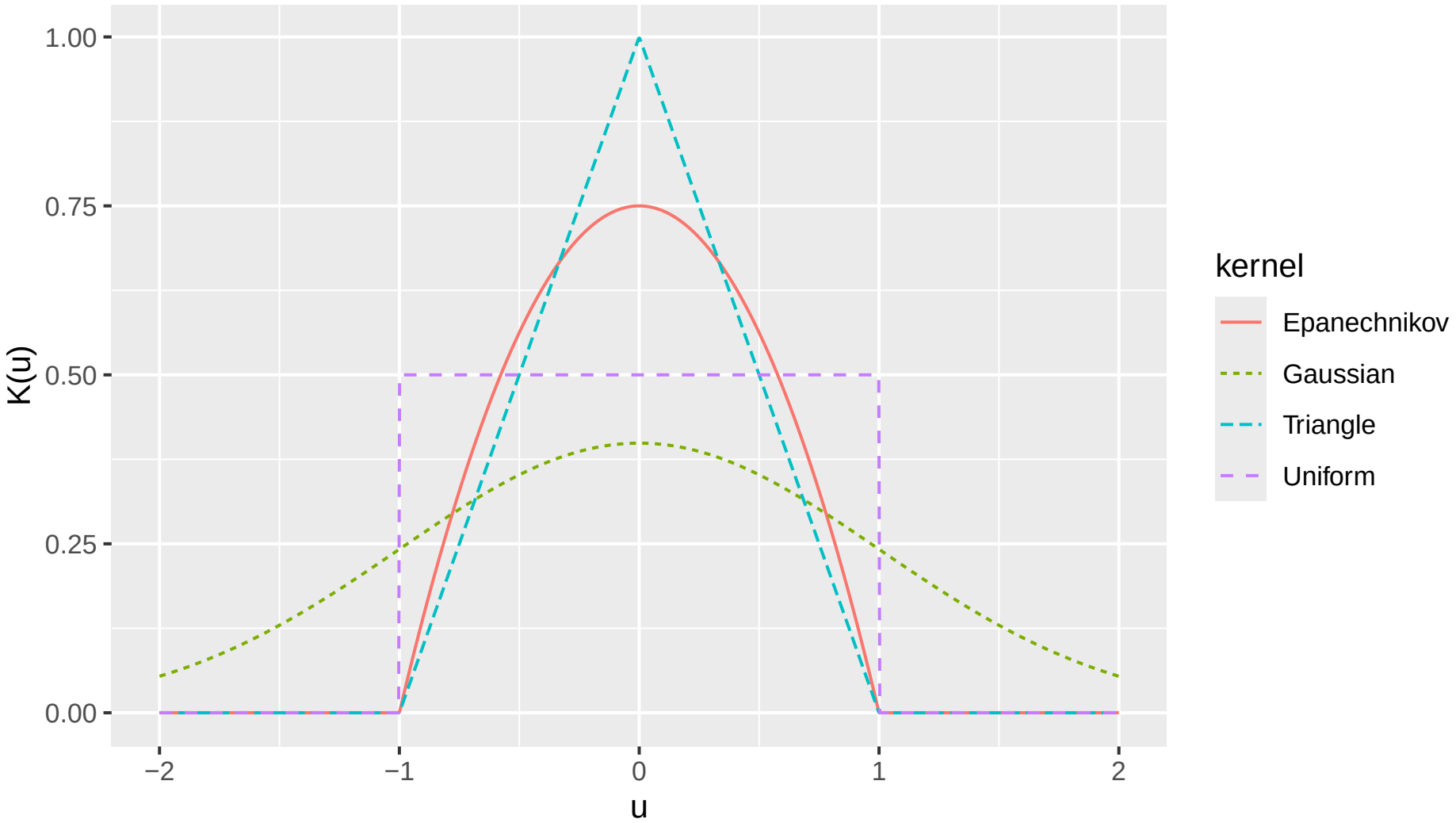
$$\text{Quartic (Biweight)}: \quad K(u) = \frac{15}{16}\left(1 - u^2\right)^2 I(|u| \leq 1).$$

$$\text{Triweight}: \quad K(u) = \frac{35}{32}\left(1 - u^2\right)^3 I(|u| \leq 1).$$

$$\text{Gaussian}: \quad K(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right).$$

$$\text{Cosine}: \quad K(u) = \frac{\pi}{4} \cos\left(\frac{\pi u}{2}\right) I(|u| \leq 1).$$

Kernel Function: Example



Kernel Density: Bias and Variance

Suppose that the true density $f(x)$ is smooth enough. Taylor expansion yields

$$f(x+ht) = f(x) + f'(x)\,ht + \frac{1}{2}f''(x)\,h^2t^2 + o\left(h^2\right), \quad \text{as } h \to 0.$$

Hence, we can show that

$$\begin{aligned} \mathrm{E}\left[\hat{f}_h(x)\right] - f(x) &= \frac{1}{2}h^2 f''(x) \int t^2 K(t)\,dt + o(h), \\ \mathrm{Var}\left[\hat{f}_h(x)\right] &= \frac{1}{nh} f(x) \int K^2(t)\,dt + o\left(\frac{1}{nh}\right), \end{aligned}$$

as $h \to 0$ and $nh \to \infty$.

- A small h yields a small bias but a large variance.
- A large h yields a large bias but a small variance.

MSE

For a fixed x, the mean squared error satisfies

$$\begin{aligned} \text{MSE}\left(\hat{f}_h(x)\right) &= \frac{1}{nh}\|K\|_2^2 f(x) + \frac{1}{4}h^4\mu_2^2(K)\left[f''(x)\right]^2 \\ &\quad + o\left(\frac{1}{nh}\right) + o\left(h^4\right), \end{aligned}$$

where

$$\|K\|_2^2 = \int K^2(t)\,dt \qquad \mu_2(K) = \int t^2 K(t)\,dt.$$

As $h \to 0$ and $nh \to \infty$, $\text{MSE}\left(\hat{f}(x)\right) \to 0$. Hence, $\hat{f}(x)$ is a consistent estimator of $f(x)$.

Optimal Binwidth

The MISE satisfies

$$\text{MISE}\left(\hat{f}_h\right) = \underbrace{\frac{1}{nh}\|K\|_2^2 + \frac{1}{4}h^4\mu_2^2(K)\left\|f''(x)\right\|_2^2}_{=\text{AMISE}\left(\hat{f}_h\right)} + o\left(\frac{1}{nh}\right) + o\left(h^4\right).$$

We can minimize AMISE to obtain the optimal binwidth.

- Minimizing $\text{AMISE}\left(\hat{f}\right)$ as a function in h yields

$$h_0 = \left[\frac{\|K\|_2^2}{n\mu_2^2(K)\left\|f''(x)\right\|_2^2}\right]^{1/5},$$

  the same order as $n^{-1/5}$.
- However, a dilemma is that we do not know $f''(x)$.

Rate of Convergence

- With the optimal h_0, the AMISE of histogram satisfies

$$\begin{aligned} \text{AMISE}\left(\hat{f}_h\right) &= \frac{1}{nh_0} + \frac{h_0^2}{12}\left\|f'(x)\right\|_2^2 \\ &= O\left(n^{-2/3}\right). \end{aligned}$$

- With the optimal h_0, the AMISE of kernel density estimation satisfies

$$\begin{aligned} \text{AMISE}\left(\hat{f}_h\right) &= \frac{1}{nh}\|K\|_2^2 + \frac{1}{4}h^4\mu_2^2(K)\left\|f''(x)\right\|_2^2 \\ &= \frac{5\left\|f''(x)\right\|_2^{2/5}}{4n^{4/5}}\mu_2^{2/5}(K)\|K\|_2^{8/5} = O\left(n^{-4/5}\right), \end{aligned}$$

converging faster than histogram.

Choice of Kernel and Bandwidth

- In theory, the optimal nonnegative kernel that minimizes AMISE $\text{AMISE}\left(\hat{f}_{h_0}\right)$ is the Epanechnikov kernel.
  - However, it is not practical, since the optimal bandwidth

$$h_0 = \left[\frac{\|K\|_2^2}{n\mu_2^2(K)\|f''(x)\|_2^2}\right]^{1/5},$$

    depends on unknown $f''(x)$.
- In practice, the effect of kernel is often minor, relative to the choice of the bandwidth.

Choice of Bandwidth: Rule-of-Thumb

Silverman's rule-of-thumb (plug-in method): Regardless of the true f, we use a Gaussian kernel and the density of $N\left(0, \sigma^2\right)$ as the true f. Then,

$$h_0 = \left[\frac{\|K\|_2^2}{n\mu_2^2(K)\left\|f''(x)\right\|_2^2}\right]^{1/5} = \left[\frac{4}{3}\right]^{1/5}\hat{\sigma}n^{-1/5},$$

where $\hat{\sigma}^2$ is the sample variance.

- It works well if the true density is not too far from normal.

But estimating σ^2 by sample variance is sensitive to outliers. A more robust estimator is based on the interquartile range. For $X \sim N\left(\mu, \sigma^2\right)$, the difference between the 75%-quantile and the 25%-quantile is 1.34σ. The bandwidth becomes

$$h_0 = \left[\frac{4}{3}\right]^{1/5}\min\left\{\hat{\sigma}, \frac{R}{1.34}\right\}n^{-1/5},$$

where R is the interquartile range.

Choice of Bandwidth: Cross Validation

Consider the integrated squared error (ISE) for the goodness of estimation:

$$\begin{aligned} \text{ISE}\left(\hat{f}\right) &= \int \left[\hat{f}(x) - f(x)\right]^2 dx \\ &= \int \hat{f}_h^2(x)\, dx - 2\int \hat{f}_h(x) f(x)\, dx + \int f^2(x)\, dx. \end{aligned}$$

We can estimate the second integral by $n^{-1}\sum_{i=1}^n \hat{f}_{h,-i}(X_i)$, where

$$\hat{f}_{h,-i}(x) = \frac{1}{(n-1)h}\sum_{j\neq i} K\left(\frac{x - X_j}{h}\right).$$

The leave-one-out cross validation criterion minimizes

$$\text{CV}(h) = \int \hat{f}_h^2(x)\, dx - \frac{2}{n}\sum_{i=1}^n \hat{f}_{h,-i}(X_i).$$

Limiting Distribution

Recall that the leading terms of bias and variance are

$$\begin{aligned} \mathrm{E}\left[\hat{f}_h(x)\right] - f(x) &\approx \frac{1}{2}h^2 f''(x)\,\mu_2(K), \\ \mathrm{Var}\left[\hat{f}_h(x)\right] &\approx \frac{1}{nh} f(x)\,\|K\|_2^2. \end{aligned}$$

For a fixed x, if the bandwidth satisfies $h = cn^{-1/5}$ for a constant c, we would expect

$$n^{2/5}\left\{\hat{f}_h(x) - f(x)\right\} \xrightarrow{d} N\left(b(x),\, v^2(x)\right),$$

where

$$\begin{aligned} b(x) &= \frac{c^2}{2} f''(x)\,\mu_2(K), \\ v^2(x) &= \frac{1}{c} f(x)\,\|K\|_2^2. \end{aligned}$$

Pointwise Confidence Band

Let λ_α be the α quantile of $N(0,1)$. Then,

$$1-\alpha \approx \mathrm{P}\left(-\lambda_{1-\alpha/2} \leq \frac{n^{2/5}\left\{\hat{f}_h(x) - f(x)\right\} - b(x)}{v(x)} \leq \lambda_{1-\alpha/2}\right).$$

An asymptotic interval for $f(x)$ is

$$\hat{f}_h(x) - \frac{h^2}{2} f''(x)\,\mu_2(K) \pm \lambda_{1-\alpha/2}\sqrt{\frac{1}{hn} f(x)\,\|K\|_2^2},$$

which depends on unknown $f''(x)$.

- One ad-hoc way is to ignore the bias term $\frac{h^2}{2} f''(x)\,\mu_2(K)$.
- An alternative is to estimate $f''(x)$, e.g., use the derivative of the kernel density estimator.

Pointwise Confidence Band

However, the central limit theorem implies that

$$\sqrt{n}\left\{\hat{f}_h(x) - \mathrm{E}\left[\hat{f}_h(x)\right]\right\} \xrightarrow{d} N\left(0, \mathrm{Var}\left[\frac{1}{h}K\left(\frac{x-X}{h}\right)\right]\right).$$

We can show that

$$\mathrm{Var}\left[\frac{1}{h}K\left(\frac{x-X}{h}\right)\right] \approx \frac{1}{h}f(x)\|K\|_2^2.$$

Hence, an asymptotic interval for $\mathrm{E}\left[\hat{f}_h(x)\right]$ is

$$\hat{f}_h(x) \pm \lambda_{1-\alpha/2}\sqrt{\frac{1}{nh}\hat{f}_h(x)\|K\|_2^2},$$

the same interval as the ad-hoc solution of the asymptotic interval for $f(x)$.

Bootstrap Confidence Band

We can also bootstrap to construct confidence band if we don't want to rely on asymptotic normality.

- Suppose that nonparametric bootstrap is used to create B bootstrap samples.
- For each bootstrap sample, we apply kernel density estimation and obtain $\hat{f}^{(b)}(x)$.
- The bootstrap confidence interval methods can be used to construct a bootstrap confidence interval using the bootstrap replicates $\hat{f}^{(1)}(x)$, ..., $\hat{f}^{(B)}(x)$.

However, the bootstrap confidence interval is still a confidence interval for $\mathrm{E}\left[\hat{f}(x)\right]$.

Bins

It is easy to generalize the histogram for uni-dimension to a histogram for multi-dimension.

- The bin for the univariate case

$$B_j(x_0, h) = [x_0 + (j-1)h,\ x_0 + jh],$$

  is simply an interval.

- The bin for the bivariate case can be a rectangle such as

$$B_j(x_{10}, h_1) \times B_k(x_{20}, h_2).$$

- In general, we have a "rectangular grid" with length h_t in the tth coordinate.

Histogram

Consider $x \in \mathbb{R}^d$. Suppose that each bin B_j is of the form

$$B_{j_1}(x_{10}, h_1) \times \cdots \times B_{j_d}(x_{d0}, h_d).$$

For $x \in B_j$, we still approximate $f(x)$ by

$$\frac{1}{\prod_{k=1}^d h_k} \mathrm{P}(X \in B_j).$$

Then, the histogram becomes

$$\begin{aligned} \hat{f}_h(x) &= \sum_j 1(x \in B_j) \frac{1}{\prod_{k=1}^d h_k} \left[\sum_{i=1}^n \frac{1}{n} 1(X_i \in B_j) \right]. \\ &= \frac{1}{n \prod_{k=1}^d h_k} \sum_{i=1}^n \sum_j 1(X_i \in B_j) 1(x \in B_j). \end{aligned}$$

Introducing Kernel Function

An alternative to the bins in histogram is to make the interval along each coordinate centered around the corresponding element as

$$\left[x_1 - \frac{h_1}{2}, x_1 + \frac{h_1}{2}\right] \times \cdots \times \left[x_d - \frac{h_d}{2}, x_d + \frac{h_d}{2}\right].$$

Then, we can approximate the density by

$$\begin{aligned} f(x) &\approx \frac{\mathrm{P}(X \in B(x))}{h_1 \cdots h_d} \\ &= \frac{1}{h_1 \cdots h_d} \int_{x_1 - \frac{h_1}{2}}^{x_1 + \frac{h_1}{2}} \cdots \int_{x_d - \frac{h_d}{2}}^{x_d + \frac{h_d}{2}} f(u)\, du_1 \cdots du_d \\ &= \int \frac{1}{h_1 \cdots h_d} f(u)\, K\left(\frac{x_1 - u_1}{h_1}, \cdots, \frac{x_d - X_d}{h_d}\right) du_1 \cdots du_d, \end{aligned}$$

where $K(\cdot)$ is the density of a uniform distribution on $B(x)$.

Multivariate Kernel density

It means that we can generalize the kernel density estimation technique to the multidimensional case. But now the kernel function operates on d arguments such as

$$K\left(\frac{x_1 - X_{i1}}{h_1}, \cdots, \frac{x_d - X_{id}}{h_d}\right).$$

In such a case, the kernel density estimator is

$$\hat{f}_h(x) = \frac{1}{n\prod_{k=1}^{d} h_k} \sum_{i=1}^{n} K\left(\frac{x_1 - X_{i1}}{h_1}, \cdots, \frac{x_d - X_{id}}{h_d}\right),$$

where $h = \begin{bmatrix} h_1 & \cdots & h_d \end{bmatrix}$ is the bandwidth vector.

Specify Kernel Function

1. The easiest way to specify a multidimensional kernel is

$$K(u) = \prod_{j=1}^{d} K_1(u_j),$$

where each $K_1(u_j)$ is the unidimensional kernel.

2. We can also use a multivariate kernel such that it is not multiplicative:

$$K(u) \propto K_1\left(u^T u\right),$$

where $\int K(u)\,du = 1$. For example,

$$\text{Epanechnikov}: \quad K(u) \propto \left(1 - u^T u\right) I\left(u^T u \leq 1\right).$$

$$\text{Gaussian}: \quad K(u) \propto \exp\left(-\frac{u^T u}{2}\right).$$

Bandwidth Matrix

A general approach is to use a bandwidth matrix H. The multivariate density estimator is

$$\hat{f}_H(x) = \frac{1}{n \det(H)} \sum_{i=1}^{n} K\left[H^{-1}(x - X_i)\right],$$

where H is a symmetric and positive definite matrix, and the kernel function $K()$ is spherical symmetric and satisfy $\int K(u)\, du = 1$.

- Suppose that $H = \text{diag}\{h_j\}$. Then, we obtain the multiplicative kernel.
- Another example, the multivariate Gaussian kernel yields

$$K\left[H^{-1}(x - X_i)\right] = \frac{1}{(2\pi)^{d/2}} \exp\left(-\frac{(x - X_i)^T H^{-2} (x - X_i)}{2}\right).$$

Bias and Variance

The derivations of bias and variance are similar to the univariate case. Let $\mu_2(K) = \int u_j^2 K(u)\, du$ for all $j = 1, ..., d$, because of spherical symmetry. Then

$$\begin{aligned} \mathrm{E}\left[\hat{f}_H(x)\right] - f(x) &= \frac{1}{2}\mu_2(K)\,\mathrm{tr}\left[H\frac{\partial^2 f(x)}{\partial x \partial x^T}H\right] + o\left(\mathrm{tr}\left[H^2\right]\right), \\ \mathrm{Var}\left[\hat{f}_H(x)\right] &= \frac{1}{n\det(H)} f(x)\,\|K\|_2^2 + o\left(\frac{1}{n\det(H)}\right), \end{aligned}$$

as $H \to 0$ and $n\det(H) \to \infty$, where $\|K\|_2^2 = \int K^2(u)\, du$.

MSE and AMISE

Hence, the leading term in MSE is

$$\text{MSE}\left(\hat{f}_H(x)\right) \approx \frac{f(x)\|K\|_2^2}{n\det(H)} + \frac{1}{4}\mu_2^2(K)\,\text{tr}^2\left[H\frac{\partial^2 f(x)}{\partial x \partial x^T}H\right].$$

The AMISE is

$$\text{AIMSE}\left(\hat{f}_H(x)\right) = \frac{\|K\|_2^2}{n\det(H)} + \frac{1}{4}\mu_2^2(K)\int \text{tr}^2\left[H\frac{\partial^2 f(x)}{\partial x \partial x^T}H\right]dx.$$

The AMISE optimal bandwidth matrix minimizes $\text{AIMSE}\left(\hat{f}_H(x)\right)$.

Curse of Dimensionality

For simplicity, suppose that $H = hI_d$, where I_d is a $d \times d$ identity matrix. Then, the AMISE optimal bandwidth matrix is

$$h_0 = \left[\frac{d \|K\|_2^2}{n\mu_2^2(K) \int \mathrm{tr}^2 \left[\frac{\partial^2 f(x)}{\partial x \partial x^T} \right] dx} \right]^{1/(d+4)} = O\left(n^{-1/(d+4)}\right).$$

The convergences rate is much slower than the rate in the unidimensional case $h_0 = O\left(n^{-1/5}\right)$, especially for large d. Hence, multidimensional density estimation is not reliable for large d.

k-Nearest Neighbor Estimator

Suppose that we have observed a random sample $X_1, ..., X_n \in \mathbb{R}^d$. Instead of considering a bin, we consider a ball $B(x, \rho)$ with the center x and radius $\rho(x)$.

- For $x \in B(x, \rho)$, we still approximate $f(x)$ by

$$\frac{1}{\text{Vol}(B(x, \rho))} \text{P}(X \in B(x, \rho)),$$

where $\text{Vol}(B(x, \rho)) = \frac{\pi^{d/2}}{\Gamma(1+2^{-1}d)} \rho^d$ is the volume of $B(x, \rho)$.

- The k-nearest neighbor (kNN) estimator is

$$\begin{aligned} \hat{f}_k(x) &= \frac{1}{n\text{Vol}(B(x, R_k(x)))} \left[\sum_{i=1}^{n} \frac{1}{n} 1(X_i \in B(x, R_k(x))) \right], \\ &= \frac{k}{n\text{Vol}(B(x, R_k(x)))}, \end{aligned}$$

where $R_k(x)$ be the distance of x to its kth nearest point.

Nearest Neighbor Estimator

Using the kernel function, the k-nearest neighbor (kNN) estimator becomes

$$\hat{f}_k(x) = \frac{1}{nR_k^d(x)} \sum_{i=1}^{n} K\left(\frac{x - X_i}{R_k(x)}\right),$$

where $K()$ is the kernel function.

In contrast, the histogram is

$$\hat{f}_h(x) = \frac{1}{n\prod_{k=1}^{d} h_k} \sum_{i=1}^{n} K\left(\frac{x_1 - X_{i1}}{h_1}, \cdots, \frac{x_d - X_{id}}{h_d}\right),$$

where $K(\cdot)$ is the density of a uniform distribution on a d-dimensional bin, and the kernel density estimator with $H = hI_d$ is

$$\hat{f}_H(x) = \frac{1}{nh^d} \sum_{i=1}^{n} K\left(\frac{x - X_i}{d}\right).$$

Let

- $\hat{\theta} = \hat{\theta}(X)$ be the MLE, where X is the observed data to estimate θ,
- θ^* be the minimizer of

$$\mathrm{KL}(p, g) = \mathrm{E}_p\left[\log\left(\frac{p(X)}{g(X \mid \theta)}\right)\right]$$

with repect to θ,

- x^* is a future observation and

$$R_n = \int p(x^*) \log g\left(x^* \mid \hat{\theta}\right) dx^*.$$

An naive estimator of $\mathrm{E}[R_n]$ is

$$\frac{1}{n}\sum_{i=1}^{n} \log g\left(x_i \mid \hat{\theta}\right).$$

But this is a biased estimator, since X^* should be independent of $\hat{\theta}$, if we assume data are independent.

The Taylor expansion yields

$$\begin{aligned}\frac{1}{n}\sum_{i=1}^{n} \log g\left(x_i \mid \hat{\theta}\right) &= \frac{1}{n}\sum_{i=1}^{n} \log g(x_i \mid \theta^*) + \frac{\partial \frac{1}{n}\sum_{i=1}^{n} \log g(x_i \mid \theta^*)}{\partial \theta^T}\left(\hat{\theta} - \theta^*\right) \\ &\quad + \frac{1}{2}\left(\hat{\theta} - \theta^*\right)^T \underbrace{\frac{\partial^2 \frac{1}{n}\sum_{i=1}^{n} \log g(x_i \mid \theta^*)}{\partial \theta \partial \theta^T}}_{\to -H(\theta^*)}\left(\hat{\theta} - \theta^*\right) + \text{Remainder},\end{aligned}$$

where $H(\theta^*) = -\mathrm{E}\left[\frac{\partial^2 \log g(x^* \mid \theta^*)}{\partial \theta \partial \theta^T}\right]$. Similarly, Taylor expansion yields

$$\log g\left(x^* \mid \hat{\theta}\right) = \log g(x^* \mid \theta^*) + \frac{\partial \log g(x^* \mid \theta^*)}{\partial \theta^T}\left(\hat{\theta} - \theta^*\right) + \frac{1}{2}\left(\hat{\theta} - \theta^*\right)^T \frac{\partial^2 \log g(x^* \mid \theta^*)}{\partial \theta \partial \theta^T}\left(\hat{\theta} - \theta^*\right) + \text{Remainder}.$$

We plug in the expansion to R_n and obtain

$$\begin{aligned}R_n &= \int p(x^*) \log g(x^* \mid \theta^*)\, dx^* + \int p(x^*) \frac{\partial \log g(x^* \mid \theta^*)}{\partial \theta^T} dx^*\left(\hat{\theta} - \theta^*\right) \\ &\quad + \frac{1}{2}\left(\hat{\theta} - \theta^*\right)^T \underbrace{\int p(x^*) \frac{\partial^2 \log g(x^* \mid \theta^*)}{\partial \theta \partial \theta^T} dx^*}_{=-H(\theta^*)}\left(\hat{\theta} - \theta^*\right) + \text{Remainder}.\end{aligned}$$

Since θ^* also minimizes $\int p(x) \log g(x \mid \theta)\, dx$, we should have

$$0 = \frac{\partial \int p(x) \log g(x \mid \theta)\, dx}{\partial \theta} = \int p(x^*) \frac{\partial \log g(x^* \mid \theta^*)}{\partial \theta} dx^*.$$

Thus, we can write R_n as

$$R_n = \int p(x^*) \log g(x^*, \theta^*)\, dx^* - \frac{1}{2n}\sqrt{n}\left(\hat{\theta} - \theta^*\right)^T H(\theta^*) \sqrt{n}\left(\hat{\theta} - \theta^*\right) + \text{Remainder}.$$

Under some regularity conditions, the MLE $\hat{\theta}$ satisfies

$$V_n = \sqrt{n}\left(\hat{\theta} - \theta^*\right) \approx N\left(0, H^{-1}\mathcal{I}H^{-1}\right),$$

where $\mathcal{I}$ is the Fisher information. Hence, using V_n we obtain

$$\frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \hat{\theta}\right) = \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \theta^*\right) + \frac{1}{\sqrt{n}}\frac{\partial \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \theta^*\right)}{\partial \theta^T}V_n - \frac{1}{2n}V_n^T H V_n + \text{Remainder},$$

and

$$R_n = \int p\left(x^*\right)\log g\left(x^* \mid \theta^*\right)dx^* - \frac{1}{2n}V_n^T H V_n + \text{Remainder}.$$

Thus, the bias becomes

$$\begin{aligned}\mathrm{E}\left(\frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \hat{\theta}\right) - \mathrm{E}\left[R_n\right]\right) &= \underbrace{\mathrm{E}\left\{\frac{1}{n}\sum_{i=1}^{n}\left[\log g\left(x_i \mid \theta^*\right) - \int p\left(x^*\right)\log g\left(x^* \mid \theta^*\right)dx^*\right]\right\}}_{\text{term 1}} \\ &\quad + \frac{1}{\sqrt{n}}\underbrace{\mathrm{E}\left\{\frac{\partial \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \theta^*\right)}{\partial \theta^T}V_n\right\}}_{\text{term 2}} + \text{Remainder}.\end{aligned}$$

1. For the first term, $\int p\left(x^*\right)\log g\left(x^* \mid \theta^*\right)dx^*$ is just the expected value of $\log g\left(x_i \mid \theta^*\right)$. Hence, the expected value of the first term is 0.

2. For the second term, the Taylor expansion yields

$$0 = \frac{\partial \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \hat{\theta}\right)}{\partial \theta} = \frac{\partial \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \theta^*\right)}{\partial \theta} + \frac{\partial^2 \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \theta^*\right)}{\partial \theta \partial \theta^T}\left(\hat{\theta} - \theta^*\right) + \text{Remainder}.$$

   Hence,

$$\begin{aligned}\frac{\partial \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \theta^*\right)}{\partial \theta} &= -\frac{\partial^2 \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \theta^*\right)}{\partial \theta \partial \theta^T}\left(\hat{\theta} - \theta^*\right) + \text{Remainder} \\ &= H\left(\hat{\theta} - \theta^*\right) + \text{Remainder},\end{aligned}$$

   and

$$\begin{aligned}\mathrm{E}\left\{\frac{\partial \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \theta^*\right)}{\partial \theta^T}V_n\right\} &= \mathrm{E}\left\{\left(\hat{\theta} - \theta^*\right)^T H V_n\right\} + \text{Remainder} \\ &= \frac{1}{\sqrt{n}}\mathrm{E}\left\{V_n^T H V_n\right\} + \text{Remainder} \\ &= \frac{1}{\sqrt{n}}\mathrm{E}\left\{\mathrm{tr}\left[V_n^T \frac{\partial^2 \frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \theta^*\right)}{\partial \theta \partial \theta^T}V_n\right]\right\} + \text{Remainder} \\ &= \frac{1}{\sqrt{n}}\mathrm{tr}\left\{H\mathrm{E}\left[V_n V_n^T\right]\right\} + \text{Remainder} \\ &= \frac{1}{\sqrt{n}}\mathrm{tr}\left\{\mathcal{I}H^{-1}\right\} + \text{Remainder}.\end{aligned}$$

This means that the bias is

$$\mathrm{E}\left(\frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i \mid \hat{\theta}\right) - \mathrm{E}\left[R_n\right]\right) = \frac{1}{n}\mathrm{tr}\left\{\mathcal{I}H^{-1}\right\} + \text{Remainder}.$$

A bias corrected estimator is

$$\frac{1}{n}\sum_{i=1}^{n}\log g\left(x_i, \hat{\theta}\right) - \frac{1}{n}\mathrm{tr}\left\{\mathcal{I}H^{-1}\right\}.$$